

SS3 TECHNICAL DRAWING LESSON NOTES

FIRST TERM

WEEK 1: ENGINEERING DESIGNS AND WORKING DRAWING

1.1 INTRODUCTION: THE LANGUAGE OF ENGINEERING

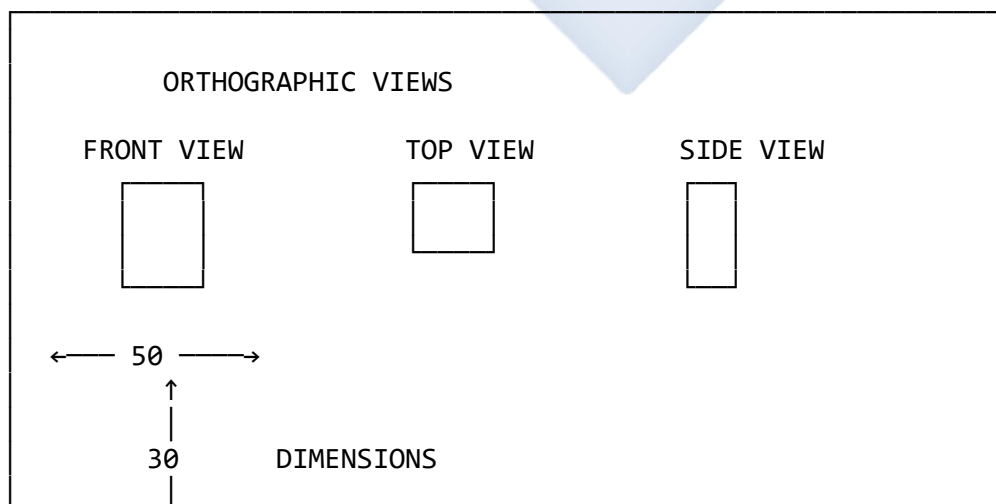
Technical Drawing is the universal language used by engineers, designers, technicians, and craftspeople to communicate ideas and information precisely and unambiguously. An **Engineering Design** is the systematic process of identifying a problem, conceptualizing a solution, and refining that solution into a viable, manufacturable product. The **Working Drawing** is the final result of that design process—a complete and detailed set of instructions on how to manufacture, build, or assemble a component or system.

Think of it this way: If you wanted to build a house in Japan but didn't speak Japanese, technical drawings would be your translator. A well-executed drawing transcends language barriers.

1.2 WHAT IS A WORKING DRAWING?

A **working drawing** (also called a "blueprint" or "manufacturing drawing") provides all the necessary information to manufacture a single part or assemble a complete machine. It must be **clear, concise, and complete**, leaving no room for guesswork.

Diagram 1: Anatomy of a Working Drawing





NOTES:

- Material: Mild Steel
- Finish: Smooth all surfaces
- Drill and tap M10 hole

TITLE BLOCK

Part Name: MOUNTING BRACKET

Part No: MB-001

Scale: 1:1

Material: MILD STEEL

Date: 10/11/2025

Drawn by: W. THOMAS

Sheet: 1 of 1

1.3 COMPONENTS OF A WORKING DRAWING (DETAILED)

1. Orthographic Views

Purpose: To show the exact 2D shape of the object from different directions.

Standard:

- **First Angle Projection** (common in Europe, Nigeria, and most of the world)
- **Third Angle Projection** (common in USA, Canada)

Diagram 2: First Angle vs Third Angle Projection

FIRST ANGLE PROJECTION
(Nigeria Standard)

Symbol:



Layout:

TOP



SIDE → FRONT

THIRD ANGLE PROJECTION
(USA Standard)

Symbol:



Layout:

FRONT



FRONT ← SIDE



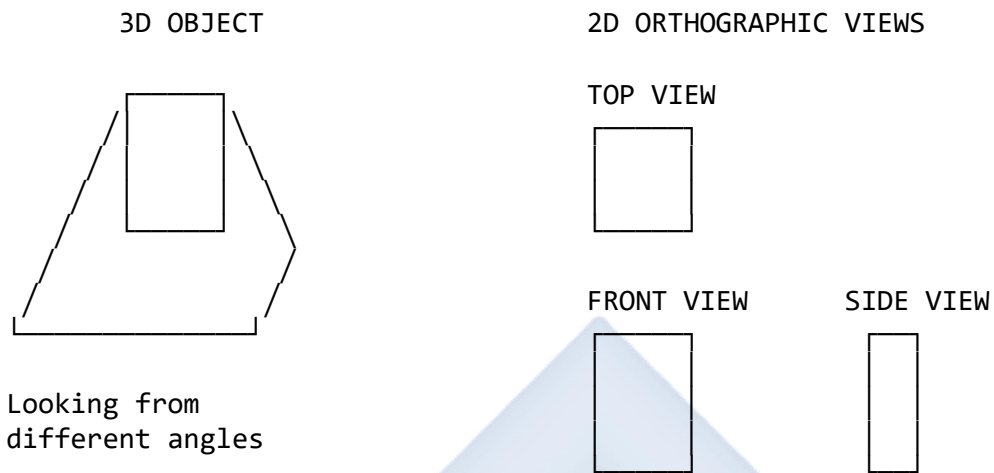
TOP

Main Views:

- **Front View (Elevation):** Shows the most descriptive features
- **Top View (Plan):** Shows the object as seen from directly above

- **Side View (Left or Right):** Shows the object from the side

Diagram 3: Understanding Orthographic Views

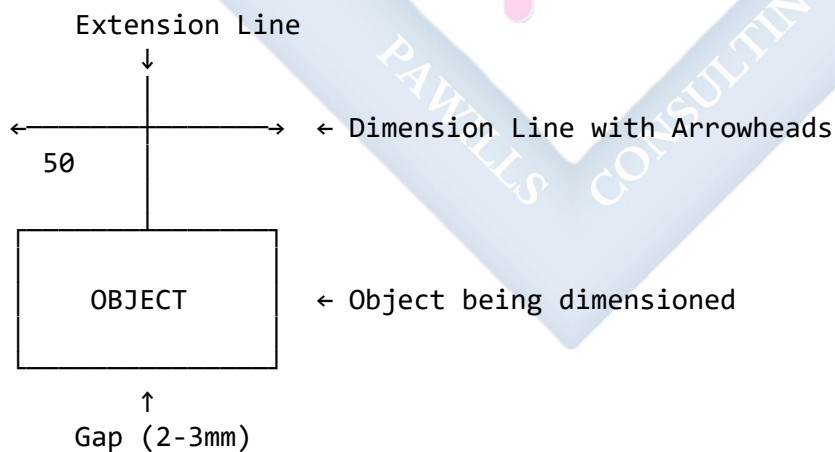


2. Dimensioning

Purpose: To specify the exact size and location of all features.

Elements:

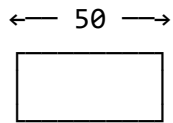
Diagram 4: Dimensioning Components



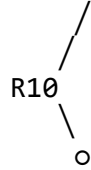
Types of Dimensions:

Diagram 5: Dimension Types

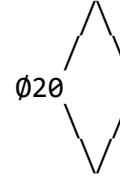
LINEAR DIMENSION



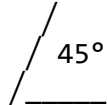
RADIAL DIMENSION



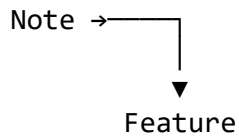
DIAMETRAL DIMENSION



ANGULAR DIMENSION



LEADER LINE



Dimensioning Rules:

1. **Don't duplicate dimensions** - each dimension should appear only once
2. **Dimension from visible outlines** - not from hidden lines
3. **Place dimensions outside the view** - for clarity
4. **Align dimensions** - for neatness
5. **Use standard symbols:**
 - **Ø** = Diameter (Ø20 = 20mm diameter)
 - **R** = Radius (R10 = 10mm radius)
 - **□** = Square (□15 = 15mm square)
 - **SR** = Spherical radius

3. Line Types

Diagram 6: Standard Line Types

LINE TYPE	APPEARANCE	USE
	Thick continuous (0.7mm)	Visible outlines
	Thin chain (0.35mm)	Centre lines
	Thin continuous (0.35mm)	Dimension lines, Extension lines
	Thin dashed (0.35mm)	Hidden details
	Thick chain (0.7mm)	Cutting plane line

4. Title Block

The title block is the "ID card" of the drawing, typically located in the **bottom right corner** of the sheet.

Diagram 7: Standard Title Block Layout

COMPANY NAME/LOGO		
PART NAME: Piston Assembly	PART NUMBER: PA-2025-001	SHEET: 1 of 3
MATERIAL: Cast Iron	SCALE: 1:2	PROJECTION: 1st Angle ✓
DRAWN BY: W. Thomas	CHECKED BY: O. Adeyemi	DATE: 10/11/2025
TOLERANCES: $\pm 0.5\text{mm}$ unless otherwise specified DO NOT SCALE DRAWING		

Key Title Block Information:

- **Part Name:** Clear description (e.g., "Piston," "Mounting Bracket")
- **Part Number:** Unique identifier for inventory/ordering
- **Material:** Exact specification (e.g., "Mild Steel BS 4360," "Aluminium Alloy 6061")
- **Scale:** Drawing ratio (1:1 = full size, 1:2 = half size, 2:1 = double size)
- **Drawn By:** Draughtsman's name/initials
- **Date:** Completion date
- **Tolerances:** Acceptable variation range
- **Projection Type:** First or Third Angle

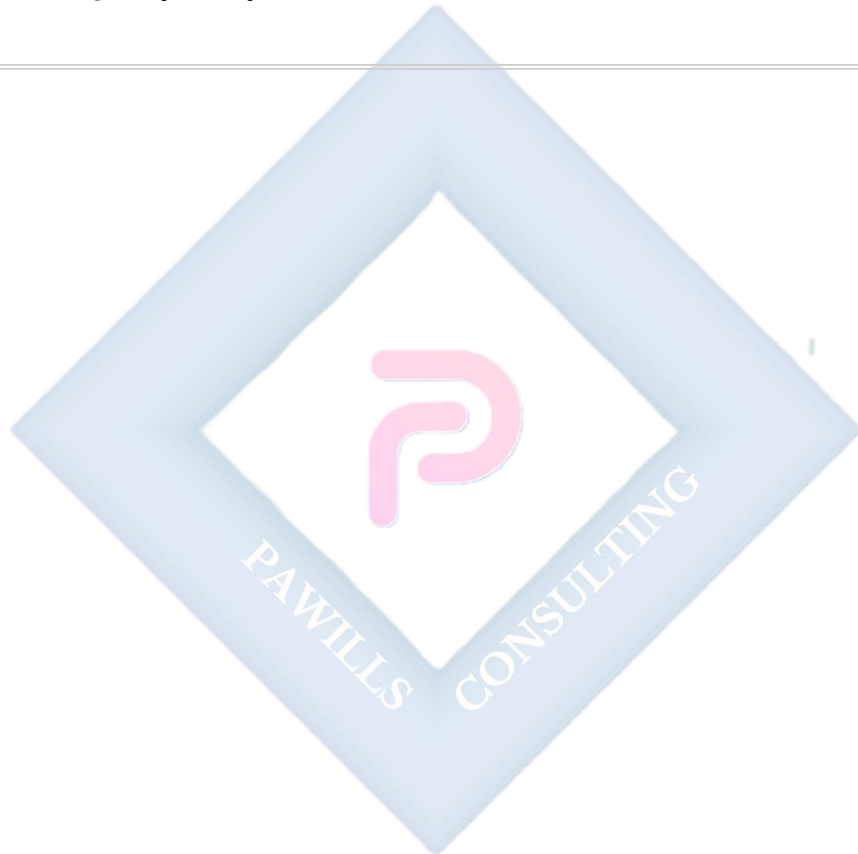
EVALUATION

1. Differentiate between "Engineering Design" and a "Working Drawing."
2. List the **six essential components** of a complete working drawing.
3. Explain the purpose of orthographic views and why multiple views are necessary.
4. What does the symbol " $\varnothing 50$ " mean on a technical drawing?
5. Why must dimensions not be duplicated on a drawing?

ASSIGNMENT

1. On an **A4 sheet** (297mm 210mm), draw:
 - A **10mm border** around the entire sheet

- A **title block** (120mm wide × 50mm high) in the bottom right corner
 - 2. Fill in the title block with the following details:
 - Part Name: "**Spacer**"
 - Part Number: **SP-001**
 - Material: "**Brass**"
 - Scale: "**1:1**"
 - Drawn By: (**Your Name**)
 - Date: (**Today's Date**)
 - Projection: **First Angle**
 - 3. Draw the First Angle Projection symbol in the title block.
-

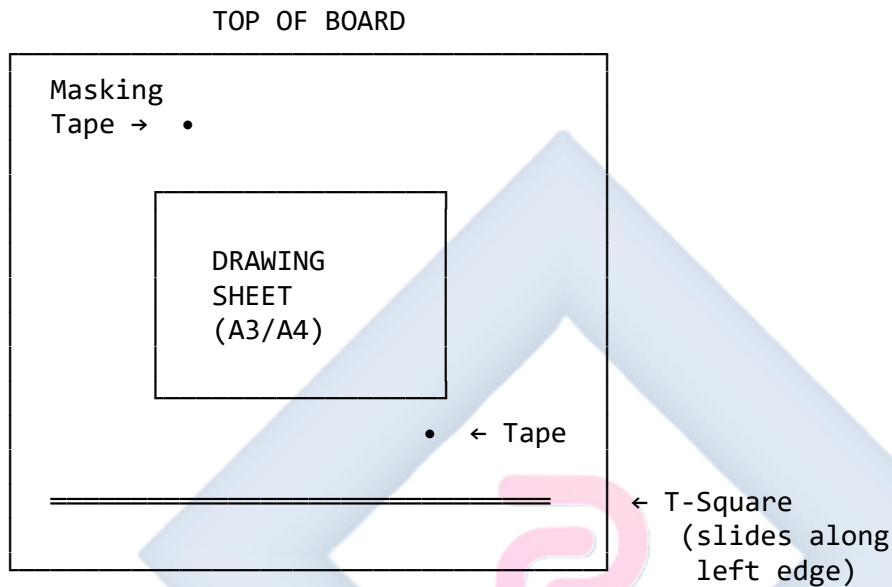


WEEK 2: ENGINEERING DESIGNS AND WORKING DRAWING (CONTINUED)

2.1 PREPARATION ON THE DRAWING TABLE (MANUAL DRAFTING)

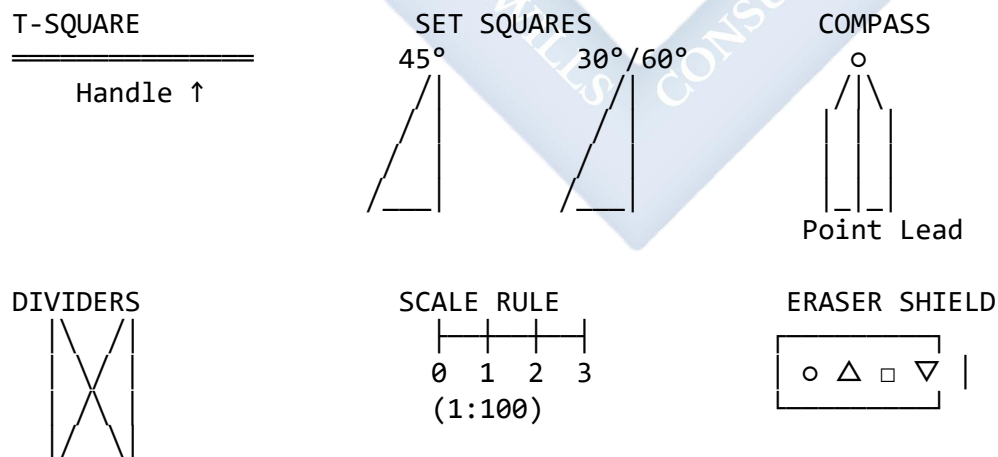
Setting up correctly is 50% of producing a good drawing!

Diagram 8: Drawing Board Setup



Essential Tools:

Diagram 9: Drawing Instruments



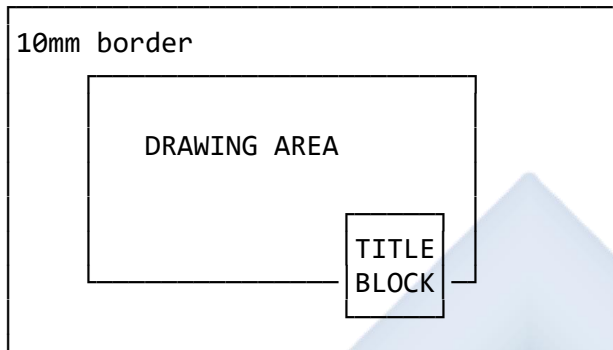
Setup Process:

1. **Clean your board** - dust and dirt will show on your drawing

2. **Align the sheet:**

- Place drawing sheet on board
- Align top edge with T-square
- Use **masking tape** (not cello tape - it leaves residue) at all four corners

3. **Draw border lines:**



4. **Construction technique:**

- Use **2H pencil** for light construction lines
- Use **HB pencil** for final dark outlines
- Keep pencils **sharp** - blunt pencils = poor drawings

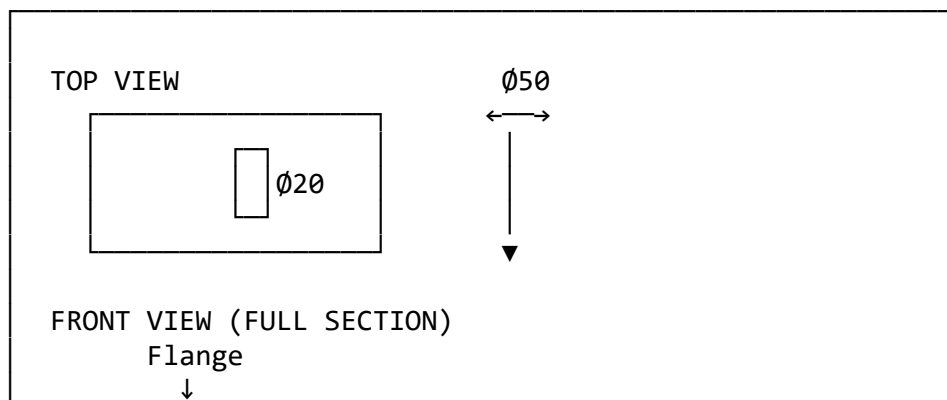
2.2 TYPES OF WORKING DRAWINGS

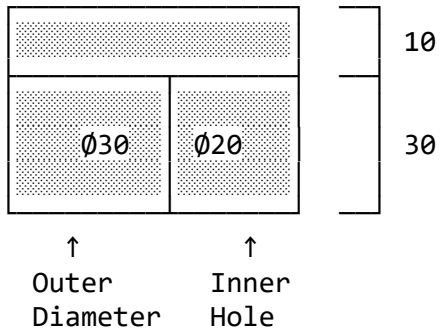
Working drawings fall into **two main categories**: Detail Drawings and Assembly Drawings.

1. *Detail Drawing*

Purpose: Provides ALL information to manufacture **ONE single component**.

Diagram 10: Sample Detail Drawing - Flanged Bush





SECTION A-A
45° hatching shows material in section

NOTES:

1. Material: Bronze
2. All surfaces to be machined smooth
3. Chamfer both ends 1×45°

FLANGED BUSH	DWG NO: FB-001
MATERIAL: BRONZE	SCALE: 1:1
DRAWN: W.THOMAS	DATE: 10/11/25

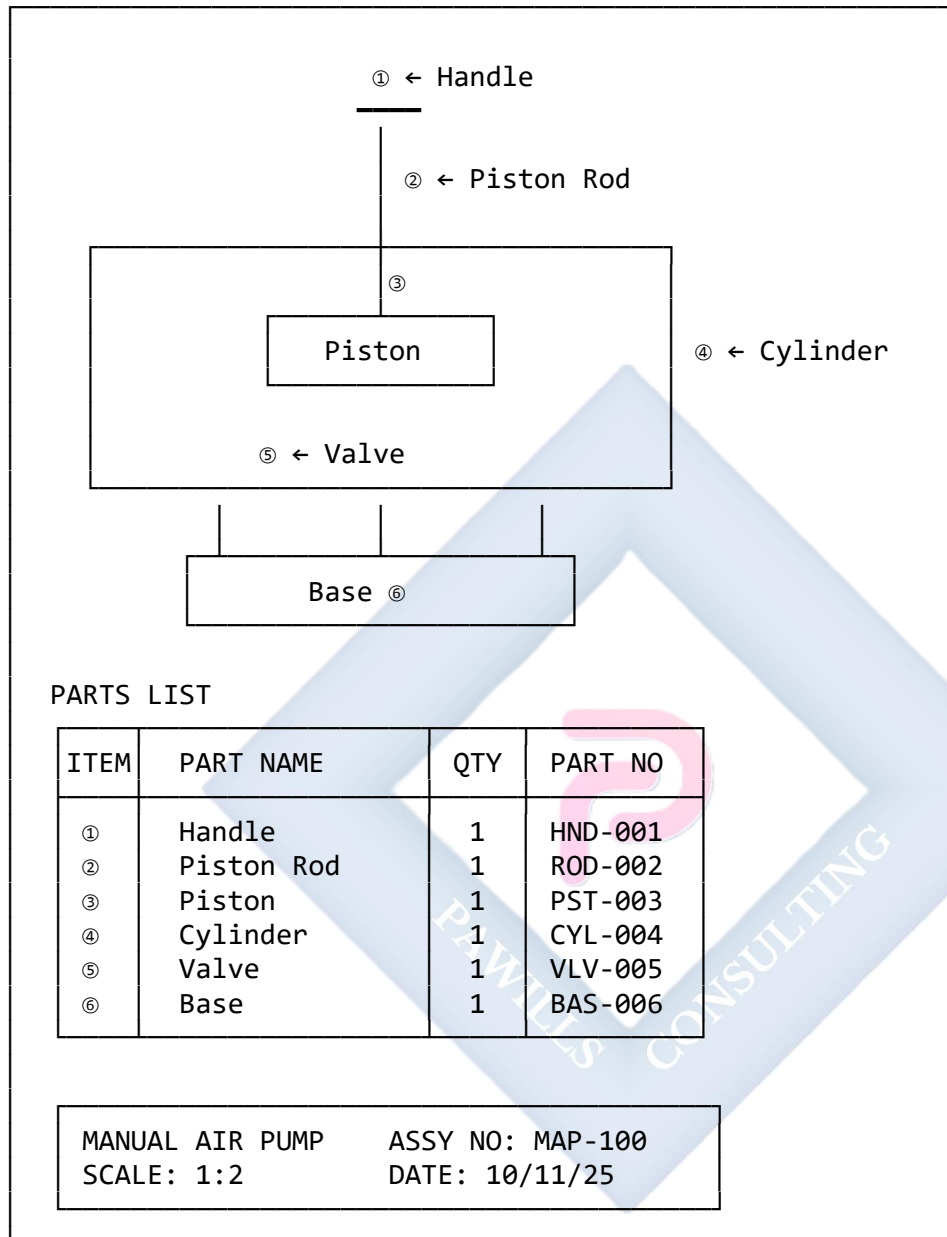
Content of a Detail Drawing:

- ✓ Multiple orthographic views
- ✓ Sectional views (if needed)
- ✓ **COMPLETE dimensions** - every size needed
- ✓ Material specification
- ✓ Surface finish notes
- ✓ Manufacturing processes (drill, tap, weld, etc.)
- ✓ Title block

2. Assembly Drawing

Purpose: Shows how all individual components fit together.

Diagram 11: Sample Assembly Drawing - Manual Air Pump

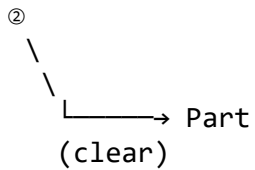


Content of an Assembly Drawing:

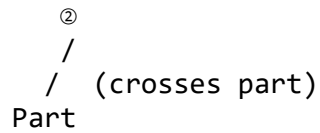
- ✓ **Overall view** of assembled product
- ✗ **NO hidden lines** (would clutter the drawing)
- ✗ **NO detailed dimensions** (only overall/critical dimensions)
- ✓ **Part balloons** (numbers in circles with leader lines)
- ✓ **Parts List (Bill of Materials)** - table listing all parts

Balloon Guidelines:

Good Balloon Placement



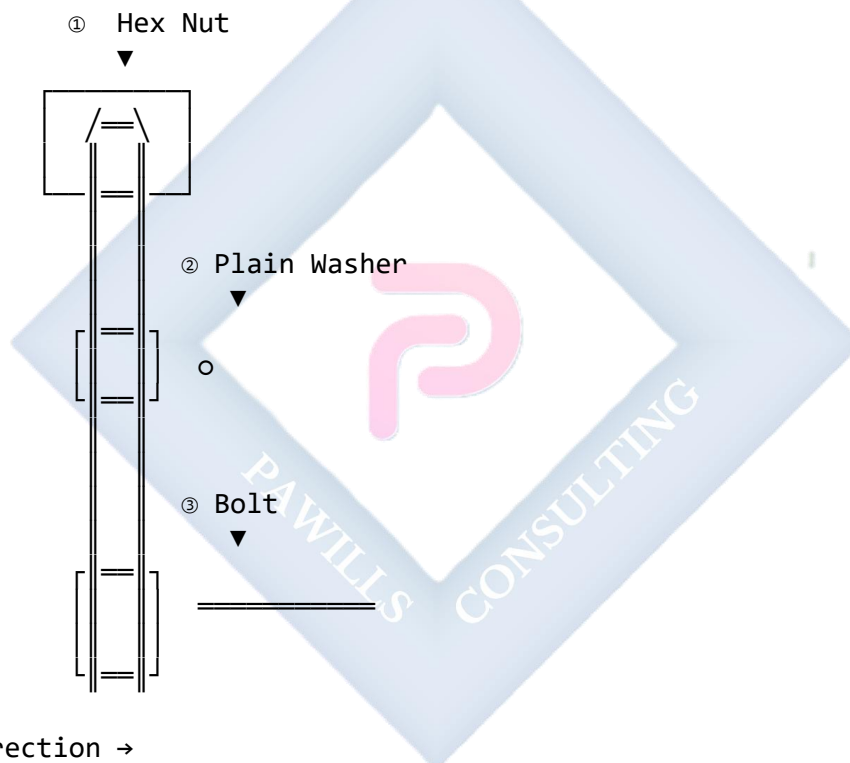
Bad Balloon Placement



2.3 EXPLODED VIEW

An **exploded view** is a special type of assembly drawing where parts are shown "pulled apart" along their assembly axis.

Diagram 12: Exploded View - Bolt and Nut Assembly



EVALUATION

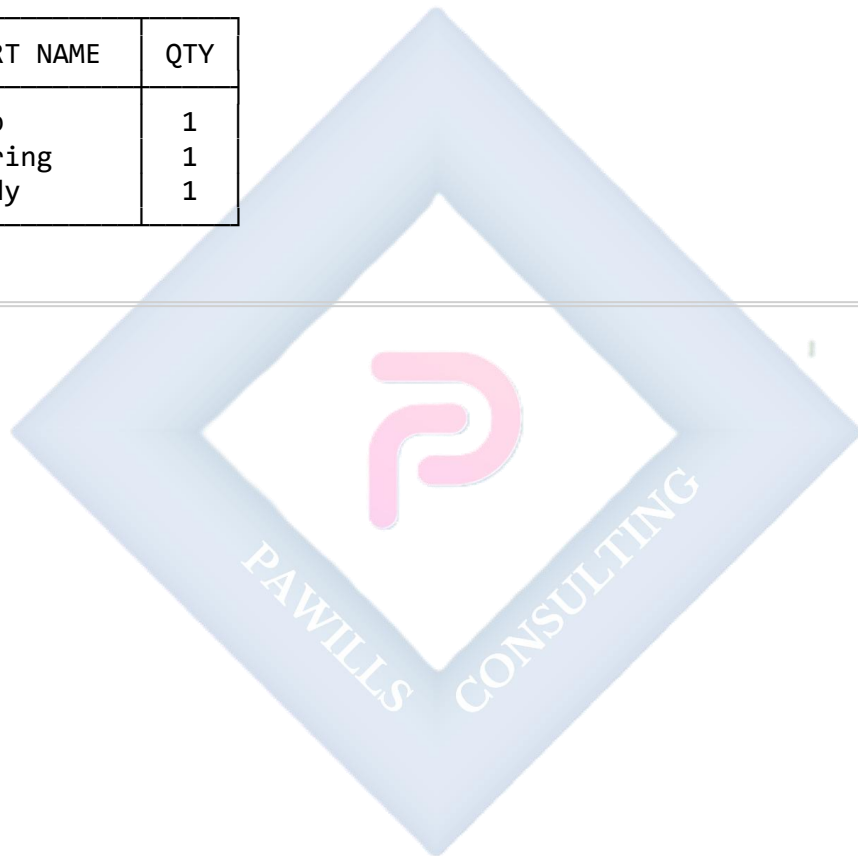
1. What is the primary difference between a "Detail Drawing" and an "Assembly Drawing"?
 2. What are "Part Balloons" and where are they used?
 3. Why are hidden lines usually omitted from assembly drawings?
 4. List **four** pieces of information found in a Parts List (BOM).
 5. In which type of drawing would you find complete dimensions for manufacturing?
-

ASSIGNMENT

1. Find a simple object at home (e.g., stapler, pen, padlock, bicycle pump).
2. **Sketch an exploded view** showing 3-5 main parts separated.
3. Add part balloons (①, ②, ③...) with leader lines.
4. Create a **Parts List table** with columns:
 - Item No.
 - Part Name
 - Quantity (QTY)

Example:

ITEM	PART NAME	QTY
①	Cap	1
②	Spring	1
③	Body	1



WEEK 3: SCREW THREADS, FASTENERS, AND LOCKING DEVICES

3.1 MEANING OF SCREW THREADS

A **screw thread** is a helical (spiral) ridge of uniform cross-section formed on the inside or outside of a cylinder or cone. It converts rotary motion into linear motion.

- **External Thread:** Thread on the outside of a cylinder (e.g., bolt, screw)
- **Internal Thread:** Thread on the inside of a hole (e.g., nut, tapped hole)

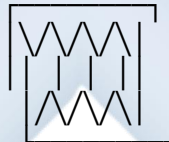
Diagram 13: External vs Internal Threads

EXTERNAL THREAD (Bolt)



Thread cut on
outside surface

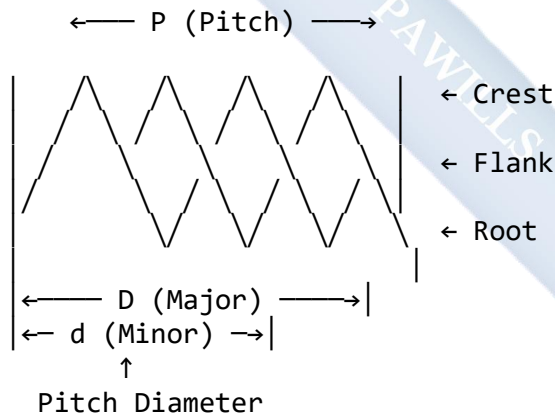
INTERNAL THREAD (Nut)



Thread cut on
inside surface

3.2 TERMINOLOGY OF SCREW THREADS

Diagram 14: Thread Profile and Terminology



Key Terms:

1. **Crest:** Top surface (highest point) of thread
2. **Root:** Bottom surface (lowest point) of thread
3. **Flank:** Sloped side connecting crest and root
4. **Major Diameter (D):** Largest diameter (crest to crest)
5. **Minor Diameter (d):** Smallest diameter (root to root)

6. **Pitch Diameter:** Effective diameter (midway between major and minor)
7. **Pitch (P):** Distance from one crest to the next crest
8. **Lead (L):** Distance nut advances in one complete rotation (360°)

Types by Start:

- **Single-Start Thread:** Lead = Pitch (most common)
- **Multi-Start Thread:** Lead = Pitch $\times$ Number of Starts
 - Used for fast action (bottle caps, ballpoint pens)

Types by Direction:

- **Right-Hand Thread:** Advances when turned clockwise (standard)
- **Left-Hand Thread:** Advances when turned counter-clockwise (special applications)

Diagram 15: Single-Start vs Double-Start Thread

SINGLE-START



One thread
helix
Lead = Pitch

DOUBLE-START



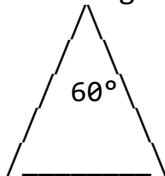
Two parallel
helices
Lead = $2 \times$ Pitch

3.3 COMMON THREAD FORMS (PROFILES)

Diagram 16: Thread Profile Types

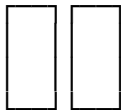
1. V-THREAD (ISO METRIC/UNIFIED)

60° angle



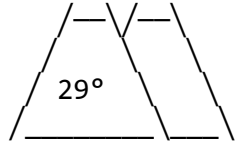
Most common
General fasteners
(bolts, screws)

2. SQUARE THREAD



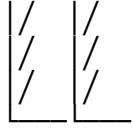
Power transmission
High efficiency
Hard to manufacture
(vice screws, lathes)

3. ACME THREAD



Trapezoidal
Stronger than square
(car jacks, clamps)

4. BUTTRESS THREAD



One-direction force
Very high load capacity
(gun breech, presses)

↑
Loading
direction

5. KNUCKLE THREAD



Rough service
Easy to cast
(light bulbs, bottles)

Uses Summary:

Thread Type	Primary Use	Key Feature
V-Thread	General fastening	Universal standard
Square	Power transmission	Maximum efficiency
Acme	Heavy-duty transmission	Stronger than square
Buttress	One-direction loads	Asymmetric profile
Knuckle	Rough service	Easy to manufacture

3.4 DRAWING REPRESENTATION OF THREADS

Drawing actual helical threads is time-consuming. We use **simplified conventions**.

Diagram 17: Thread Representation Methods

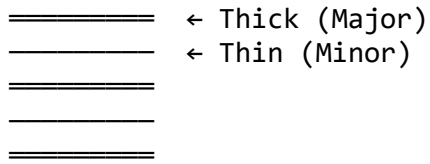
1. DETAILED REPRESENTATION (Rarely used - large threads only)



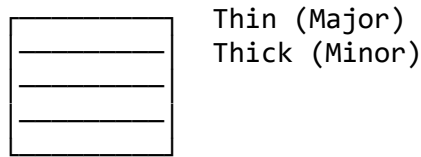
Shows actual
V-shape profile

2. SCHEMATIC REPRESENTATION (Most common)

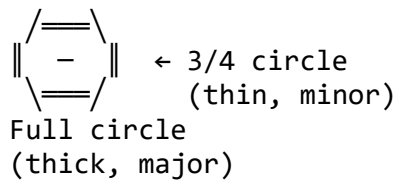
EXTERNAL THREAD



INTERNAL THREAD (Section)



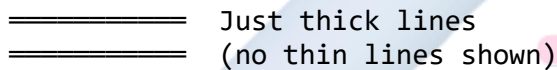
END VIEW OF BOLT



END VIEW OF NUT

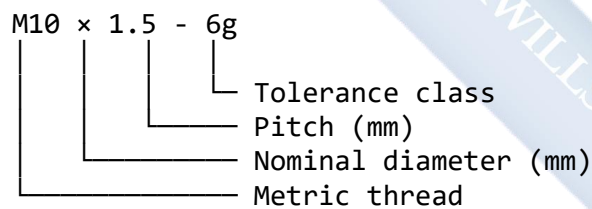


3. SIMPLIFIED REPRESENTATION (Very simple)



3.5 STANDARD THREAD DESIGNATIONS

ISO Metric Thread:



Examples:

- **M6** = 6mm diameter, standard (coarse) pitch
- **M10 × 1.5** = 10mm diameter, 1.5mm pitch (fine)
- **M20** = 20mm diameter, standard pitch

EVALUATION

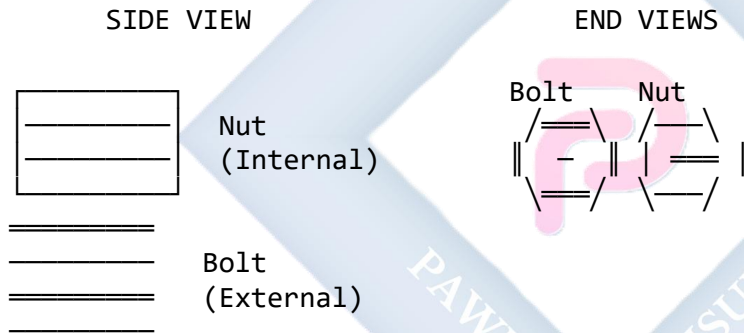
1. Define "Pitch" and "Lead." How are they related in a single-start thread?
2. Sketch the profile of a **Square thread** and a **Buttress thread**. Label key features.
3. What is the main difference between a right-hand and a left-hand thread?

4. What is the **major diameter** of a screw thread, and where is it measured?
5. Why is the Square thread used for power transmission rather than V-thread?
6. Explain why we use schematic representation instead of drawing actual helical threads.

ASSIGNMENT

1. **Sketch a simple bolt and nut assembly** (side view).
2. Using **schematic representation**, show:
 - External threads on the bolt (use thick and thin parallel lines)
 - Internal threads on the nut in section
 - Label: Major Diameter, Minor Diameter, Pitch
3. Draw the **end view** of:
 - The bolt (showing major and minor diameter circles)
 - The nut (showing hexagon with thread circles inside)

Example format:



WEEK 4: SCREW THREADS, FASTENERS, AND LOCKING DEVICES (CONTINUED)

4.1 MEANING OF FASTENERS

A **fastener** is any hardware device that mechanically joins or affixes two or more objects together.

Classification:

- **Permanent Fasteners:** Cannot be easily removed (rivets, welds)
- **Non-Permanent (Removable) Fasteners:** Can be disassembled (bolts, screws)

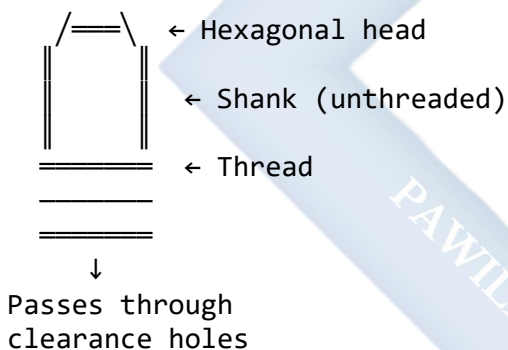
4.2 IDENTIFICATION AND SKETCHING OF FASTENERS

1. BOLTS

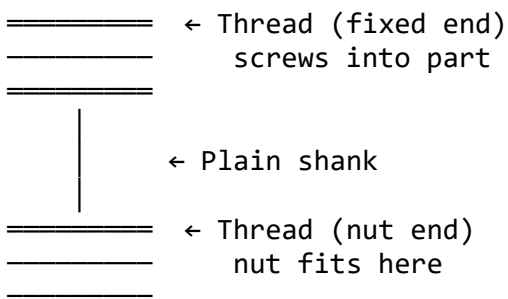
A bolt passes through holes in both parts and uses a nut to apply clamping force.

Diagram 18: Types of Bolts

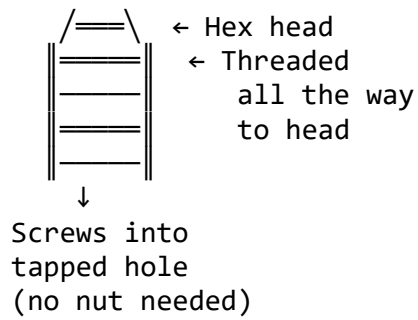
A. HEX HEAD BOLT (Most Common)



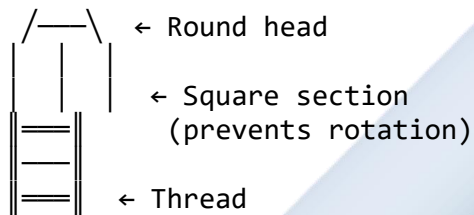
B. STUD BOLT



C. TAP BOLT (MACHINE SCREW)



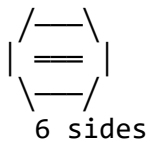
D. CARRIAGE BOLT



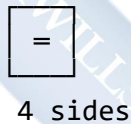
2. NUTS

Diagram 19: Types of Nuts

A. HEXAGONAL NUT (Most common)



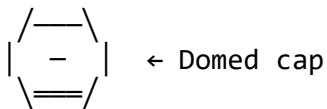
B. SQUARE NUT (Heavy duty)



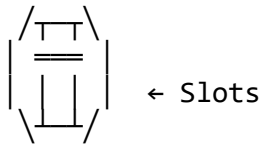
C. WING NUT (Hand tightening)



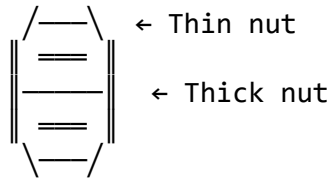
D. CAP NUT (ACORN NUT) (Decorative/protective)



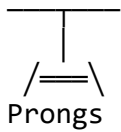
E. CASTLE NUT
(With split pin)



F. LOCK NUT (JAM NUT)
(Double nut system)



G. T-NUT
(Furniture/wood)



H. DOME NUT
(Smooth finish)



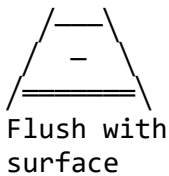
3. SCREWS

Screws create their own threads or fit into pre-tapped holes.

Diagram 20: Types of Screws and Head Styles

HEAD STYLES:

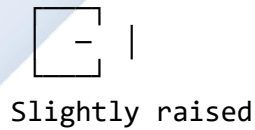
A. FLAT HEAD
(Countersunk)



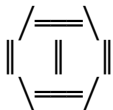
B. ROUND HEAD
(Raised)



C. PAN HEAD
(Low profile)



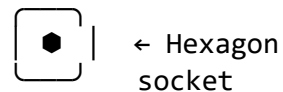
D. HEX HEAD



E. BUTTON HEAD
(Domed)



F. SOCKET HEAD
(Allen key)

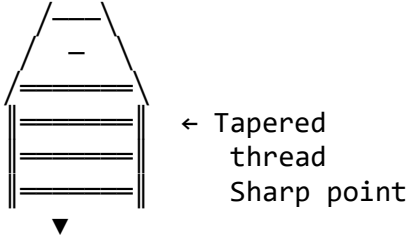


DRIVE TYPES:

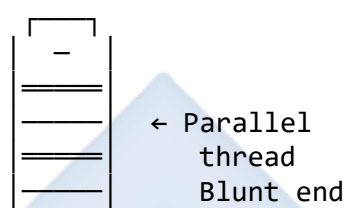
— +   
Slot Phillips Hex Pozidriv Torx

SCREW TYPES:

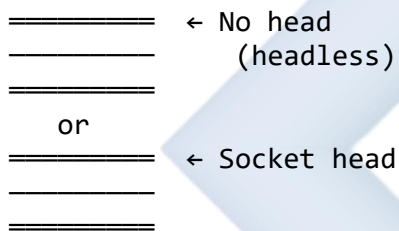
A. WOOD SCREW



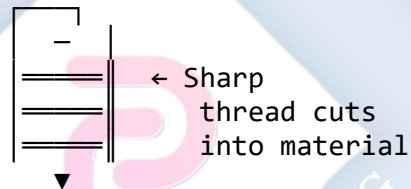
B. MACHINE SCREW



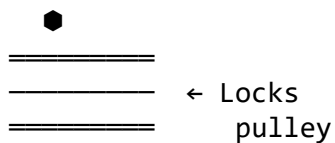
C. SET SCREW



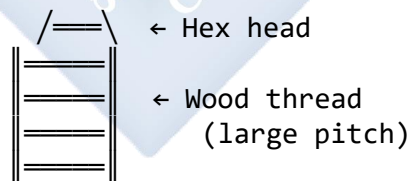
D. SELF-TAPPING SCREW



E. GRUB SCREW (Same as set screw)



F. COACH SCREW (Heavy wood)

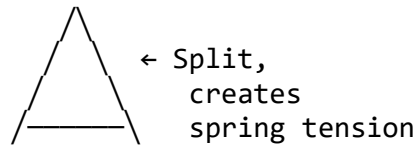
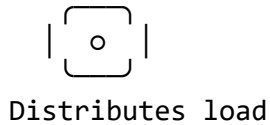


4. WASHERS

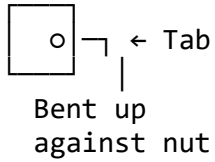
Diagram 21: Types of Washers

A. PLAIN WASHER (Flat washer)

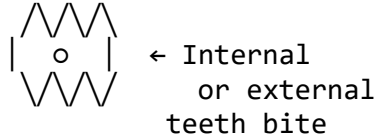
B. SPRING WASHER (Lock washer)



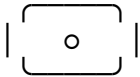
C. TAB WASHER



D. TOOTHED WASHER



E. FENDER WASHER (Extra large OD)



F. SQUARE WASHER (Timber/metal)

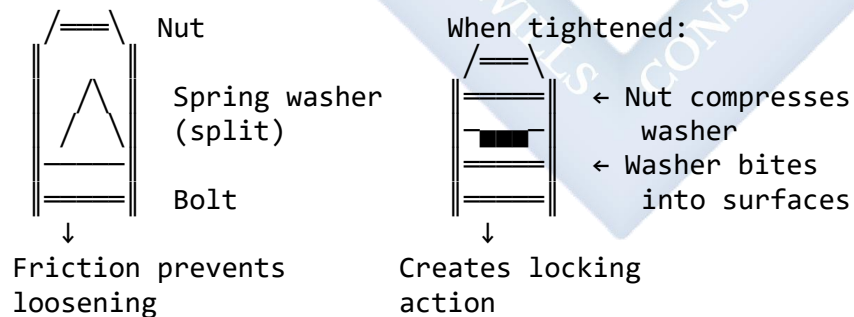


4.3 LOCKING DEVICES

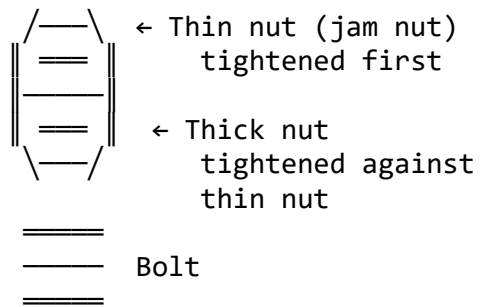
Vibration can loosen fasteners. Locking devices prevent this.

Diagram 22: Locking Device Methods

1. SPRING WASHER METHOD

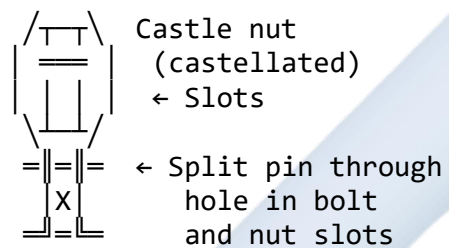


2. LOCK NUT METHOD (Double Nut)



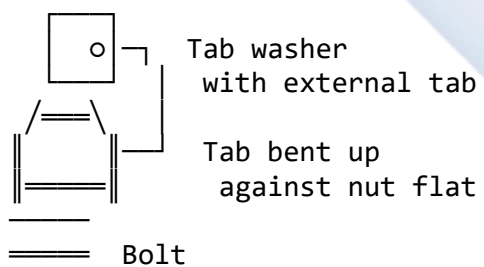
↓
Nuts jam together

3. CASTLE NUT AND SPLIT PIN



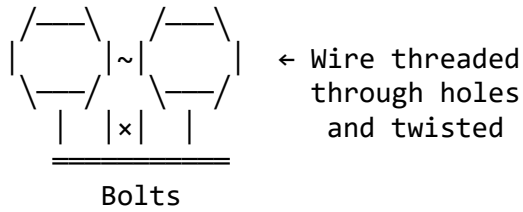
↓
Pin bent over to lock

4. TAB WASHER METHOD



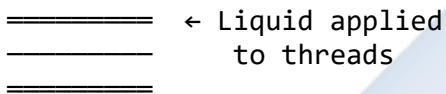
↓
Prevents rotation

5. LOCK WIRE (SAFETY WIRE)



↓
Prevents loosening
(aircraft/motorcycles)

6. THREAD LOCKING COMPOUND



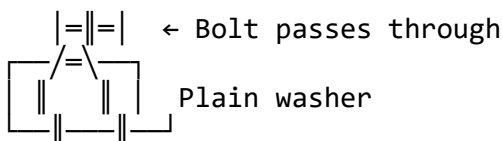
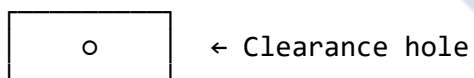
↓
Chemical bond
prevents rotation
(e.g., Loctite)

4.4 PRACTICAL APPLICATIONS

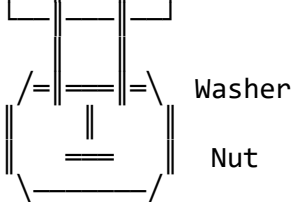
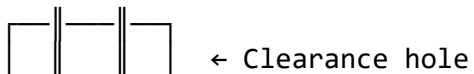
Diagram 23: Common Fastener Applications

A. THROUGH BOLT ASSEMBLY (Two parts with clearance holes)

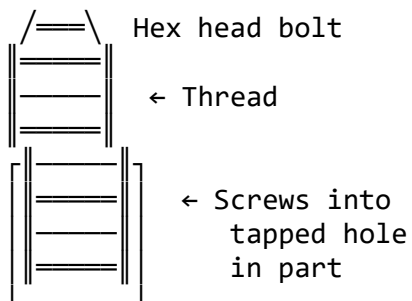
Part 1



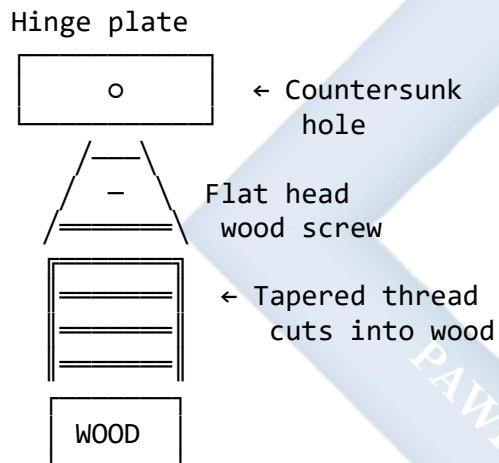
Part 2



B. TAP BOLT ASSEMBLY
(One tapped hole, no nut)



C. WOOD SCREW ASSEMBLY
(Door hinge to wood)



EVALUATION

1. What is the difference between a "bolt" and a "screw"?
 2. Sketch: (a) hex nut, (b) wing nut, (c) castle nut
 3. Explain the function of a **plain washer**.
 4. Describe how a **castle nut and split pin** work together to lock a nut.
 5. What is a **stud bolt** and where is it commonly used?
 6. Why would you use a **lock nut** instead of a single nut?
 7. Sketch a **wood screw** and label: head, shank, thread, point
-

ASSIGNMENT

1. **Draw a Through Bolt Assembly** (side view in section), showing:

- Two wooden blocks being joined
- Hex bolt passing through
- Plain washer under nut
- Hex nut
- Label all parts

2. **Sketch a Wood Screw Application** showing:

- Door hinge plate
- Two flat head wood screws
- Cross-section view through screw and wood
- Label parts

3. **Create a comparison table:**

FASTENER TYPE	REQUIRES NUT?	TYPICAL USE
Bolt Machine Screw Wood Screw Set Screw Stud		

WEEK 5: MIDTERM TEST

COVERAGE:

- Working Drawing Principles (Weeks 1-2)
- Detail vs Assembly Drawings
- Thread Terminology and Forms (Week 3)
- Fasteners: Bolts, Nuts, Screws (Week 4)
- Locking Devices and Applications

SAMPLE QUESTIONS:

SECTION A: THEORY (40 Marks)

1. Define the following terms: (8 marks)
 - a) Working Drawing
 - b) Orthographic Projection
 - c) Detail Drawing
 - d) Parts List (BOM)
2. Sketch and label the parts of a screw thread profile showing: Crest, Root, Flank, Major Diameter, Minor Diameter, Pitch (10 marks)
3. Explain THREE advantages of an assembly drawing over detail drawings when showing how a machine works. (6 marks)
4. Describe how a spring washer acts as a locking device. Include a sketch. (8 marks)
5. List FOUR types of information that must be included in a title block. (4 marks)
6. State TWO differences between a bolt and a screw. (4 marks)

SECTION B: PRACTICAL (60 Marks)

1. Draw the following using schematic representation: (20 marks)
 - a) External thread (side view and end view)
 - b) Internal thread in section (side view and end view)
 - Label: Major Diameter, Minor Diameter
2. Sketch the following fasteners: (15 marks)
 - a) Hex head bolt
 - b) Castle nut
 - c) Wing nut
 - d) Flat head wood screw
 - e) Set screw
3. Draw a simple assembly showing a bolt, plain washer, and nut joining two metal plates. Include part balloons and a parts list. (25 marks)



WEEK 6: ENGINEERING WORKING DRAWING

6.1 READING AND INTERPRETING ENGINEERING DRAWINGS

Reading a drawing is visualizing the 3D object from 2D views. This is a crucial skill!

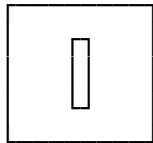
Diagram 24: Systematic Drawing Reading Process

STEP 1: START WITH TITLE BLOCK

What is it?	→ BRACKET
Material?	→ MILD STEEL
Scale?	→ 1:1 (Full size)
How many?	→ QTY: 2



STEP 2: IDENTIFY MAIN VIEW (Front View)

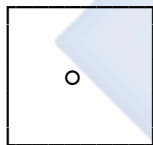


What shapes do I see?

- Rectangle
- Small circle ($\varnothing 10$)



STEP 3: RELATE TO TOP VIEW



Does circle become:

- Full circle = Hole?
- Part of circle = Edge?



STEP 4: CHECK SIDE VIEW



Confirms:

- Thickness
- Depth



STEP 5: FOLLOW HIDDEN LINES



Dashed lines show
internal or
hidden features



STEP 6: READ DIMENSIONS & NOTES

← 50 →



$\varnothing 10$ ← Hole diameter



Drill and tap M10

6.2 DECODING DRAWING SYMBOLS AND ANNOTATIONS

Diagram 25: Common Drawing Symbols

DIAMETER SYMBOLS

$\varnothing 20$
—
Diameter
20mm

RADIUS SYMBOLS

R15
┌
Radius
15mm

SQUARE SYMBOLS

□15
┌
Square
15mm side

SPHERICAL SYMBOLS

SR10
○
Spherical
radius 10mm

COUNTERBORE

□15
┌
Flat bottom
recess

COUNTERSINK

V90°
┌
Tapered hole
for flat head
screw

SPOTFACE

○ SF12
Shallow flat
surface around
hole

DEPTH SYMBOL

┌10
↓15
Hole 10mm dia
15mm deep

THREAD CALLOUT

M10 × 1.5
Metric thread
10mm dia
1.5mm pitch

TAPER SYMBOL

1:10
1 unit drop
per 10 units length

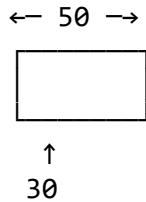
6.3 DIMENSIONING GUIDELINES

Diagram 26: Good vs Bad Dimensioning Practice

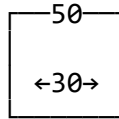
✓ GOOD PRACTICE

✗ BAD PRACTICE

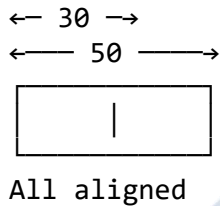
Dimensions outside views



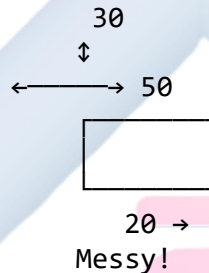
Dimensions crossing parts



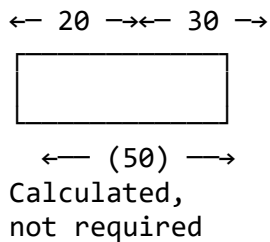
Aligned dimensions



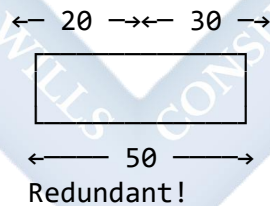
Scattered dimensions



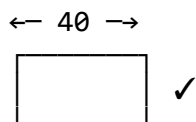
Reference dimensions
(enclosed in brackets)



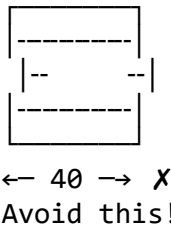
Over-dimensioned



Dimension to visible lines



Dimension to hidden lines

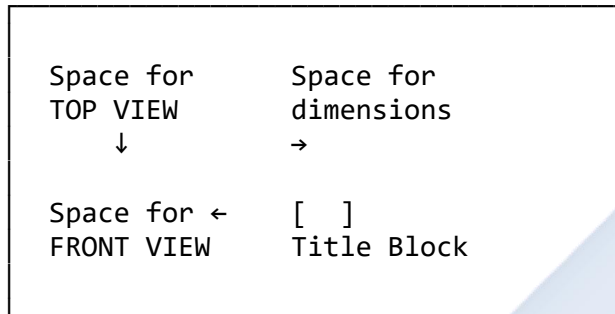


6.4 PREPARING WORKING DRAWINGS - PRACTICAL EXERCISE

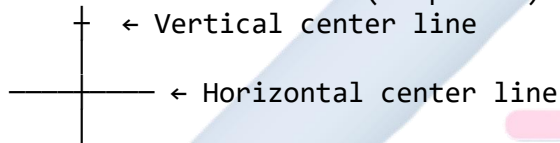
Example: Flanged Bush Detail Drawing

Diagram 27: Step-by-Step Construction

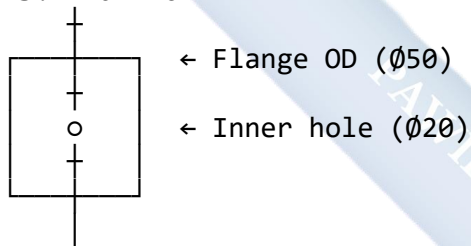
STEP 1: Plan the layout



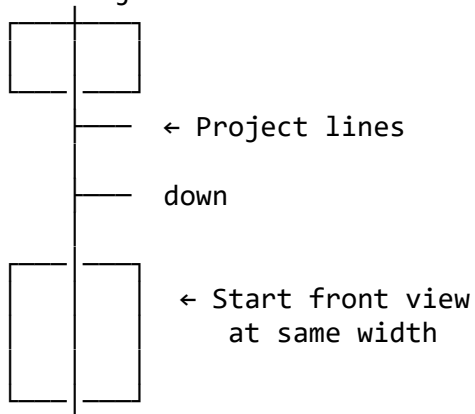
STEP 2: Draw center lines (2H pencil)



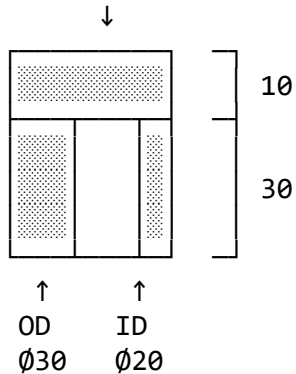
STEP 3: Draw TOP VIEW



STEP 4: Project to FRONT VIEW

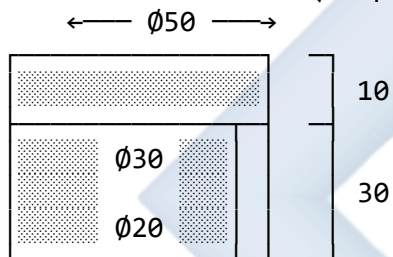


STEP 5: Complete FRONT VIEW (as section)
Flange



45° hatching = SECTION A-A

STEP 6: Add dimensions (HB pencil)



STEP 7: Add notes and title block

NOTES:

1. Material: Bronze
2. All surfaces machined
3. Chamfer 1×45° both ends
4. Remove all burrs

FLANGED BUSH
PART NO: FB-001
SCALE: 1:1 DATE:

6.5 CREATING ASSEMBLY DRAWINGS - PRACTICAL EXERCISE

Example: Simple Pulley Assembly

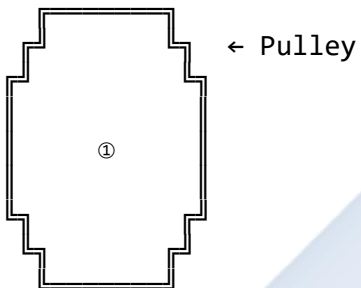
Diagram 28: Assembly Drawing Construction

COMPONENTS NEEDED:

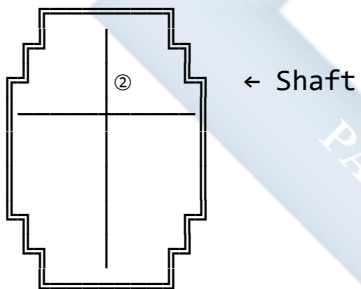
- ① Pulley wheel
- ② Shaft
- ③ Key
- ④ Bearing (x2)
- ⑤ Retaining ring (x2)

ASSEMBLY PROCESS:

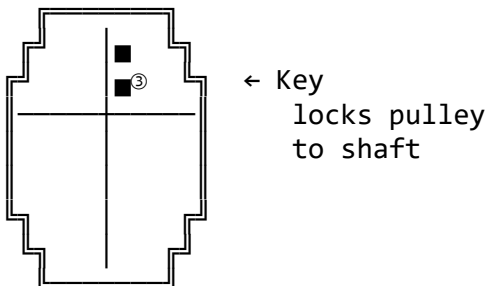
STEP 1: Draw main component (Pulley)



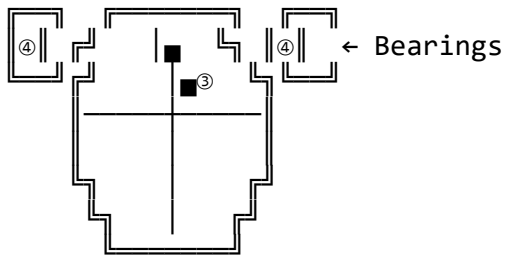
STEP 2: Add shaft through center



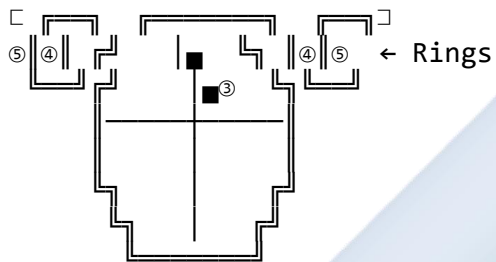
STEP 3: Add key in keyway



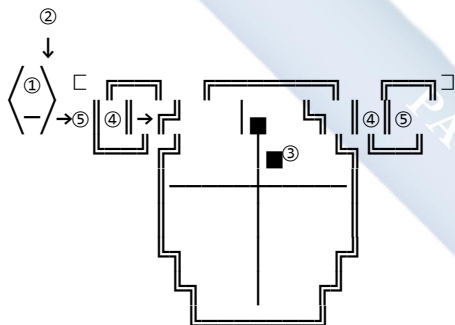
STEP 4: Add bearings on each end



STEP 5: Add retaining rings



STEP 6: Add balloons and parts list



ITEM	PART NAME	QTY
①	Pulley Wheel	1
②	Shaft	1
③	Key	1
④	Ball Bearing	2
⑤	Retaining Ring	2

EVALUATION

1. What is the **first** thing you should look at when reading a new engineering drawing?
 2. Why is it important to **relate** orthographic views to each other?
 3. What is the key difference in dimensioning a **detail drawing** versus an **assembly drawing**?
 4. What does the symbol "Ø25" mean on a drawing?
 5. Why do we use dashed lines in engineering drawings?
 6. What information must be included in a Parts List (BOM)?
-

ASSIGNMENT

Scenario: You are given two detail drawings:

- **Part 1:** Hex Bolt (M10 × 40mm long)
- **Part 2:** Hex Nut (M10)
- **Part 3:** Plain Washer (for M10)

Task:

1. Create a simple **Assembly Drawing** showing:
 - Nut screwed onto bolt
 - Washer between nut and the joined parts
 - Two metal plates being fastened together
 2. Include:
 - Part balloons (①, ②, ③) with leader lines
 - A **Parts List** table
 - Title block with: "BOLT ASSEMBLY," your name, date, scale 1:1
 3. **Do NOT include:**
 - Hidden lines
 - Detailed dimensions (except maybe overall length)
-

WEEK 7: MIDTERM BREAK

This week is for the midterm break. Students should:

- Review all work from Weeks 1-6
 - Practice sketching fasteners and thread representations
 - Organize drawing sheets and instruments for second half of term
-



WEEK 8: SECTIONS AND SECTIONAL VIEWS

8.1 MEANING OF SECTIONING

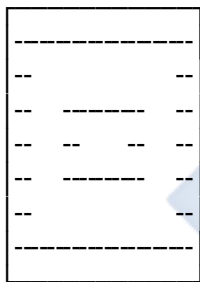
Many machine parts and assemblies have **internal features** (holes, cavities, passages) that cannot be clearly shown using only external views. **Sectioning** is a drafting technique that "cuts open" an object to reveal these internal details.

Why Section?

- **Clarity:** Easier to understand than many hidden lines
- **Manufacturing:** Shows machinist what's inside
- **Inspection:** Quality control can verify internal features

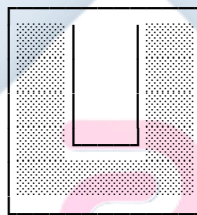
Diagram 29: Hidden Lines vs Section

WITHOUT SECTION
(Confusing)



↑
Many hidden lines
Hard to interpret

WITH SECTION
(Clear)

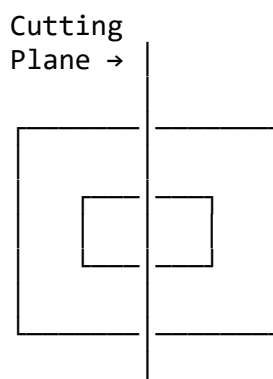


↑
Internal hole
clearly visible
with hatching

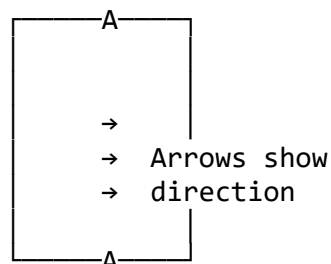
8.2 THE CUTTING PLANE AND SECTION LINES

Diagram 30: Cutting Plane Concept

3D VISUALIZATION



TOP VIEW
(Showing cutting plane)



↑
Cutting plane line

Remove this
half → |

(thick chain line)
SECTION A-A

Components of Sectioning:

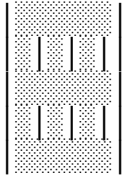
Diagram 31: Section Line Standards

1. CUTTING PLANE LINE

A - - - - - A
↑ ↑
Arrows show Letters identify
viewing the section
direction (SECTION A-A)

Line type: Thick (0.7mm) chain line
Position: On the view being "cut"

2. SECTION LINES (Cross-Hatching)

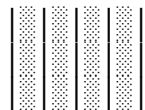
Standard Pattern:
 ← Thin lines (0.35mm)
at 45° angle
evenly spaced
(2-3mm apart)

Rules:

- 45° is standard (or 30°/60° if conflicts)
- Different parts = different directions
- Don't section: bolts, nuts, shafts, pins, keys, ribs (if cut lengthwise)

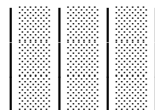
Diagram 32: Material Representation in Section

CAST IRON
(General symbol)



Single lines
45° angle

STEEL
(Denser)



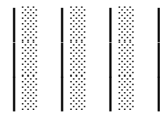
Double lines
45° angle

BRASS/BRONZE
(Crosshatch)



Crossed lines
45° angles

ALUMINUM
(Wide spacing)



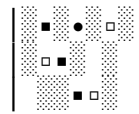
Wider spacing

RUBBER/PLASTIC
(Solid/thick)



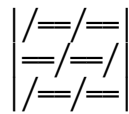
Solid black

CONCRETE
(Random)



Small shapes

WOOD
(Grain pattern)



Flowing lines

GLASS
(Very sparse)



Sparse lines

LIQUID
(No hatch)



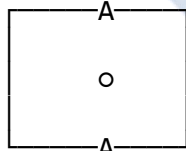
Wavy line

8.3 TYPES OF SECTIONS

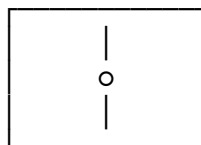
Diagram 33: Full Section

BEFORE SECTIONING

TOP VIEW



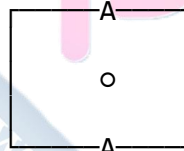
FRONT VIEW



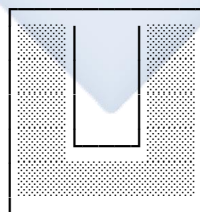
Cutting plane
passes completely
through object

AFTER SECTIONING

TOP VIEW



SECTION A-A

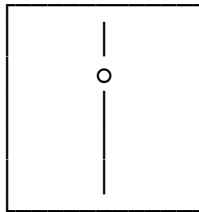


Half removed,
section replaces
front view

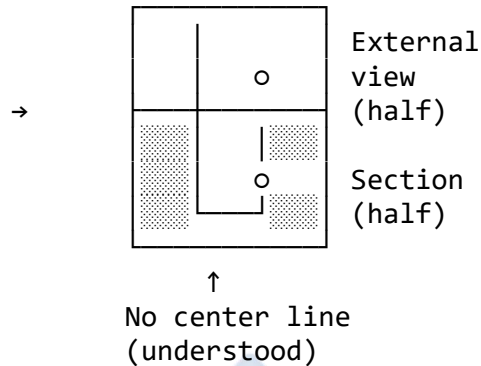
Use: Most common type. Shows complete internal profile. Replaces one standard view.

Diagram 34: Half Section

SYMMETRICAL OBJECT



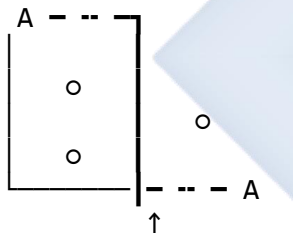
HALF SECTION VIEW



Use: For symmetrical objects. Shows both internal (sectioned) and external (not sectioned) in ONE view. Saves space!

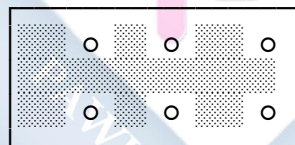
Diagram 35: Offset Section

TOP VIEW
(Showing offset plane)



OFFSET SECTION A-A

Features that are NOT
in line shown AS IF
they were in line

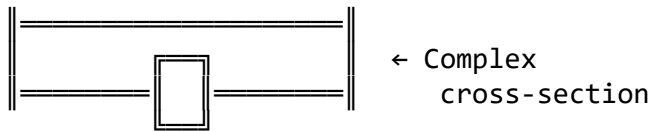


Use: When features (holes, pockets) are NOT aligned. Cutting plane "steps" to pass through all features.

Important: The bends/steps in the cutting plane are NEVER shown in the section view!

Diagram 36: Revolved Section

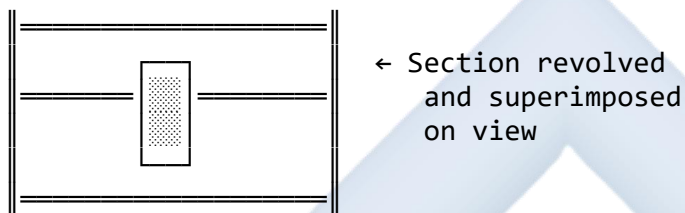
OBJECT (Beam with I-section)



↓ Slice taken here

REVOLVED SECTION

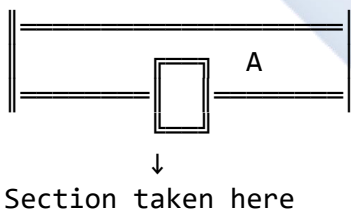
(Rotated 90° and drawn ON the view)



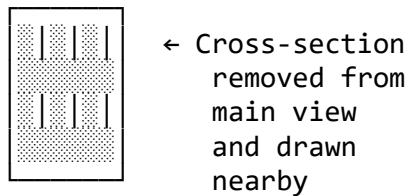
Use: To show the cross-sectional shape of arms, spokes, ribs, or beams. Drawn directly on the main view.

Diagram 37: Removed Section

OBJECT



REMOVED SECTION A-A
(Drawn separately)

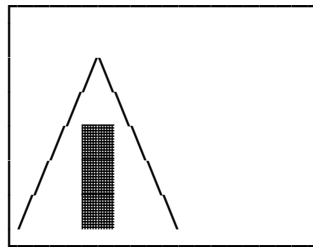


SECTION A-A

Use: Same as revolved, but when there's no space on the main view. Drawn elsewhere on sheet.

Diagram 38: Broken-Out Section

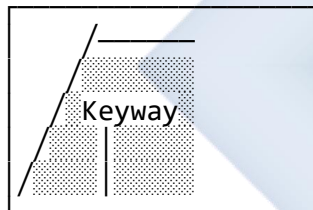
OBJECT WITH LOCAL FEATURE



← Only this area
needs to be shown

↑
Break line (irregular,
jagged, freehand)

BROKEN-OUT VIEW



← Small area
"broken away"
to show
internal feature

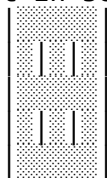
Use: To show a small, localized internal feature without creating a full section. Break line is **irregular** and hand-drawn.

8.4 WHAT NOT TO SECTION

Diagram 39: Standard Parts NOT Sectioned

WRONG ✗

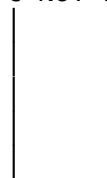
Shaft in section:



↓
Confusing!

CORRECT ✓

Shaft NOT sectioned:



↓
Clear!

← Show whole,
even though
cutting plane
passes through

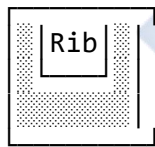
Parts NOT sectioned even if cutting plane passes through:

- Bolts, nuts, screws, washers
- Pins, keys, cotters
- Shafts, spindles, rods
- Balls, rollers (bearings)
- Ribs, webs, spokes (if cut lengthwise/radially)
- Gear teeth, splines

Reason: These are standard parts.
Sectioning them adds no useful information
and makes drawing harder to read.

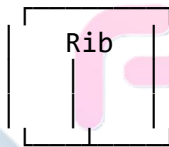
Diagram 40: Ribs and Webs - Special Case

CUT ACROSS RIB
(Section it)



↑
Section across
(hatch the rib)

CUT ALONG RIB
(DON'T section it)



← NOT
hatched!

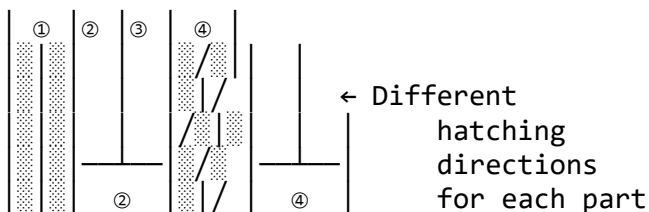
↑
Section along
(no hatch)

Why? Hatching along length makes rib look SOLID,
which is misleading. Rib is thin!

8.5 SECTION ASSEMBLY DRAWINGS

Diagram 41: Assembly in Section

ASSEMBLY WITH MULTIPLE PARTS





- ① Cast iron base (45° left)
- ② Steel shaft (not sectioned)
- ③ Bronze bushing (45° right)
- ④ Aluminum cover (30°)

Rule: Adjacent parts have DIFFERENT hatching directions for clarity!

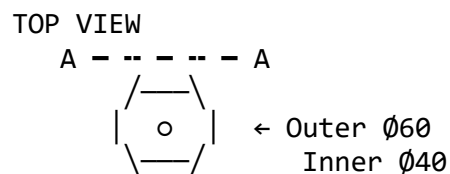
EVALUATION

1. What is the purpose of sectioning in technical drawing?
2. What is the difference between a **cutting plane line** and **section lines**?
3. List **three parts** that are NOT sectioned even if the cutting plane passes through them.
4. Differentiate between a "**Full Section**" and a "**Half Section**." When would you use each?
5. What type of section would you use to show features that are NOT in a straight line?
6. Why are ribs not sectioned when cut lengthwise?

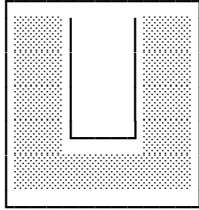
ASSIGNMENT

1. **Draw a simple hollow pipe** (cylinder with thick walls):
 - Outer Diameter: 60mm
 - Inner Diameter: 40mm
 - Length: 100mm
2. **Show:**
 - **TOP VIEW** (circles)
 - **FRONT VIEW** as a **FULL SECTION A-A**
 - Include:
 - Cutting plane line on top view
 - Section lines (45° hatching) on front view
 - All dimensions
 - Title: "PIPE SECTION"

Example sketch:



SECTION A-A



← 45° hatching



WEEK 9: BLOCK SECTIONING (ARCHITECTURAL DRAWING)

9.1 INTRODUCTION TO BUILDING DRAWINGS

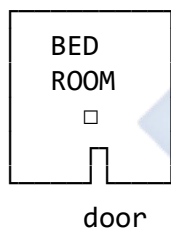
This week marks a transition from **Mechanical Engineering Drawing** to **Architectural Drawing** (Building Drawing). While the principles remain the same, the application and symbols differ.

Building drawings are specialized working drawings that describe the construction of a building. They communicate the architect's design to:

- **Clients** (for approval)
- **Quantity Surveyors** (for costing)
- **Contractors** (for construction)
- **Authorities** (for building permits)

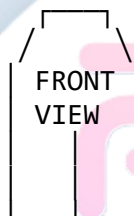
Diagram 42: Three Main Types of Building Drawings

1. FLOOR PLAN
(Horizontal cut)



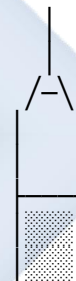
Shows layout
from above

2. ELEVATION
(External view)



Shows appearance
from outside

3. SECTION
(Vertical cut)



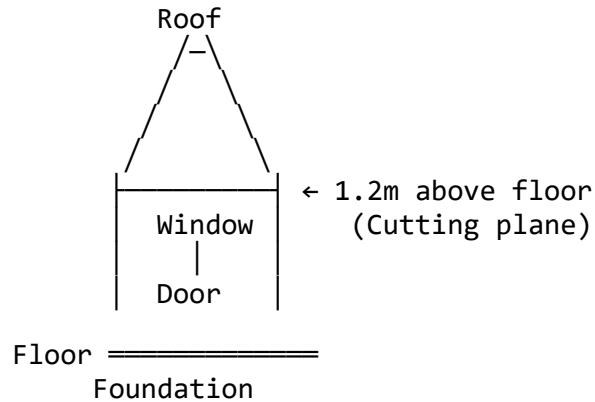
Shows
construction

9.2 DESIGN OF BUILDING FLOOR PLANS

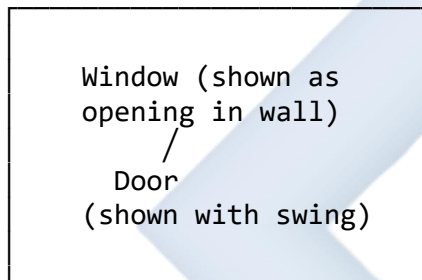
A **Floor Plan** is a **horizontal section** taken approximately **1.2 meters (4 feet)** above the finished floor level. This height ensures doors and windows are "cut through" and shown clearly.

Diagram 43: Floor Plan - Conceptual View

SIDE VIEW SHOWING CUTTING PLANE



RESULTING FLOOR PLAN (Looking down)

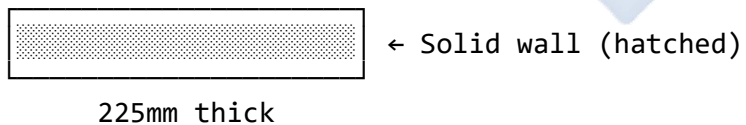


Floor Plan Elements:

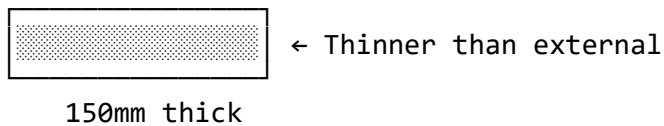
Diagram 44: Floor Plan Symbol Library

WALLS (Shown with thickness)

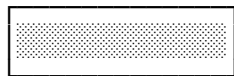
EXTERNAL WALL (225mm/9" block)



INTERNAL WALL (150mm/6" block)



PARTITION WALL (100mm/4")

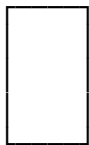


← Lightweight

100mm thick

DOORS (Multiple types)

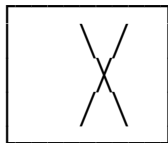
SINGLE SWING DOOR



← Door leaf
(thin line)

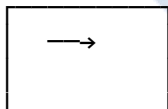
90° arc shows
door swing

DOUBLE DOORS



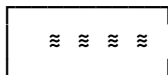
← Both leaves swing

SLIDING DOOR



← Arrow shows
slide direction

FOLDING DOOR



← Zigzag pattern

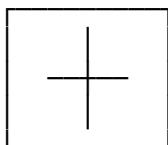
OVERHEAD DOOR (Garage)



← Up-and-over

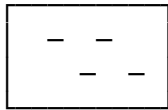
WINDOWS (Shown as openings)

CASEMENT WINDOW



← Thin lines show
frame/mullions

SLIDING WINDOW



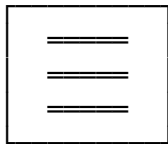
← Overlapping panels

FIXED WINDOW



← Just frame
(no opening parts)

LOUVRE WINDOW



← Horizontal slats

FIXTURES AND FITTINGS

WATER CLOSET (WC)



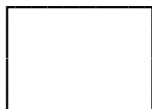
← Toilet bowl and cistern

WASH HAND BASIN (WHB)



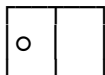
← Basin with drain

BATH TUB



← Long rectangle with rounded ends

KITCHEN SINK



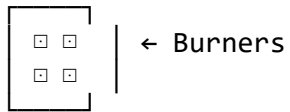
← Double bowl with drainer

SHOWER

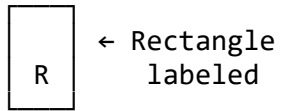


← Drain cross

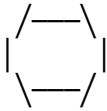
STOVE/COOKER



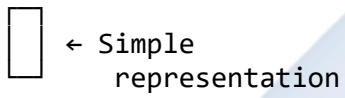
REFRIGERATOR



DINING TABLE



CHAIR



BED

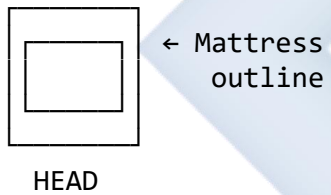
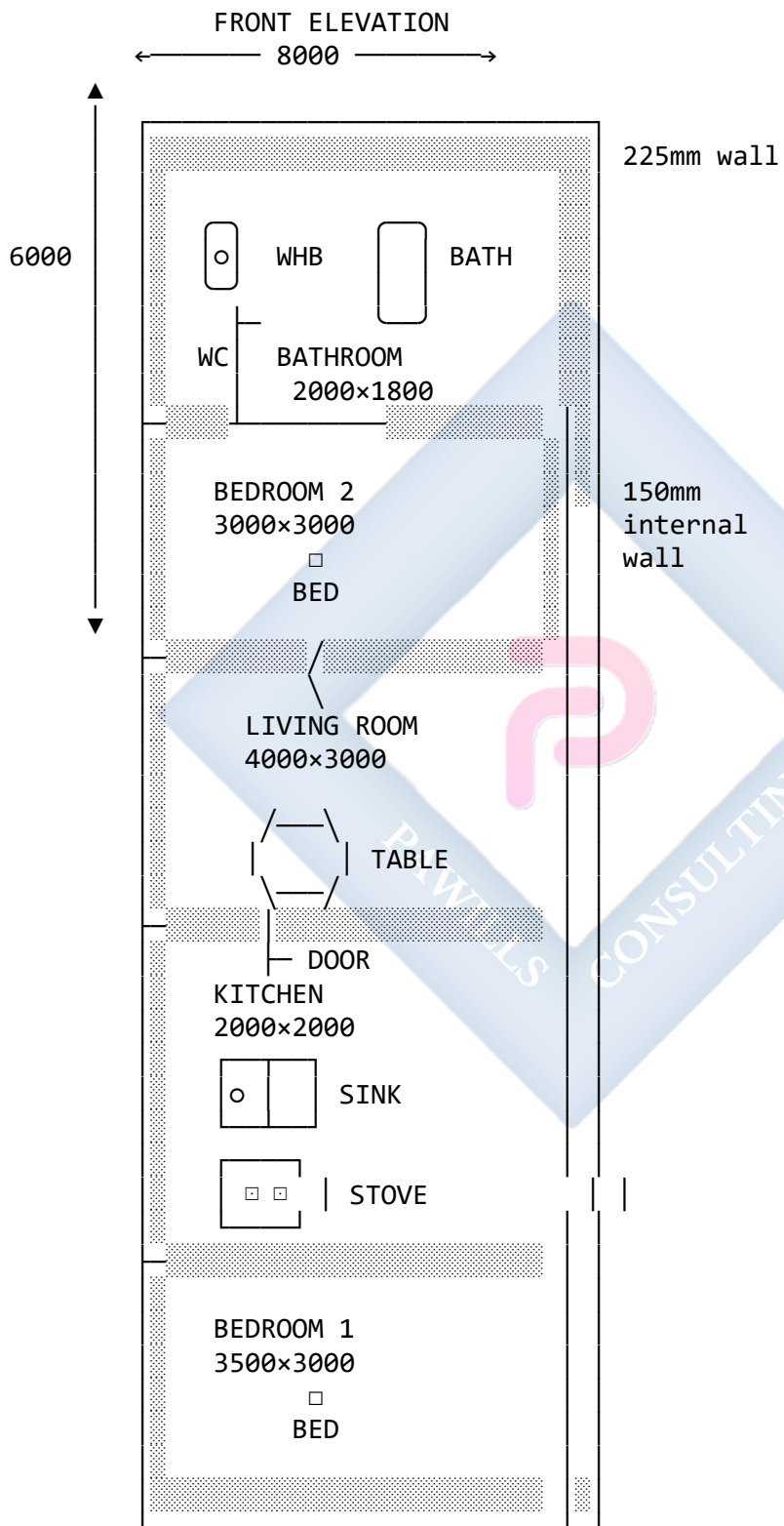


Diagram 45: Complete Floor Plan Example - 2-Bedroom Flat

SCALE: 1:100



LEGEND:

- == External wall (225mm)
- Internal wall (150mm)
- / Door swing
- Furniture
- Plumbing fixture
- ▤ Solid wall (section)

NOTES:

1. All dimensions in millimeters
2. Floor level: +0.00
3. Ceiling height: 2700
4. Windows: 1200x1200 (typical)

9.3 BUILDING ELEVATIONS

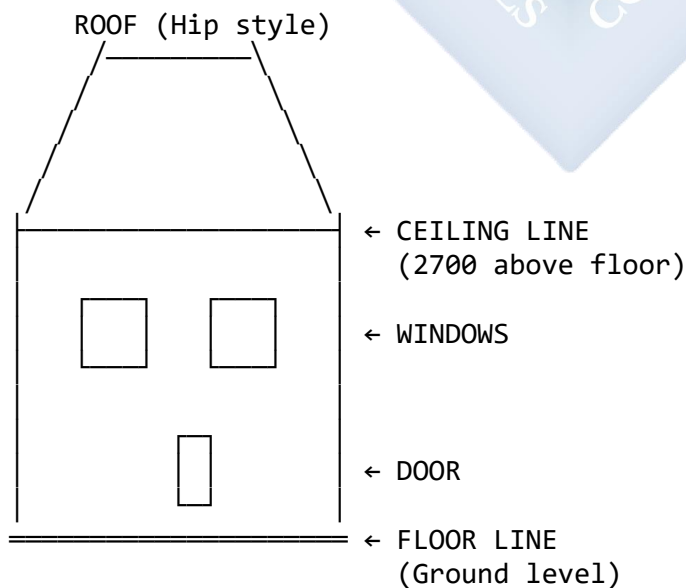
An **Elevation** is an external orthographic view of the building. It shows what you see when standing on the ground looking at one side of the building.

Four Standard Elevations:

1. **Front Elevation** - Main entrance side
2. **Rear Elevation** - Back of building
3. **Left Side Elevation** - Left side
4. **Right Side Elevation** - Right side

Diagram 46: Elevation View - Front Elevation

FRONT ELEVATION OF 2-BEDROOM FLAT
SCALE: 1:100



————— GROUND LINE
(Natural ground)

← 8000 →

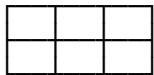
KEY FEATURES TO SHOW:

- Roof shape and pitch
- Window and door locations
- Building height
- Ground level
- Materials (textures)
- Eaves overhang

Material Indication on Elevations:

Diagram 47: Material Representation in Elevation

BRICK WALL



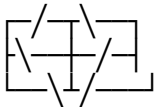
Small rectangles
(running bond)

BLOCK WALL



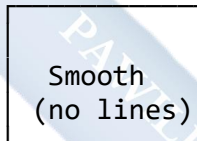
Larger rectangles
(stack bond)

STONE WALL



Irregular shapes

PLASTERED WALL



Smooth
(no lines)

TIMBER CLADDING



Vertical boards

GLASS



Reflections shown

CORRUGATED SHEET



Wave pattern

TILES



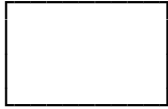
Overlapping pattern

9.4 ROOF STRUCTURES AND DESIGNS

Common Roof Types:

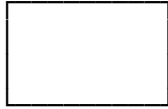
Diagram 48: Roof Designs in Elevation and Plan

1. FLAT ROOF
ELEVATION



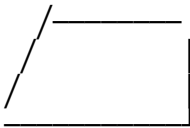
Horizontal

PLAN VIEW



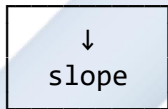
Simple rectangle

2. SHED ROOF (Lean-to)
ELEVATION



Single slope

PLAN VIEW



Arrow shows
slope direction

3. GABLE ROOF (Pitched)
ELEVATION



Two slopes

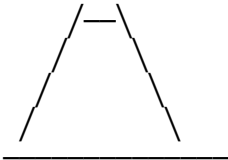
PLAN VIEW



Ridge line
(center)

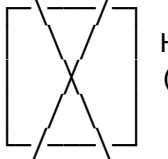
Gable ends
(triangular)

4. HIP ROOF
ELEVATION



Four slopes

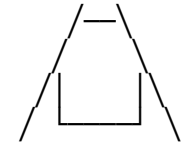
PLAN VIEW



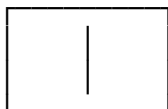
Hip lines
(diagonal)

No gable ends

5. GAMBREL ROOF
ELEVATION



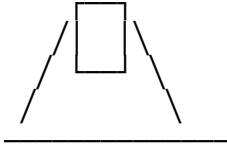
PLAN VIEW



Similar to
gable in plan

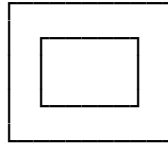
Barn style

6. MANSARD ROOF
ELEVATION



Four-sided gambrel

PLAN VIEW

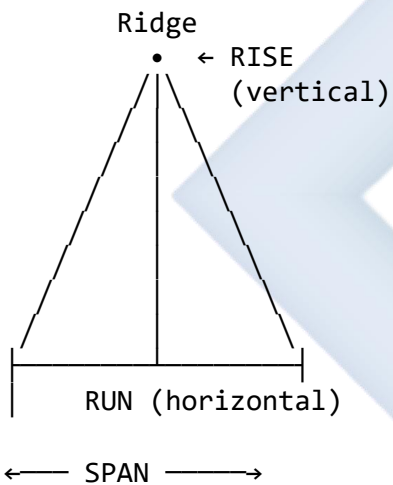


All sides
have double
slope

Roof Pitch Calculation:

Diagram 49: Understanding Roof Pitch

SIDE VIEW OF ROOF



PITCH CALCULATION:

$$\text{Pitch} = \text{Rise}/\text{Run}$$

Example 1:

Rise = 3m

Run = 6m

Pitch = $3/6 = 0.5$ (or 1:2 or 50%)

Example 2:

Rise = 2m

Run = 5m

Pitch = $2/5 = 0.4$ (or 2:5 or 40%)

EXPRESSING PITCH:

- As ratio: 1:2, 2:5, 1:3
- As fraction: $\frac{1}{2}$, $\frac{2}{5}$, $\frac{1}{3}$
- As angle: 26.6° , 21.8° , 18.4°
- As percentage: 50%, 40%, 33%
- As "X in 12": 6:12, 5:12, 4:12

9.5 SKETCHING INTERNAL PARTS OF MOULDED HOLLOW BLOCK

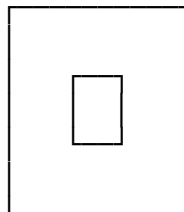
Hollow blocks (sandcrete blocks) are the primary walling material in Nigeria.

Diagram 50: Standard Hollow Block Types

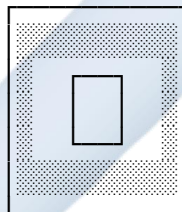
9-INCH BLOCK (225mm × 225mm × 450mm)

FRONT VIEW

TOP VIEW (Section)



225mm
(9")



450mm
(18")

← Hollow
cores

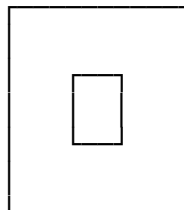
FEATURES:

- Two hollow cores
- Three vertical webs
- Top and bottom solid sections
- Reduces weight
- Provides insulation

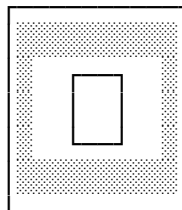
6-INCH BLOCK (150mm × 225mm × 450mm)

FRONT VIEW

TOP VIEW (Section)



150mm
(6")



450mm
(18")

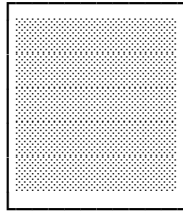
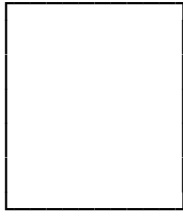
Thinner than 9" block

Used for partitions

SOLID BLOCK (No hollows)

FRONT VIEW

TOP VIEW (Section)



← Completely solid

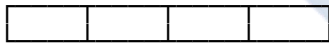
Very heavy
Used for foundations
or load-bearing walls

Diagram 51: Block Laying Pattern

STRETCHER BOND (Most common)
(Running bond - overlap by half)

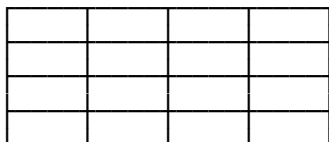


← Course 2
(offset)



← Course 1
(base)

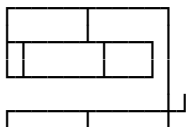
STACK BOND (Not recommended)
(Joints align vertically - weak!)

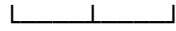


← Weak
vertical
joints

CORNER DETAIL

Course 2





Course 1

Blocks interlock
at corners for
strength

EVALUATION

1. What is a floor plan, and at what height above the finished floor is the cutting plane taken?
 2. What is the difference between a **floor plan** and an **elevation**?
 3. List **four common types** of roof designs.
 4. Sketch the floor plan symbol for: a) A single swing door b) A water closet (WC) c) A window
 5. What three features are typically shown in a floor plan?
 6. What is the purpose of hollow cores in sandcrete blocks?
 7. Define "roof pitch" and explain how it is calculated.
-

ASSIGNMENT

Part 1: Simple Floor Plan

1. On an A4 sheet, draw a **floor plan** for a single room (e.g., your bedroom or a bathroom).
2. **Dimensions:** Approximately $3\text{m} \times 4\text{m}$
3. **Include:**
 - External walls (225mm thick, hatched)
 - One door (show swing with arc)
 - One window (show opening in wall)
 - At least two furniture items or fixtures (use appropriate symbols)
 - Label the room name
 - Add dimensions for length and width
 - 10mm border and simple title block

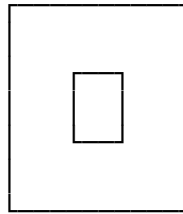
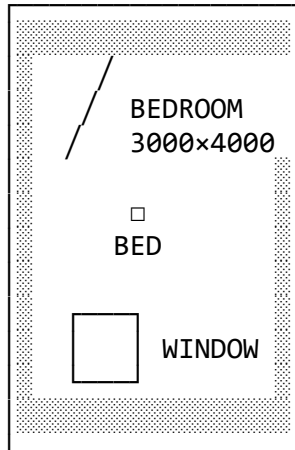
Part 2: Hollow Block Sketch 4. Draw a 9-inch hollow block showing:

- **Front view** (simple rectangle)
- **Top view in section** (showing the two hollow cores and webs)
- Label: Overall dimensions, web thickness, core dimensions
- Use hatching (45°) to show solid material

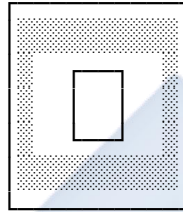
Example format:

BEDROOM FLOOR PLAN
SCALE: 1:50

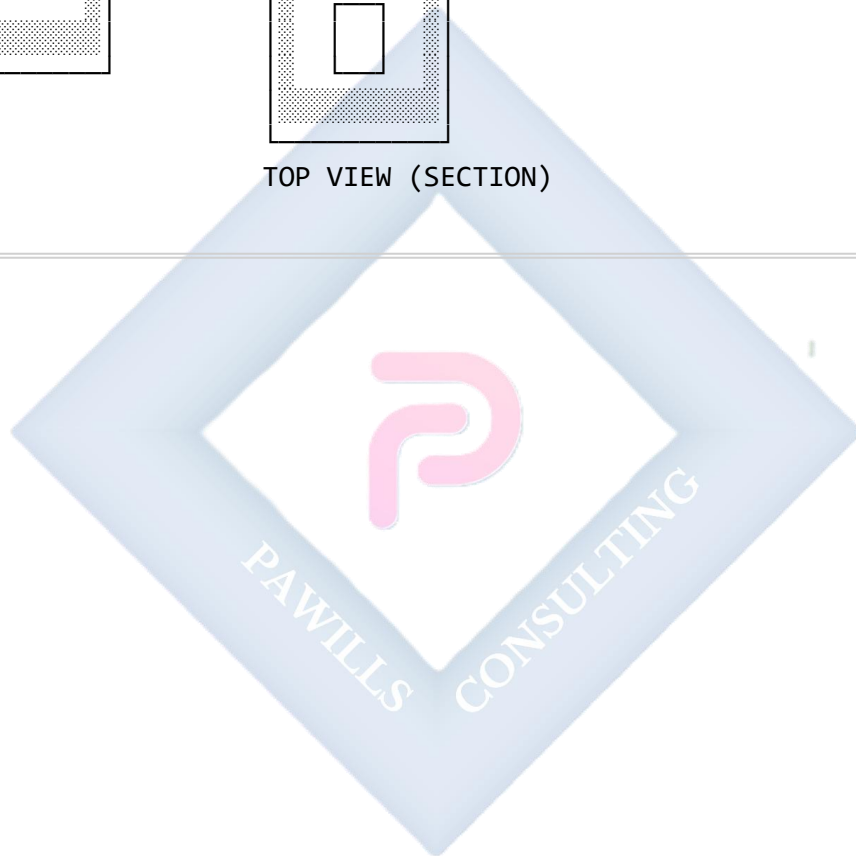
BLOCK DETAILS
SCALE: 1:5



FRONT VIEW



TOP VIEW (SECTION)



WEEK 10: BLOCK SECTIONING (CONTINUED)

10.1 BUILDING SECTIONAL VIEWS

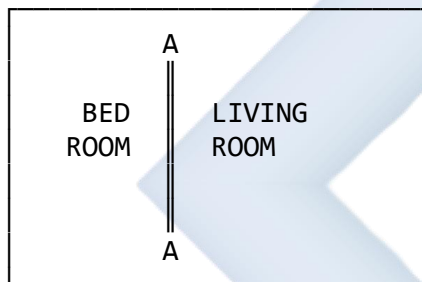
While a Floor Plan is a **horizontal section**, a **Building Section** is a **vertical section** that cuts through the building from roof to foundation, revealing the internal construction.

Purpose:

- Shows **how the building is constructed**
- Reveals **structural elements** (foundation, floor slab, wall construction, roof structure)
- Indicates **heights and levels**
- Shows **materials** in section
- Guides contractors during construction

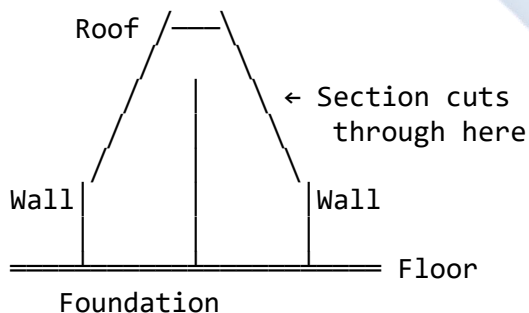
Diagram 52: Building Section - Conceptual View

FLOOR PLAN (showing cutting plane location)



↑
Cutting plane line
(SECTION A-A taken here)

3D VIEW (showing where section cuts)

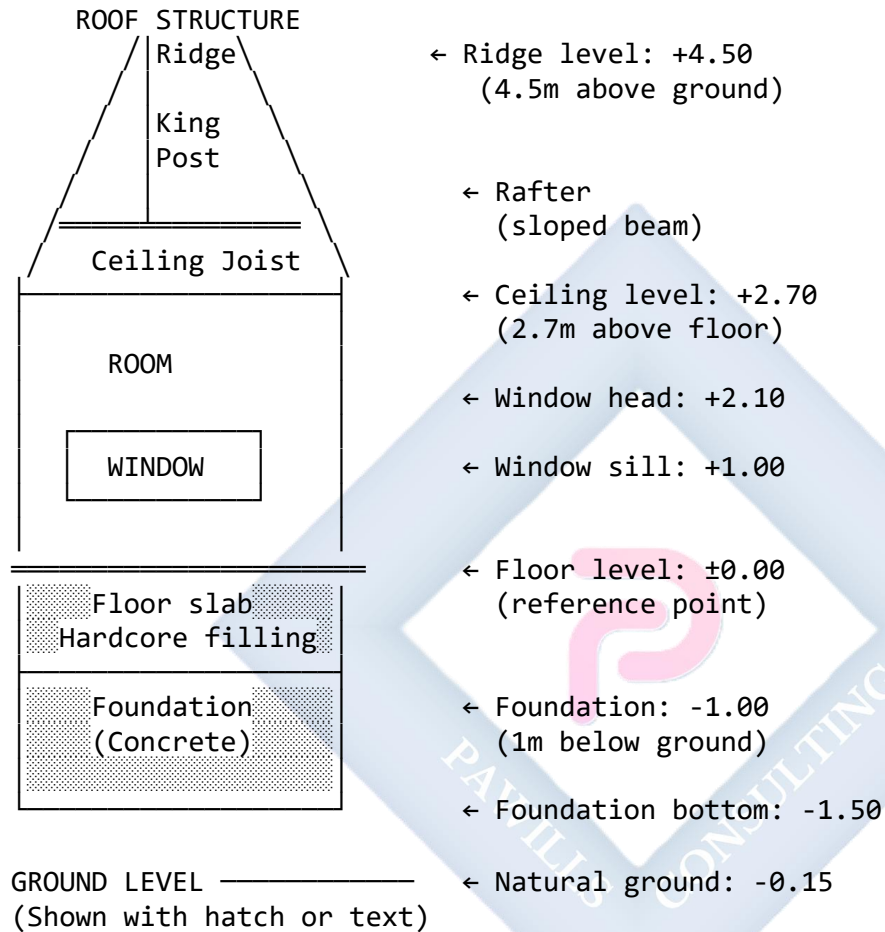


10.2 COMPONENTS OF A BUILDING SECTION

Diagram 53: Complete Building Section - Single Story

SECTION A-A

SCALE: 1:50



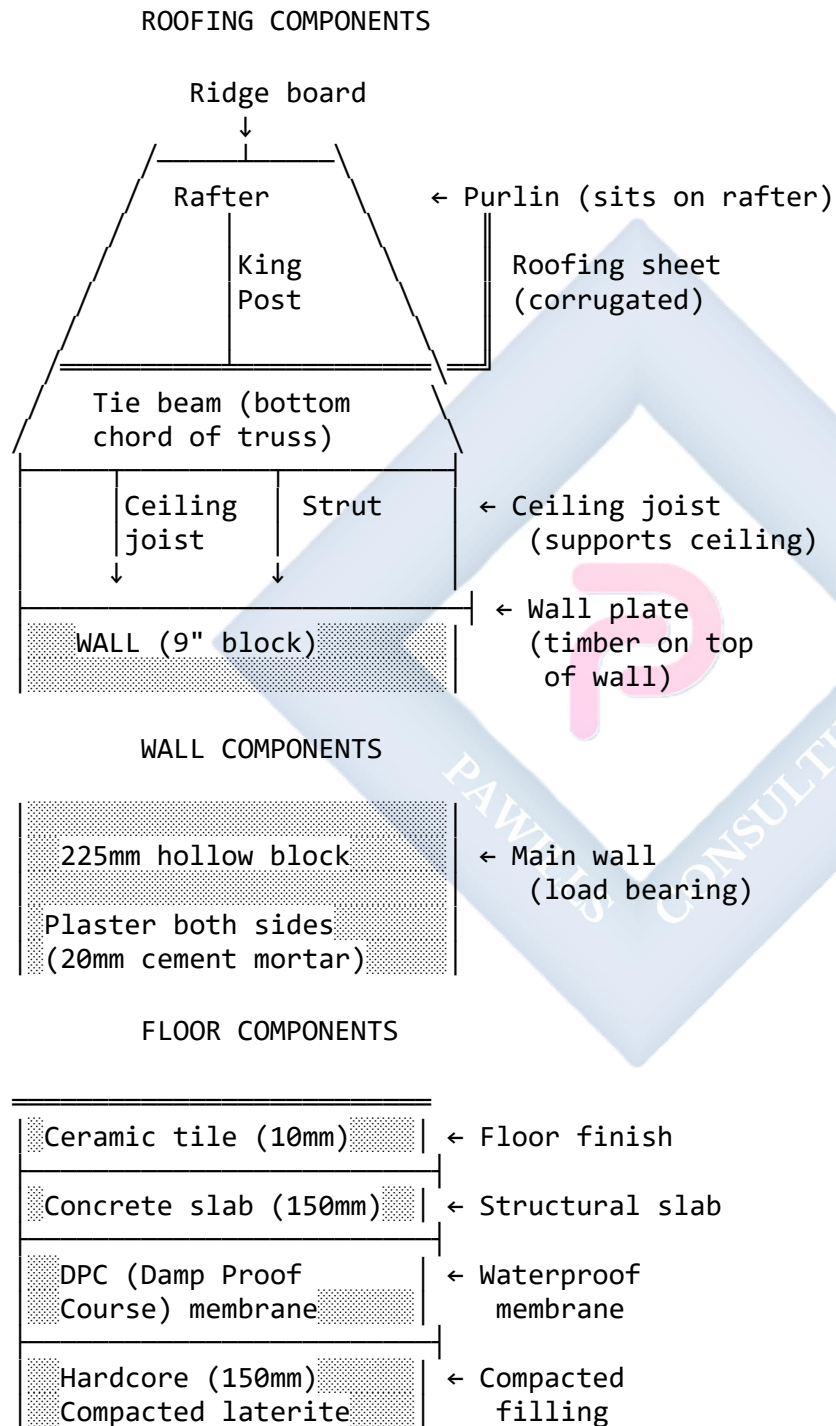
TYPICAL ANNOTATIONS:

MATERIALS & NOTES:

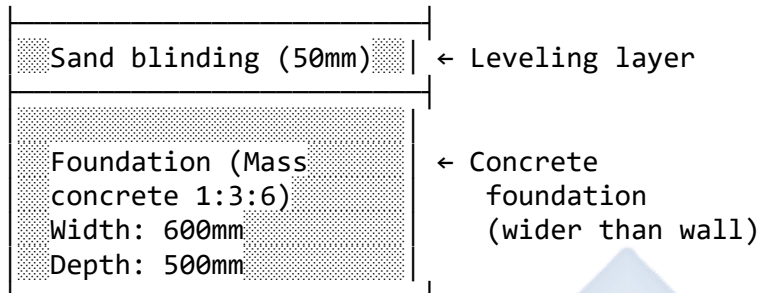
- Roof: Aluminum sheet on timber trusses
- Ceiling: POP finish
- Walls: 225mm blocks, plastered both sides
- Floor: Concrete slab on hardcore, with ceramic tile finish
- Foundation: Mass

concrete 1:3:6 mix

Diagram 54: Building Section - Key Components Labeled



FOUNDATION COMPONENTS

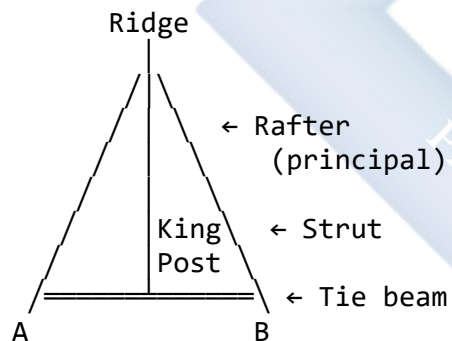


10.3 ROOF TRUSS DESIGNS

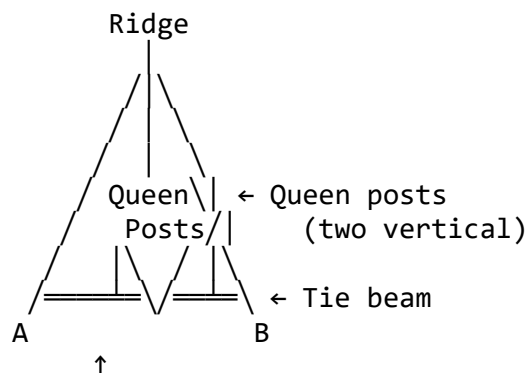
A **truss** is a rigid triangular framework that spans across the building to support the roof.

Diagram 55: Common Truss Types

1. KING POST TRUSS (Simple, span up to 8m)

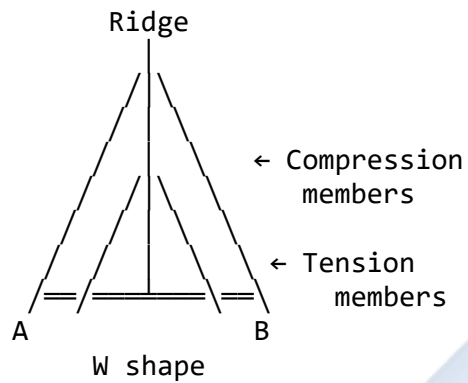


2. QUEEN POST TRUSS (Span 8-12m)

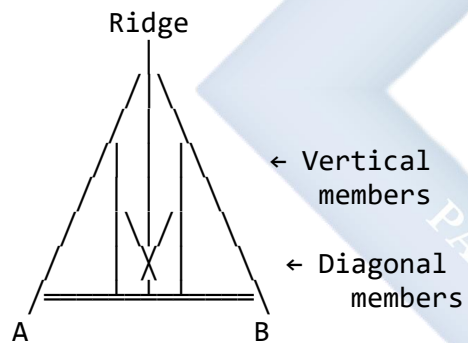


Allows storage
space in center

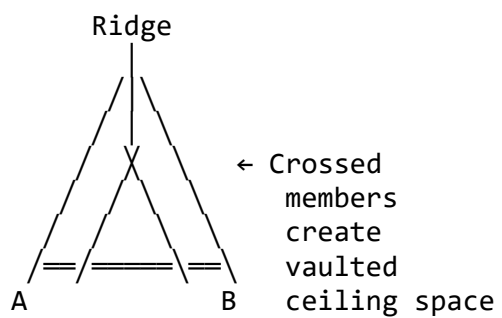
3. FINK TRUSS (Modern, span 8-15m)
(W-shape web)



4. HOWE TRUSS (Heavy loads, span 15-30m)

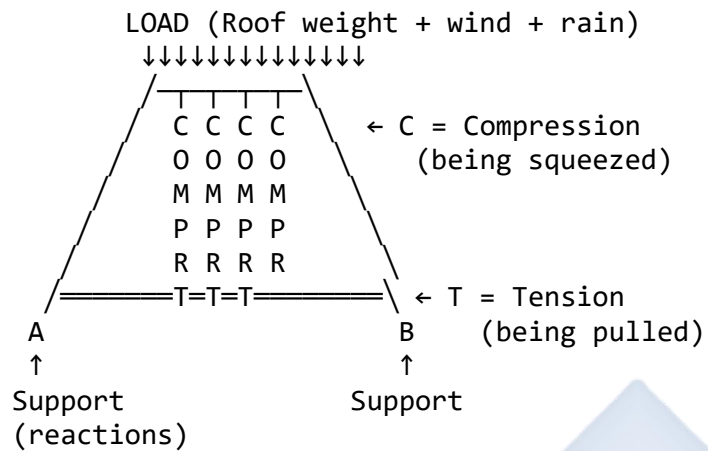


5. SCISSORS TRUSS (Cathedral ceiling)



Truss Member Forces:

Diagram 56: Force Distribution in King Post Truss



KEY PRINCIPLES:

- Rafters in COMPRESSION (pushed inward)
- Tie beam in TENSION (pulled apart)
- King post in COMPRESSION (squeezed)
- Struts in COMPRESSION
- Triangular shape prevents collapse

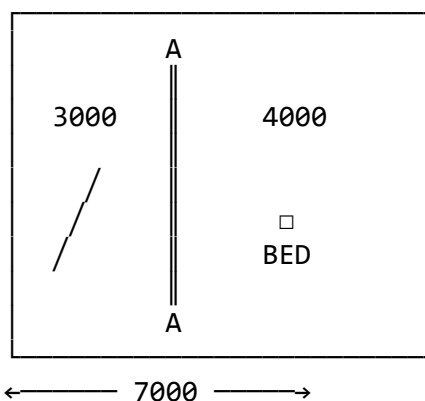
10.4 INTEGRATION OF VIEWS

All three views (Plan, Elevation, Section) must work together to fully describe the building.

Diagram 57: View Relationship

FLOOR PLAN (Top view)

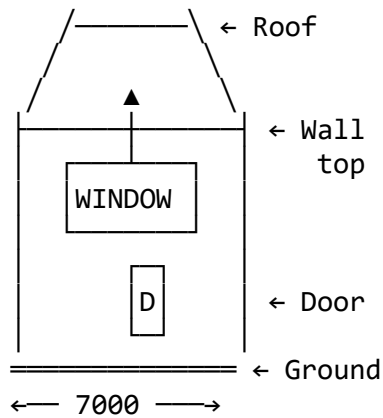
Shows: Room layout, walls, doors, windows
WHERE section A-A is taken



FRONT ELEVATION (External view)

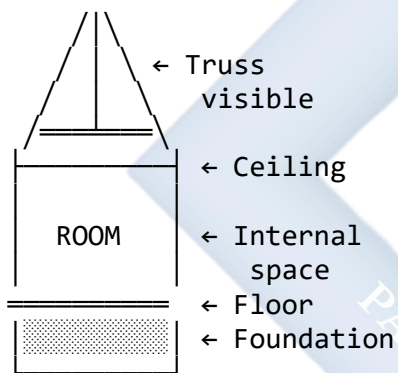
Shows: What building looks like from outside

Height, roof shape, windows, doors



SECTION A-A (Vertical cut)

Shows: Internal construction
Foundation, floor, roof structure



All three views use SAME WIDTH (7000)
Plan and Section use SAME HEIGHT dimensions
Elevation and Section show SAME vertical heights

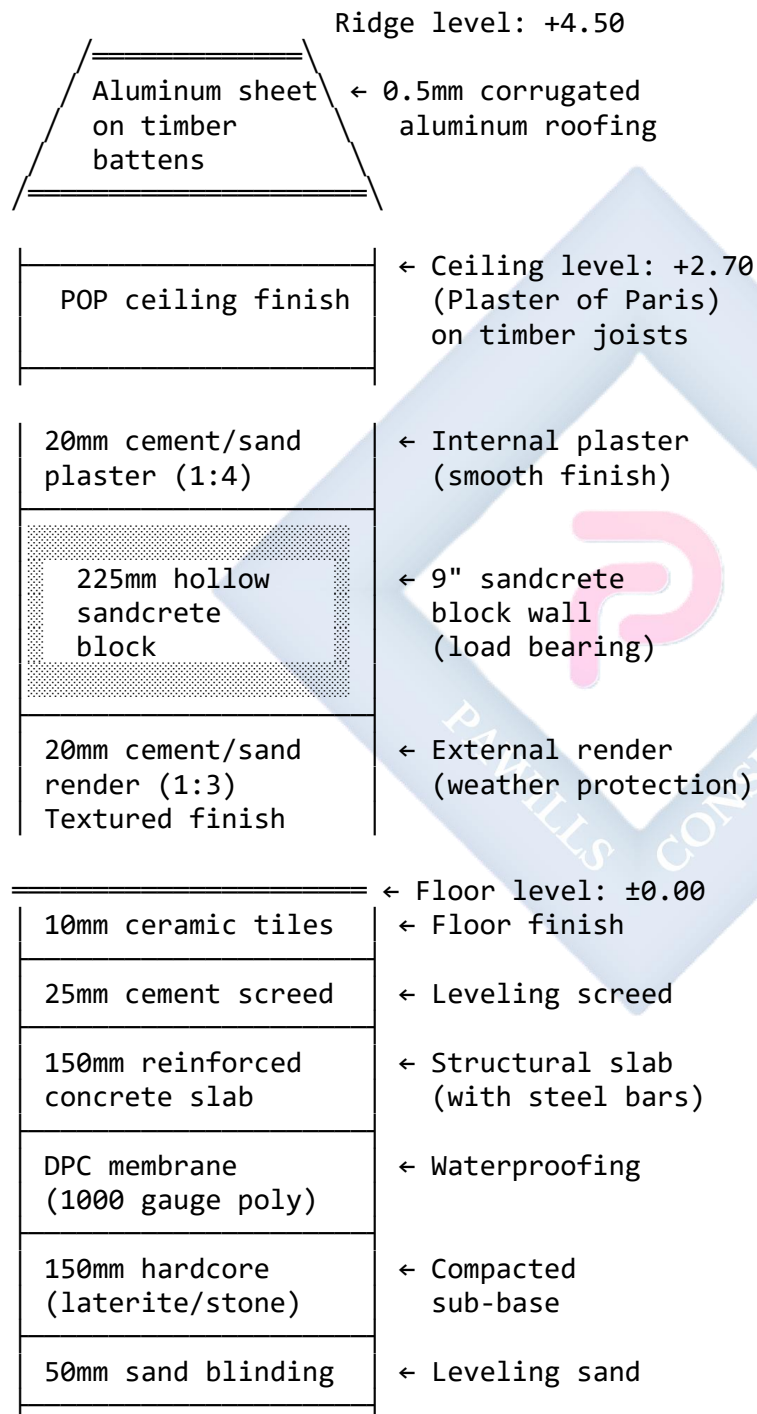
10.5 DETAILED SECTION THROUGH WALL

Diagram 58: Wall Section Detail (Typical at 1:20 scale)

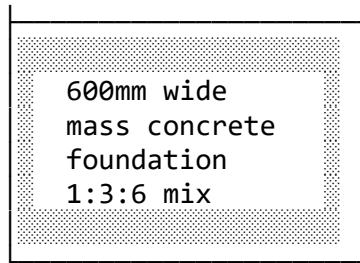
EXTERNAL WALL DETAIL
(Section showing all layers)

OUTSIDE

INSIDE



—— GROUND LEVEL —— -0.15



← Strip foundation
(concrete grade
1:3:6 mix)

Depth: 500mm
Base level: -1.50

MATERIAL KEY:

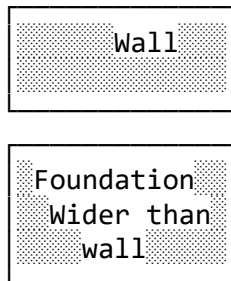
- = Solid material (block, concrete)
- = Horizontal surface
- = Vertical surface
- = Sloped surface (roof)

10.6 FOUNDATION TYPES

Diagram 59: Common Foundation Types

1. STRIP FOUNDATION (For walls)

PLAN VIEW



Continuous under
entire wall length

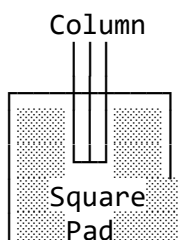
SECTION



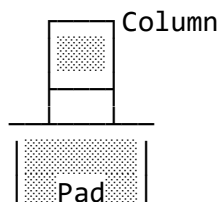
Distributes
wall load

2. PAD FOUNDATION (For columns)

PLAN VIEW



SECTION

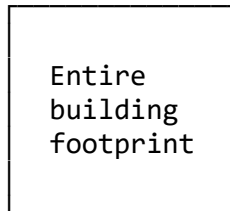


Isolated
square pad

Ground
Spreads
point load

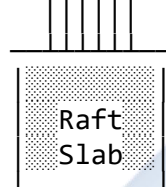
3. RAFT FOUNDATION (Weak soil)

PLAN VIEW



Covers whole
building area

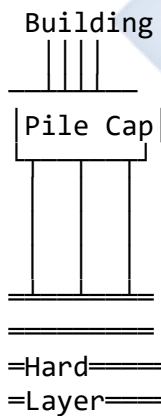
SECTION
Building



Ground
Spreads load
over large area

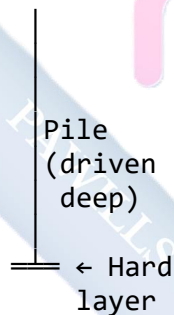
4. PILE FOUNDATION (Very weak soil)

ELEVATION



Long piles
driven/drilled
to hard layer

DETAIL



Transfer load
to deep stable
layer

EVALUATION

1. What is the main difference between a **Floor Plan** and a **Building Section**?
2. What key component of the roof structure is made clearly visible in a building section?
3. Where is the **cutting plane line** for a "Section A-A" shown on the drawings?

4. In a building's plan, what does a rectangle with diagonal lines from the corners represent (what roof type)?
 5. List the main components shown in a typical wall section detail (from roof to foundation).
 6. What is the purpose of a **DPC** (Damp Proof Course) in a floor construction?
 7. Sketch a simple **King Post Truss** and label: Ridge, Rafter, King Post, Tie Beam.
-

ASSIGNMENT

Scenario: You have drawn a floor plan of a simple rectangular room last week.

Task: Now create a **Building Section** through that room.

Requirements:

1. **Draw SECTION A-A** (scale 1:50) showing:
 - **Foundation** (below ground level -1.00m)
 - Width: 600mm
 - Depth: 500mm (from -1.00 to -1.50)
 - Material: Mass concrete (hatch 45°)
 - **Floor Construction** (at ground level ± 0.00)
 - 150mm hardcore (compacted)
 - 150mm concrete slab
 - Ceramic tile finish
 - **Walls** (both sides of room)
 - 225mm hollow block (hatched)
 - Height: 2.7m to ceiling level
 - Show plaster finish both sides
 - **Ceiling** (at +2.70m level)
 - Show ceiling joists
 - POP finish
 - **Roof Structure** (above ceiling)
 - Simple King Post Truss
 - Show: Ridge, Rafters, King Post, Tie Beam, Struts
 - Roof pitch: approx 30°
 - Ridge level: +4.50m
 - Aluminum roofing sheets
2. **Add dimensions** showing:
 - Room width
 - Floor level (± 0.00)

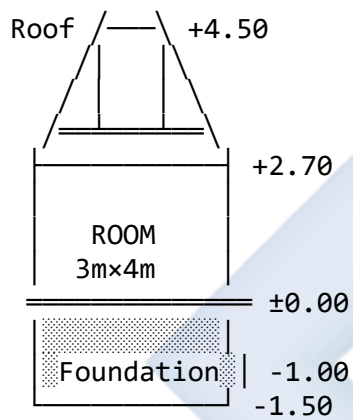
- Ceiling level (+2.70)
- Ridge level (+4.50)
- Foundation depth (-1.50)

3. **Include notes:**

- Material specifications
- "SECTION A-A"
- Scale
- Title block

Example outline:

SECTION A-A SCALE 1:50



NOTES:

- Foundation: Mass concrete 1:3:6
- Walls: 225mm hollow block
- Floor: Concrete on hardcore
- Roof: Aluminum sheet on timber truss

WEEK 11: REVISION

COMPREHENSIVE FIRST TERM REVIEW

This week consolidates all topics covered during the first term. Students should review, practice, and prepare for final examinations.

11.1 REVISION SCHEDULE

Day 1-2: Working Drawing Fundamentals (Weeks 1-2)

- Components of working drawings
- Detail vs Assembly drawings
- Orthographic projection (1st vs 3rd angle)
- Title blocks and dimensioning
- Line types and their uses

Day 3-4: Threads and Fasteners (Weeks 3-4)

- Thread terminology and profiles
- Thread representation (schematic)
- Types of bolts, nuts, screws, washers
- Locking devices and applications

Day 5-6: Sectioning (Week 8)

- Types of sections (Full, Half, Offset, etc.)
- Cutting planes and section lines
- What NOT to section
- Material representation

Day 7-8: Architectural Drawing (Weeks 9-10)

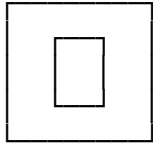
- Floor plans, Elevations, Sections
- Building symbols (doors, windows, fixtures)
- Roof types and trusses
- Foundation types

11.2 PRACTICE EXERCISES

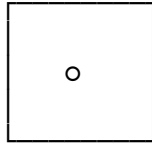
Exercise 1: Complete the Drawing

Given these orthographic views, identify the object:

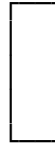
FRONT VIEW



TOP VIEW



SIDE VIEW



Object: _____ (Flanged bush? Pulley? Spacer?)

Exercise 2: Thread Representation

Draw the schematic representation of: a) External thread (M10 × 20mm long) - side view and end view b) Internal thread (M10) in section - side view and end view

Exercise 3: Fastener Identification

Sketch and label:

1. Hex head bolt
 2. Castle nut
 3. Spring washer
 4. Flat head wood screw
 5. Set screw
-

Exercise 4: Section Type Identification

For each description, name the section type:

- a) Cutting plane passes completely through object, removing half Answer: _____
 - b) Shows internal features in one half, external in other half Answer: _____
 - c) Cutting plane "steps" to pass through multiple non-aligned features Answer: _____
 - d) Cross-section rotated 90° and drawn on the main view Answer: _____
 - e) Small area "broken away" to show local feature Answer: _____
-

Exercise 5: Floor Plan Symbols

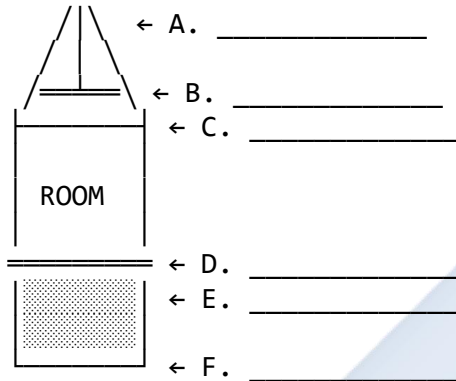
Draw the floor plan symbols for:

1. Single swing door (opening inward to right)
2. Water closet (WC)
3. Wash hand basin (WHB)

4. Casement window (1200mm wide)
5. Kitchen sink with double bowl

Exercise 6: Building Section

Label all components on this simplified building section:



Options: Foundation, Floor, Ceiling, Wall, Truss, Tie Beam

11.3 PAST EXAMINATION QUESTIONS

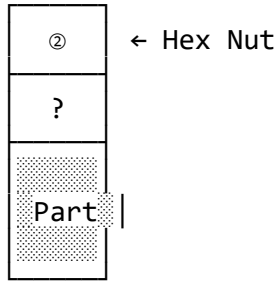
Question 1: (WAEC Style) State FOUR components that must be included in a complete working drawing. (4 marks)

Answer:

1. Orthographic views (Front, Top, Side)
2. Dimensions (all sizes and locations)
3. Title block (part name, material, scale, etc.)
4. Notes and specifications (manufacturing instructions)

Question 2: (WAEC Style) a) Sketch the profile of a V-thread. (3 marks) b) Label: Crest, Root, Major Diameter, Pitch. (4 marks) c) State ONE use of V-thread. (1 mark)

Question 3: (WAEC Style) The diagram below shows an incomplete assembly drawing.



a) What component is missing at "?"? (2 marks) b) State the function of this component. (2 marks) c) Complete the assembly by sketching the missing part. (4 marks)

Answer: a) Plain washer (or flat washer) b) Distributes clamping load / Protects surface / Prevents damage to part c) [Student draws washer between nut and part]

Question 4: (WAEC Style) a) Differentiate between a "Detail Drawing" and an "Assembly Drawing." (4 marks) b) State TWO pieces of information found in a Parts List. (2 marks)

Answer: a)

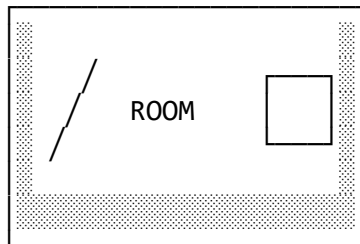
- **Detail Drawing:** Shows one component only with complete dimensions for manufacturing
- **Assembly Drawing:** Shows how multiple components fit together, no detailed dimensions, includes parts list with balloons

b)

- Part Name
- Quantity (QTY)
- Part Number
- Material (any TWO)

Question 5: (Practical - 20 marks) Draw the following using proper schematic representation: a) A hex bolt (M12 × 40mm long) with external thread b) A hex nut (M12) shown in section with internal thread c) Show both the side view and end view of each d) Label: Major Diameter, Minor Diameter e) Include a simple title block

Question 6: (WAEC Style) The floor plan below shows a room with doors and windows.



a) What does the arc symbol (↗) represent? (1 mark) b) Redraw the door symbol showing it opening outward instead. (3 marks) c) What is the typical thickness of the external wall shown? (1 mark)

Answer: a) Door swing / Direction door opens b) [Student redraws with arc on outside of wall] c) 225mm (or 9 inches)

Question 7: (Practical - 25 marks) Using a scale of 1:50, draw a simple floor plan for a bathroom (2.5m × 2.0m) including:

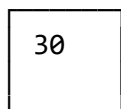
- External walls (225mm thick)
- One door (900mm wide)
- One window (600mm wide)
- Water closet (WC)
- Wash hand basin (WHB)
- Shower cubicle
- All dimensions
- Border and title block

11.4 COMMON MISTAKES TO AVOID

Diagram 60: Common Drawing Errors

DIMENSIONING ERRORS

X WRONG
50



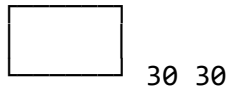
Dimension inside
view (cluttered)

✓ CORRECT
50



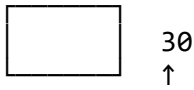
30
Dimensions
outside view

X WRONG
50 50



Duplicate dimensions

✓ CORRECT
← 50 →



Each dimension shown once

THREAD REPRESENTATION ERRORS

X WRONG
===== No thin lines shown

Missing minor diameter

✓ CORRECT
===== Thin line for minor diameter

Correct

SECTION LINE ERRORS

X WRONG
[Solid fill] Solid fill Too dense

✓ CORRECT
[Evenly spaced lines] Evenly spaced lines Proper spacing

FLOOR PLAN ERRORS

X WRONG
[Door without swing] Door without swing Missing swing

✓ CORRECT
[Door with swing arc] Door with swing arc Shows opening

X WRONG
[Window missing frame] Window missing frame No detail

✓ CORRECT
[Window properly shown] Window properly shown Clear opening

11.5 EXAMINATION TIPS

For Theory Questions:

1. **Read carefully** - Understand what is being asked
2. **Answer directly** - Don't write unnecessary information
3. **Use correct terms** - "Major diameter" not "big size"
4. **Be specific** - "225mm hollow block" not "block"
5. **State units** - Always include mm, m, degrees, etc.

For Drawing Questions:

1. **Use correct scale** - If asked for 1:50, maintain it throughout
2. **Draw faintly first** - Use 2H pencil for construction
3. **Darken final lines** - Use HB pencil for final outlines
4. **Include all requirements:**
 - Border (10mm)
 - Title block (properly filled)
 - Dimensions (where required)
 - Labels (where asked)
 - Scale indication
5. **Keep drawing clean** - Erase construction lines
6. **Check your work** - Review before submission

FINAL EVALUATION

Answer ALL questions:

1. Define the following terms: (10 marks) a) Working Drawing b) Detail Drawing c) Assembly Drawing d) Cutting Plane e) Floor Plan
2. List FIVE types of information that must appear in a title block. (5 marks)
3. Sketch the schematic representation (side view and end view) of: a) External thread (5 marks) b) Internal thread in section (5 marks)
4. Draw and label FOUR types of nuts. (8 marks)
5. Name and sketch THREE types of sections. (9 marks)
6. Draw the floor plan symbols for: (8 marks) a) Single swing door b) Water closet c) Casement window d) Kitchen sink
7. Sketch a simple King Post Truss and label ALL members. (10 marks)

TOTAL: 60 MARKS

FINAL ASSIGNMENT

PROJECT: Complete Working Drawing Set

Create a complete set of working drawings for a **simple tool box** consisting of:

Components:

1. Base (rectangular tray)
2. Two side panels
3. Two end panels
4. Handle (wooden dowel with end caps)
5. 8 × Wood screws (to join panels)
6. 2 × Small hinges (optional lid)

Required Drawings:

1. **Detail Drawing** for each component (6 sheets minimum):
 - Fully dimensioned orthographic views
 - Material specification
 - Notes on finishing
 - Proper title block
2. **Assembly Drawing** (1 sheet):
 - Exploded view showing how parts fit together
 - Part balloons (①, ②, ③, etc.)
 - Complete Parts List (BOM)
 - Overall dimensions only
 - Assembly notes/instructions

Specifications:

- Use A4 sheets
- Scale: 1:2 or 1:5 (as appropriate)
- First Angle Projection
- Include all standard conventions

SS3 TECHNICAL DRAWING LESSON NOTES

SECOND TERM

WEEK 1: INTRODUCTION TO COMPUTER

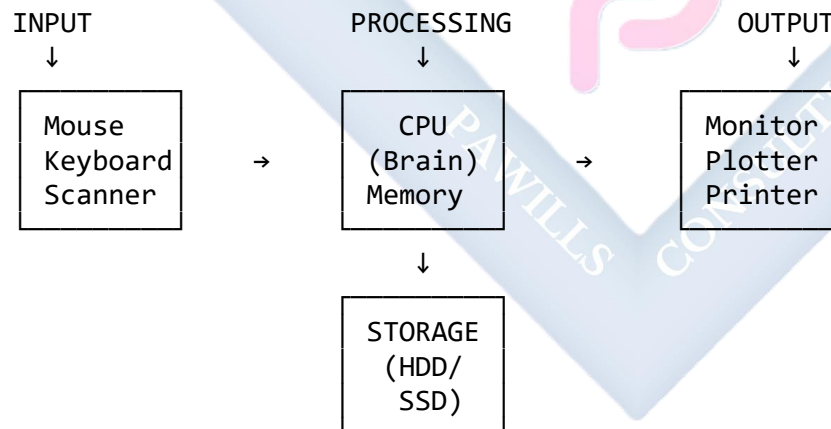
1.1 MEANING OF COMPUTER

A **computer** is an electronic device that can:

- **Accept data** (Input) - through keyboard, mouse, scanner, etc.
- **Process data** (Processing) - according to program instructions
- **Produce results** (Output) - through monitor, printer, plotter
- **Store information** (Storage) - for future retrieval and use

In Technical Drawing Context: Computers have **revolutionized** the design industry, transforming how we create, modify, store, and share technical drawings.

Diagram 61: The Computing Process



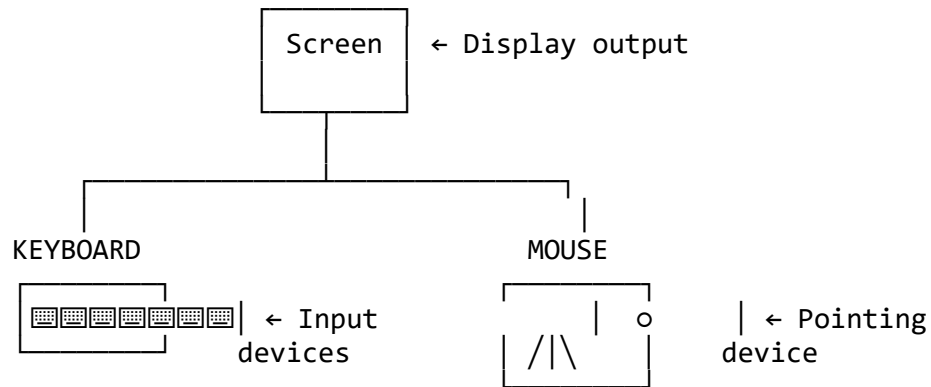
1.2 COMMON HARDWARE COMPONENTS

Hardware refers to the **physical, tangible** parts of a computer system.

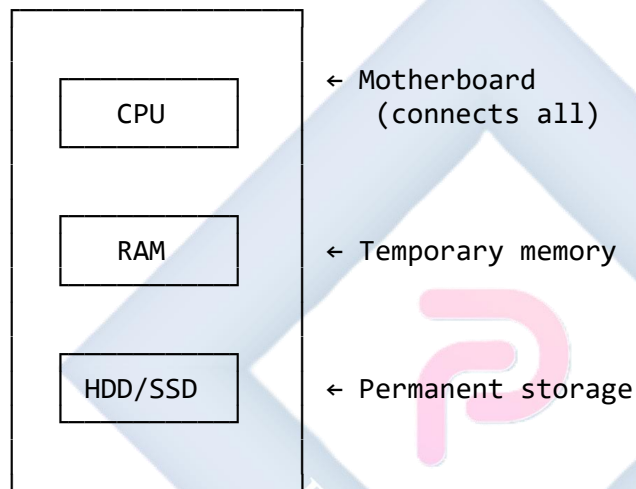
Diagram 62: Computer Hardware Components

COMPLETE COMPUTER SYSTEM

MONITOR

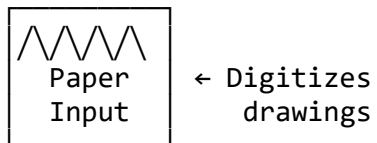


SYSTEM UNIT (Tower)

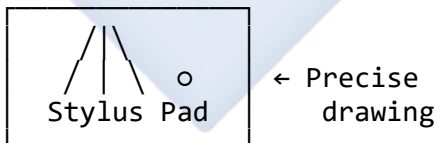


ADDITIONAL DEVICES

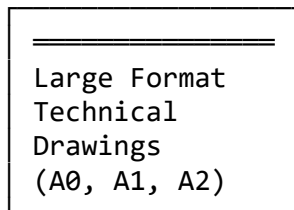
SCANNER



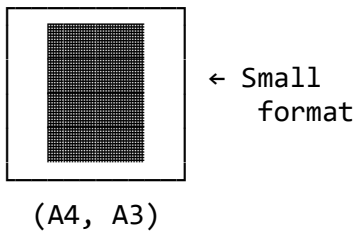
GRAPHICS TABLET



PLOTTER



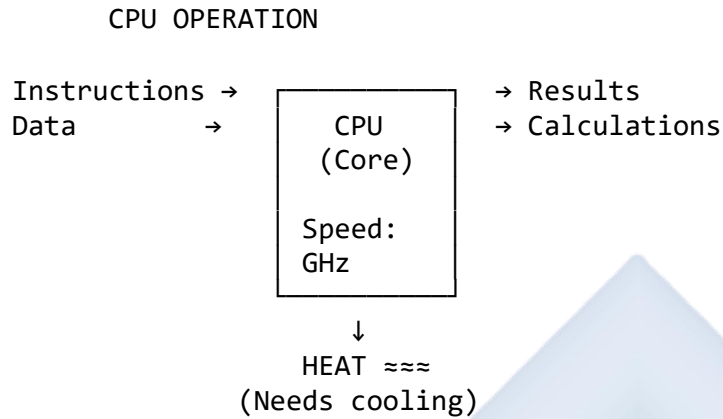
PRINTER



A. CENTRAL PROCESSING UNIT (CPU)

The "**brain**" of the computer that executes all instructions.

Diagram 63: CPU Function



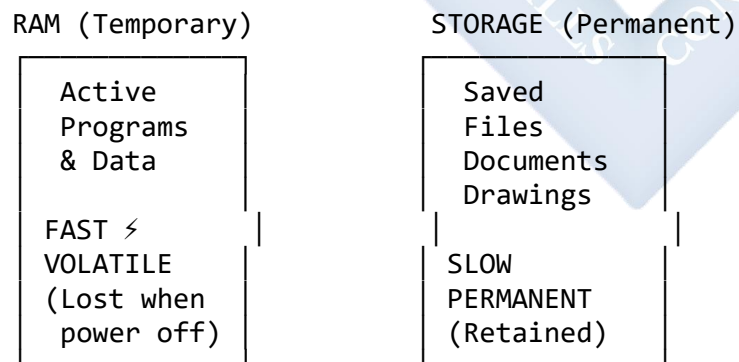
EXAMPLE SPECIFICATIONS:

- Intel Core i7
- AMD Ryzen 9
- Speed: 3.5 GHz
- Cores: 8 (multi-tasking)

B. MEMORY (RAM)

Random Access Memory - temporary "workspace" for active programs and data.

Diagram 64: RAM vs Storage



MORE RAM =

- Smoother performance
- Handle larger drawings
- Run multiple programs
- Less lag/freezing

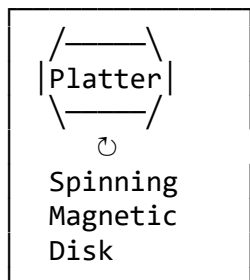
TYPICAL SIZES:

- Basic: 4-8 GB
- Professional CAD: 16-32 GB
- Advanced 3D: 64+ GB

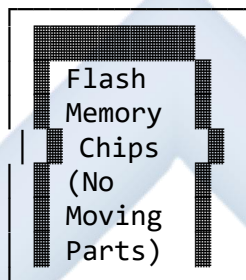
C. STORAGE DEVICES

Diagram 65: HDD vs SSD Comparison

HARD DISK DRIVE (HDD)
Traditional



SOLID STATE DRIVE (SSD)
Modern



SPEED:
DURABILITY: Fragile
COST: Cheap
CAPACITY: Large
NOISE: Audible

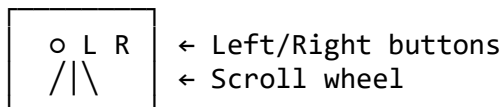
SPEED: ⚡ ⚡ ⚡
DURABILITY: Robust
COST: Expensive
CAPACITY: Moderate
NOISE: Silent

BEST FOR CAD: SSD (Operating System + CAD Software)
HDD (Archived drawings storage)

D. INPUT DEVICES

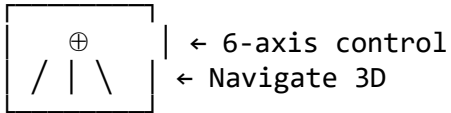
Diagram 66: Input Devices for Technical Drawing

1. STANDARD MOUSE



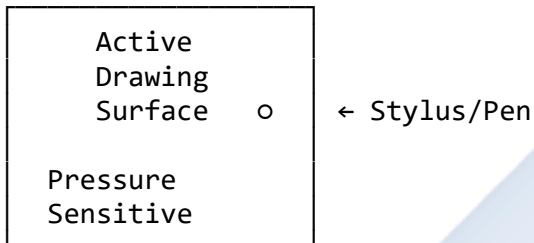
- Point and click
- Basic drawing
- Budget-friendly

2. 3D MOUSE (SpaceMouse)



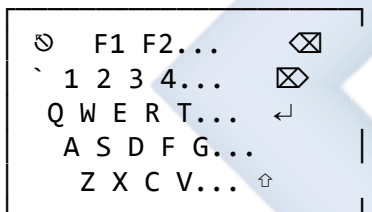
- 3D navigation
- Professional CAD
- Rotate/pan/zoom

3. GRAPHICS TABLET



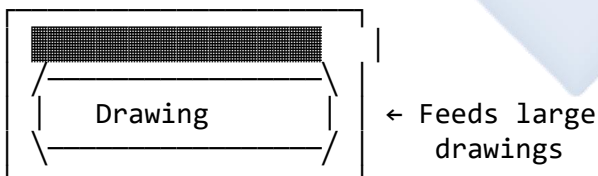
- Freehand sketching
- Precise control
- Pressure levels: 2048-8192

4. KEYBOARD



- Command shortcuts
- Text input
- CAD hotkeys

5. SCANNER (Wide-Format)



- Digitize paper drawings
- Convert to CAD
- Archive old drawings

E. OUTPUT DEVICES

Diagram 67: Output Devices Comparison

1. MONITOR (Display)

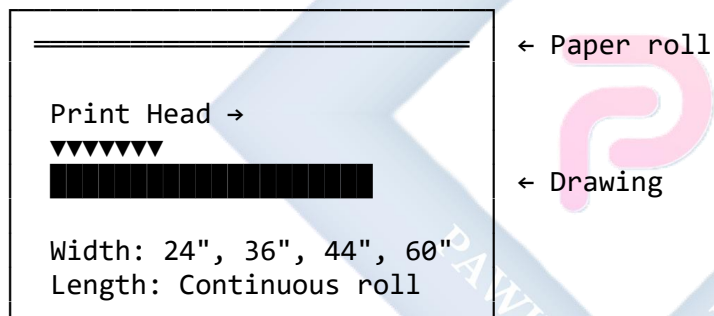
TYPES:

Standard 16:9 24-27"	Wide Ultrawide 21:9 34-38"	Curved Immersive 32-49"
--------------------------------	-------------------------------------	-----------------------------------

KEY SPECIFICATIONS:

- Resolution: 1920×1080 (Full HD)
3840×2160 (4K)
- Panel: IPS (best colors)
TN (fast response)
- Size: 27" minimum for CAD
Dual monitors recommended

2. PLOTTER (Large-Format Printer)



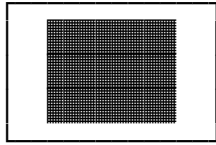
TYPES:

- Inkjet Plotter (Color, detailed)
- Pen Plotter (Traditional, precise lines)
- Laser Plotter (Fast, monochrome)

PAPER SIZES SUPPORTED:

SIZE	DIMENSIONS	USE
A4	210 × 297mm	Letters
A3	297 × 420mm	Small
A2	420 × 594mm	Plans
A1	594 × 841mm	Details
A0	841×1189mm	Large

3. REGULAR PRINTER



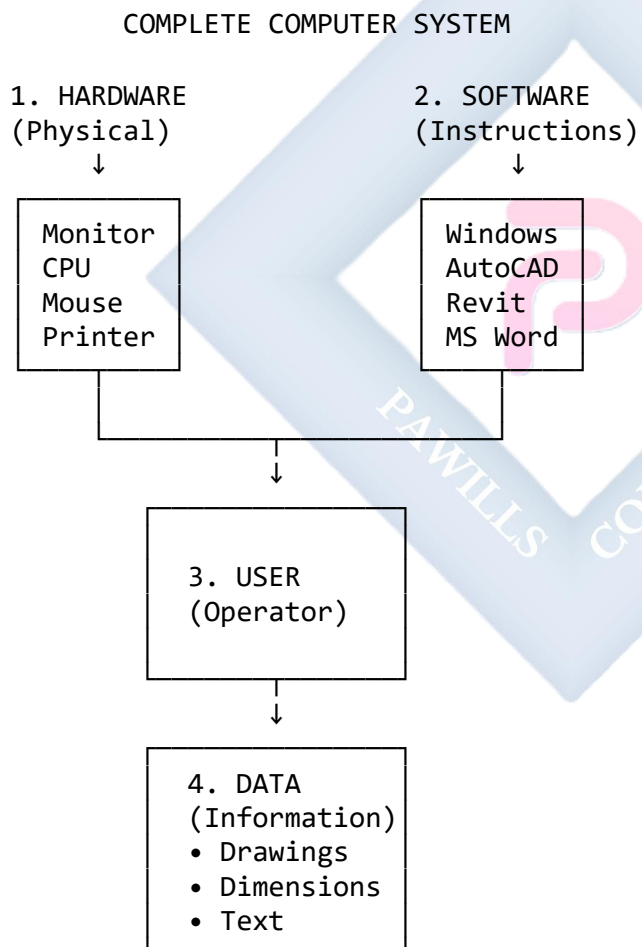
← A4/A3 only

- Office documents
- Small printouts
- Not for technical drawings

1.3 PARTS OF A COMPUTER SYSTEM

A complete system consists of **four components** working together:

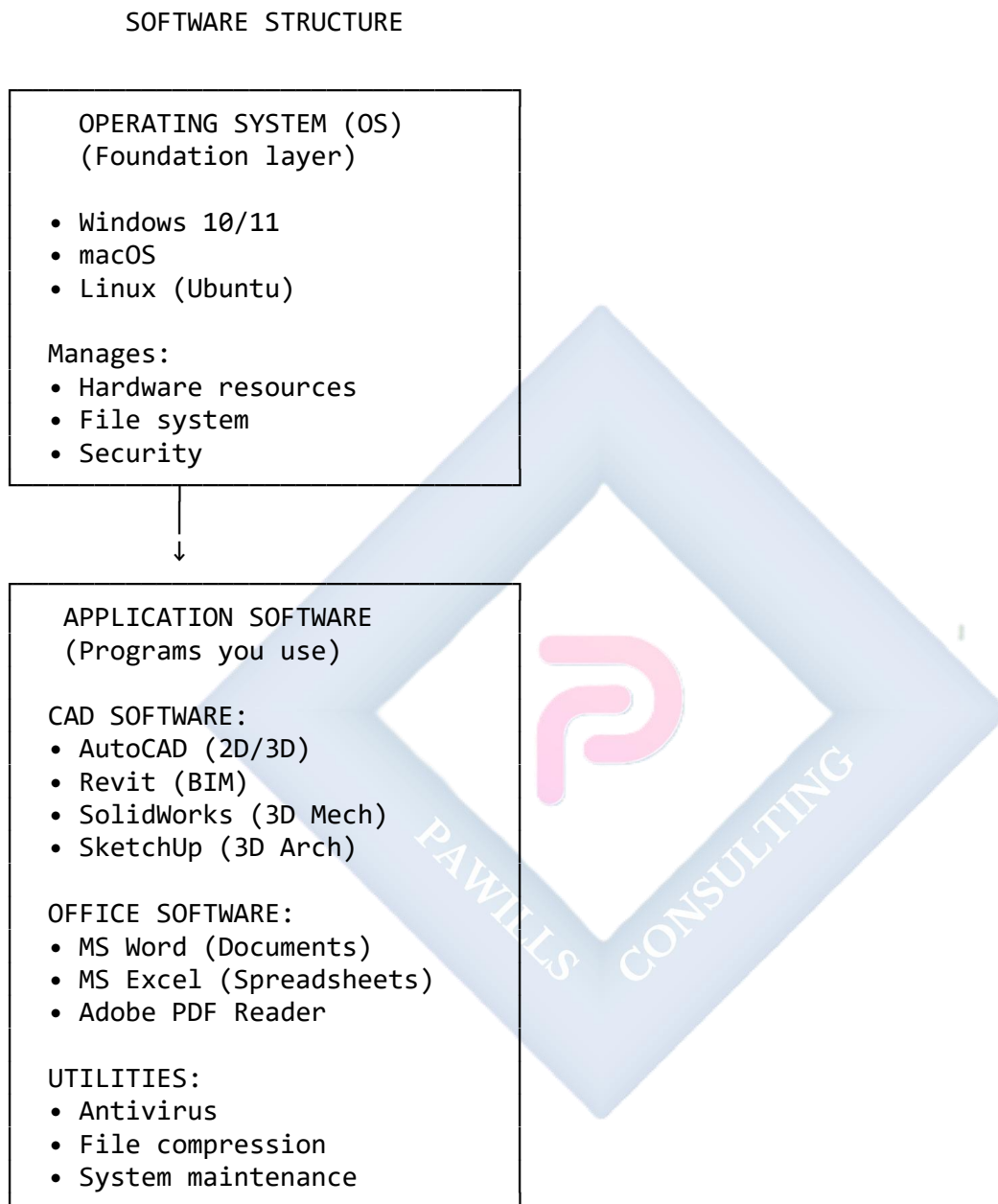
Diagram 68: The Four Components



ALL FOUR MUST WORK TOGETHER!

Software Categories:

Diagram 69: Software Hierarchy



1.4 USING COMPUTERS FOR DESIGN

Diagram 70: Design Applications by Field

ARCHITECTURAL DESIGN
(Buildings)

MECHANICAL ENGINEERING
(Machines/Parts)

Floor Plans
Elevations
Sections
3D Models

SOFTWARE:

- Revit
- ArchiCAD
- AutoCAD Arch
- SketchUp

Part Design
Assemblies
3D Models
Simulations

SOFTWARE:

- SolidWorks
- Inventor
- CATIA
- Fusion 360

CIVIL ENGINEERING
(Infrastructure)

Roads
Bridges
Dams
Site Plans

SOFTWARE:

- Civil 3D
- Bentley
- 12d Model
- Infraworks

ELECTRICAL DESIGN
(Circuits/Systems)

Schematics
PCB Design
Wiring
Layouts

SOFTWARE:

- AutoCAD Elec
- EPLAN
- Eagle
- Altium

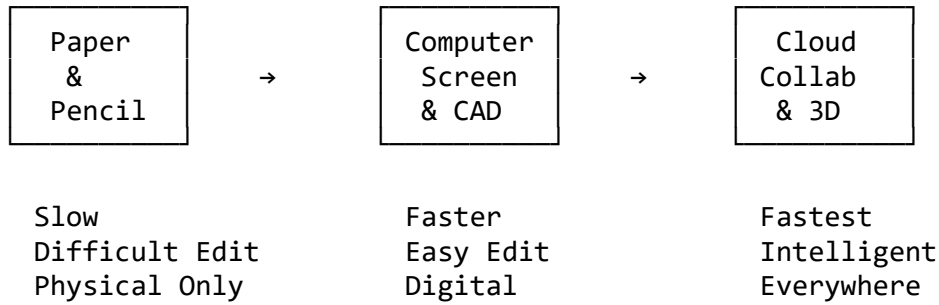
1.5 INTRODUCTION TO CAD APPLICATIONS

CAD = Computer-Aided Design (or Computer-Aided Drafting)

Diagram 71: CAD Evolution

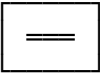
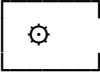
DRAWING EVOLUTION

1950s-1980s MANUAL DRAWING	→	1980s-2000s 2D CAD	→	2000s-Present 3D CAD + BIM
Drawing Board T-Square Pencils/Pens Physical Storage		AutoCAD 2D Lines Layers Digital Files		Revit/BIM 3D Models Database Cloud Storage



Popular CAD Software:

Diagram 72: CAD Software Comparison

MAJOR CAD PLATFORMS	
AutoCAD 	• Industry standard • 2D drafting + 3D modeling • General purpose • Architecture, Mech, Civil • .DWG file format
Revit 	• Building Information Modeling • Intelligent 3D building design • Automatic coordination • Architecture focused • .RVT file format
SolidWorks 	• 3D mechanical design • Parametric modeling • Assembly & parts • Manufacturing focus • .SLDPRT/.SLDASM files
SketchUp 	• Easy 3D modeling • Intuitive interface • Quick conceptual design • Free version available • .SKP file format
ArchiCAD 	• BIM for architects • European standard • Team collaboration • Detailed documentation • .PLA file format

Benefits of CAD:

Diagram 73: CAD vs Manual Drawing

CAD ADVANTAGES

1. SPEED $\angle \angle \angle$

Manual: Draw window 100 times $\rightarrow$ 100x effort

CAD: Draw once, COPY 100 times $\rightarrow$ 1x effort

2. ACCURACY

Manual: $\pm 0.5\text{mm}$ (at best)

CAD: $\pm 0.001\text{mm}$ (perfect precision)

3. EDITING

Manual: Erase $\rightarrow$ Messy $\rightarrow$ Redraw

[Before] [After - damaged]



CAD: SELECT $\rightarrow$ MOVE $\rightarrow$ DONE

[Before] [After - perfect]



4. STORAGE

Manual: 100 drawings = Large cabinet

- Fire risk
- Water damage
- Pest damage
- Takes space

CAD: 1000s drawings = Small USB

- Cloud backup
- Searchable
- Portable

5. SHARING

Manual: Plot $\rightarrow$ Package $\rightarrow$ Courier $\rightarrow$ Days

CAD: Email/Cloud $\rightarrow$ Instant $\rightarrow$ Seconds

6. VISUALIZATION

Manual: 2D only (Plan, Elevation, Section)

Isometric = Very difficult

- CAD: Easy 3D viewing
- Rotate any angle
 - Walkthroughs
 - Realistic rendering
 - VR experiences

7. COLLABORATION

- Manual: One person, one time
- CAD: Multiple people, simultaneously
- Cloud-based
 - Version control
 - Track changes

8. STANDARDIZATION

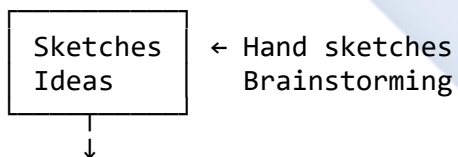
- Manual: Varies by draughtsman
- CAD:
- Blocks/Templates
 - Consistent symbols
 - Company standards
 - Quality control

Real-World CAD Workflow:

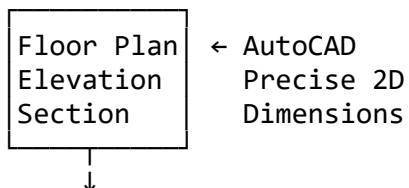
Diagram 74: Typical CAD Project Workflow

PROJECT LIFECYCLE

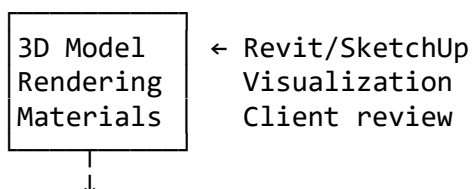
1. CONCEPTUAL DESIGN



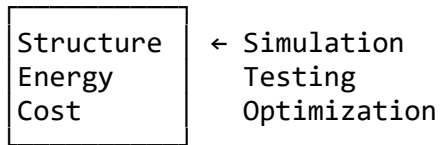
2. 2D CAD DRAFTING



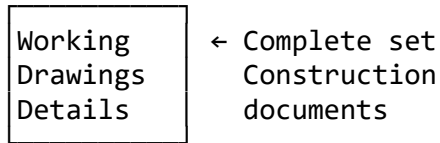
3. 3D MODELING



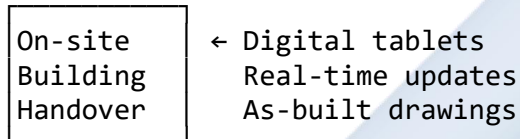
4. ANALYSIS & TESTING



5. DOCUMENTATION



6. CONSTRUCTION



EVALUATION

1. What is the difference between **hardware** and **software**? Give two examples of each.
2. What is the specific name for a **large-format printer** used for technical drawings?
3. Define "CAD" in full and list **four major benefits** it has over manual drawing.
4. Explain the difference between **RAM** and **Storage** (HDD/SSD).
5. What is the function of the **CPU** in a computer system?
6. Name **three** CAD software programs and state what they are primarily used for.
7. Why is an **SSD** preferred over an **HDD** for running CAD software?

ASSIGNMENT

Part 1: Research Task

1. Research **ONE** CAD application (choose from: AutoCAD, Revit, SolidWorks, or SketchUp).
2. Write a **paragraph (8-10 lines)** covering:
 - What is the software used for?
 - Key features (at least 3)
 - File formats it uses

- Industries that use it
- One advantage over other software

Part 2: Hardware Identification

3. Create a **table** listing:

DEVICE	TYPE	I/O	FUNCTION
Example: Mouse	Hard	Input	Pointing
1. Monitor 2. Keyboard 3. CPU 4. Scanner 5. Plotter 6. RAM 7. HDD/SSD 8. Graphics Tab			

TYPE: Hardware or Software

I/O: Input, Output, Processing, or Storage

Part 3: Visit & Report

4. If possible, visit a **computer lab** or **design office** in your area.
 5. List the hardware you observed being used for CAD work.
 6. Note: What CAD software was installed?
-

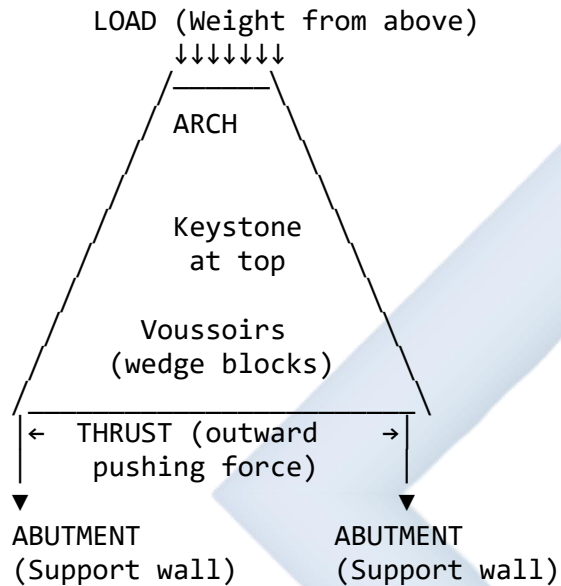
WEEK 2: ARCHES

2.1 MEANING OF ARCHES

An **arch** is a **curved structural element** that spans an opening and supports the load above it. It is one of the oldest and most elegant structural solutions in architecture and civil engineering.

Diagram 75: Arch Concept

HOW AN ARCH WORKS



PRINCIPLE:

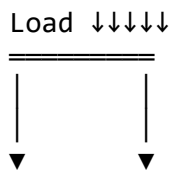
Vertical load → Converts to →
Diagonal compression through arch →
Horizontal thrust to supports

2.2 IMPORTANCE OF ARCHES

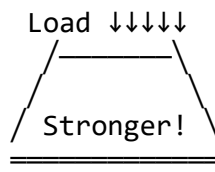
Diagram 76: Why Arches Matter

1. STRUCTURAL STRENGTH

LINTEL (Straight beam)



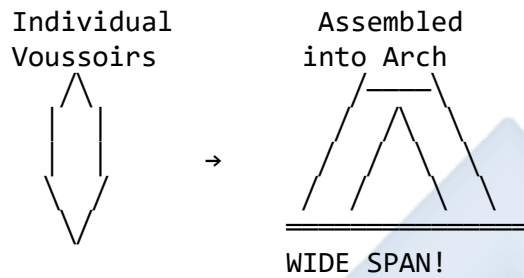
ARCH (Curved)



- Bends in middle
- Limited span
- Weak
- Distributes load
- Wider span possible
- Very strong

2. SPANNING CAPABILITY

Small blocks/bricks → Large opening



3. AESTHETIC BEAUTY

Famous Arches:

- Roman Colosseum (72 AD)
- Gothic Cathedrals (1200s)
- Islamic Architecture
- Modern bridges

4. MATERIAL EFFICIENCY

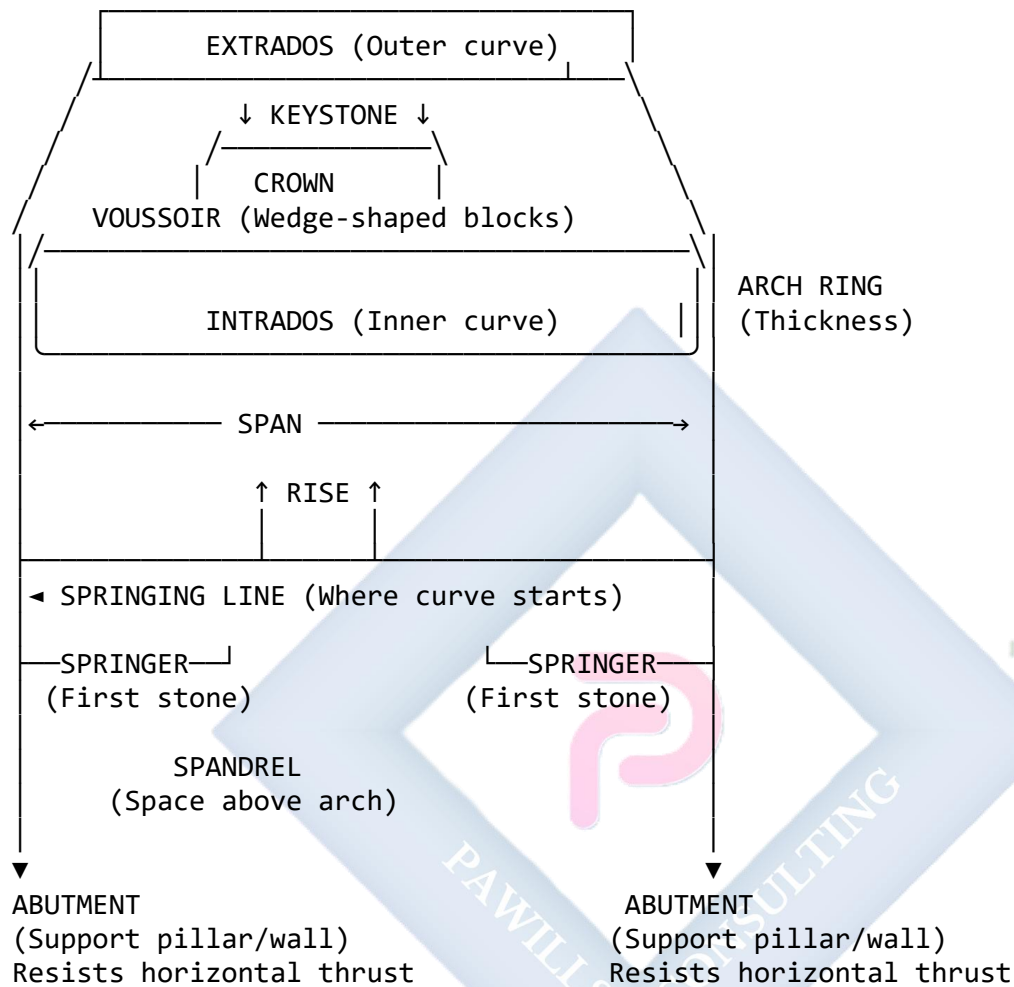
Uses: • Stone • Brick • Concrete

No need for:

- Large single beams
- Complex reinforcement
- Expensive materials

2.3 KEY TERMINOLOGY OF A STONE ARCH

Diagram 77: Arch Terminology (Comprehensive)



DEFINITIONS:

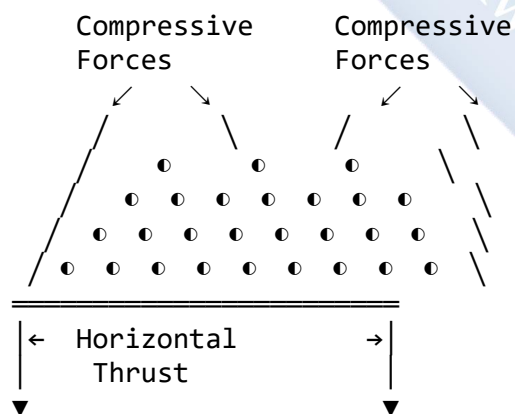
1. **SPAN:** Clear horizontal distance of opening
(Measured at springing line)
2. **RISE:** Vertical height from springing line
to highest point (crown/keystone)
3. **SPRINGING LINE:** Imaginary horizontal line
where arch curve begins
4. **SPRINGER:** First voussoir at each end
(Sits on abutment)
5. **VOUSSOIR:** Wedge-shaped stone/brick block

(French: "wedge")

6. KEYSTONE: Central top voussoir
Locks all others in place
(Often decorated)
7. INTRADOS: Inner concave surface
(Underside of arch)
8. EXTRADOS: Outer convex surface
(Top side of arch)
9. ABUTMENT: Vertical support structure
Resists horizontal thrust
10. CROWN: Highest point of arch
(At keystone)
11. ARCH RING: Structural thickness
(Depth of voussoirs)
12. SPANDREL: Triangular space between
arch and rectangular frame

Diagram 78: Forces in an Arch

STRUCTURAL BEHAVIOR



KEY PRINCIPLES:

- All forces are COMPRESSION
(stones being squeezed)
- NO tension (no pulling apart)
- NO bending (like a beam)

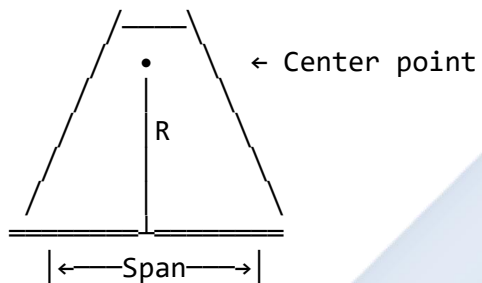
- Very efficient structure
- Can last thousands of years!

2.4 TYPES OF ARCHES

Diagram 79: Arch Types - Complete Guide

1. SEMICIRCULAR (ROMAN) ARCH

Most common traditional form



$$\text{Rise} = \text{Radius} = \text{Span}/2$$

CHARACTERISTICS:

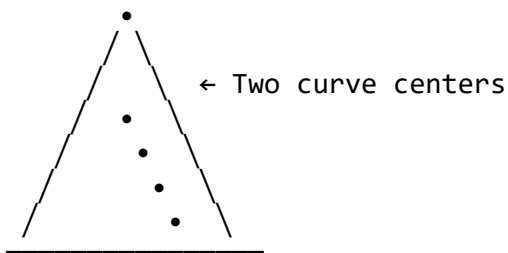
- Perfect semicircle (180°)
- Rise = 50% of span
- Very stable
- Even load distribution

USES:

- Roman aqueducts
- Colosseum
- Traditional bridges
- Classical buildings

2. POINTED (GOTHIC) ARCH

Medieval innovation



Two arcs meet at point
Centers outside arch

CHARACTERISTICS:

- Sharp point at apex
- $\text{Rise} > \text{Span}/2$
- Directs loads vertically
- Less horizontal thrust

USES:

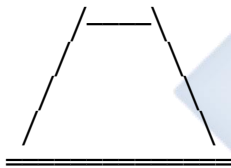
- Gothic cathedrals
- Churches
- Tall buildings
- Narrow openings

ADVANTAGE:

- Allows thinner walls
- Taller structures
- Large windows

3. SEGMENTAL ARCH

Shallow curve



← Segment of circle
(less than 180°)

- ← Center below
springing line

$\text{Rise} < \text{Span}/2$

CHARACTERISTICS:

- Shallow profile
- $\text{Rise} = 15\text{-}30\%$ of span
- Greater horizontal thrust
- Economy of height

USES:

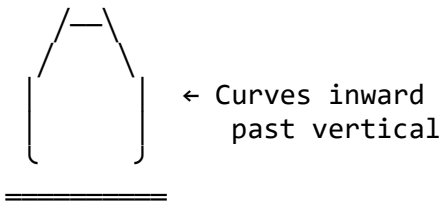
- Bridge arches
- Low doorways
- Windows
- Modern applications

ADVANTAGE:

- Saves vertical space
- Less material needed

4. HORSESHOE ARCH

Islamic/Moorish style



Curves more than 180°

CHARACTERISTICS:

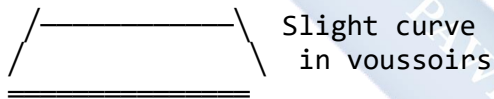
- Exceeds semicircle
- Curves $>180^\circ$
- Decorative appearance
- Cultural significance

USES:

- Islamic mosques
- Moorish palaces
- Alhambra, Spain
- North African arch.

5. FLAT (JACK) ARCH

Appears flat



CHARACTERISTICS:

- Looks horizontal
- Wedge-shaped blocks
- Flat intrados
- Weak structure

USES:

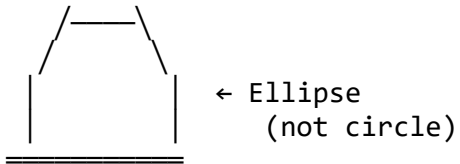
- Decorative lintels
- Windows
- Doorways
- Non-load-bearing

NOTE:

- Not true arch
- Less structural
- Mainly aesthetic

6. ELLIPTICAL ARCH

Oval shape



CHARACTERISTICS:

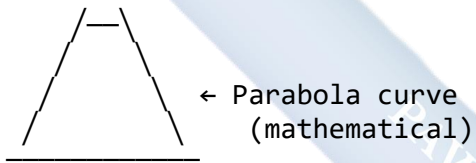
- Oval/ellipse shape
- Rise can vary
- Aesthetic appeal
- Complex construction

USES:

- Renaissance bridges
- Decorative features
- Modern architecture

7. PARABOLIC ARCH

Modern engineering



CHARACTERISTICS:

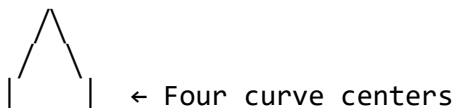
- Mathematical curve
- Ideal load distribution
- Modern materials
- Efficient structure

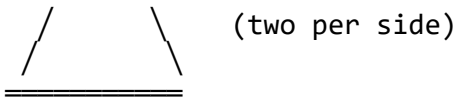
USES:

- Modern bridges
- Long spans
- Concrete arches
- Steel structures

8. TUDOR ARCH (Four-CENTERED)

English style





CHARACTERISTICS:

- Flattened point
- Four arc centers
- Wide, low opening
- Decorative

USES:

- Tudor buildings
- English architecture
- 15th-16th century

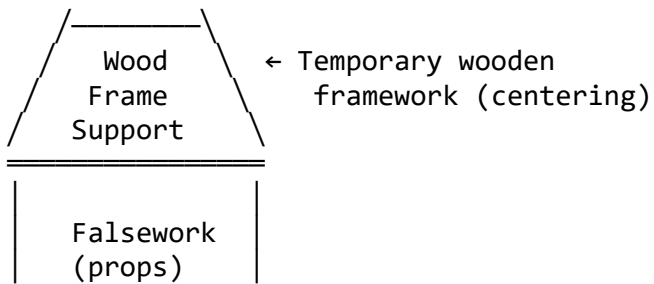
Diagram 80: Arch Comparison Chart

TYPE	RISE /SPAN	STRENGTH RATING	TYPICAL USE
Semicircular (Roman)	50% (1:2)	★★★★★	Traditional buildings
Pointed (Gothic)	>50% (varies)	★★★★★	Cathedrals, tall openings
Segmental	15-30% (<1:2)	★★★	Bridges, windows
Horseshoe	>50% (>180°)	★★★★★	Islamic architecture
Flat (Jack)	~5% (minimal)	★★	Decorative lintels
Elliptical	30-40% (varies)	★★★★★	Renaissance, bridges
Parabolic	varies (optimized)	★★★★★	Modern bridges

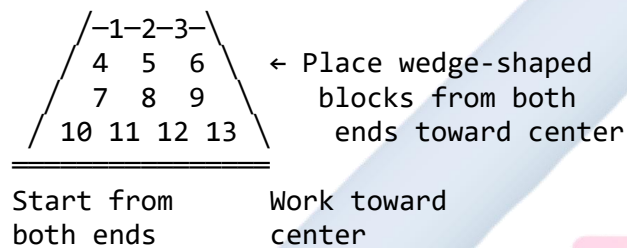
2.5 CONSTRUCTION OF ARCHES

Diagram 81: Arch Construction Process

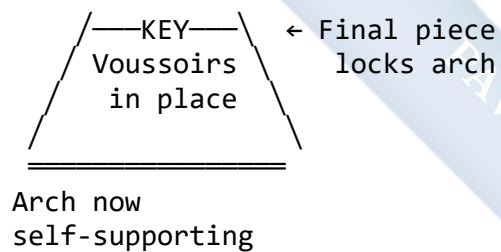
STEP 1: BUILD CENTERING (Temporary support)



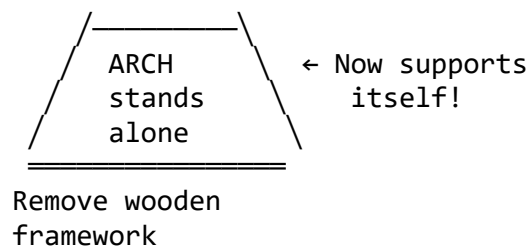
STEP 2: LAY VOUSSOIRS



STEP 3: INSERT KEYSTONE



STEP 4: REMOVE CENTERING



2.6 IDENTIFYING ARCHES IN BUILDINGS

Practical Exercise:

Look for arches in your community:

- **Churches/Mosques** - Often have pointed or horseshoe arches
- **Old bridges** - Usually semicircular or segmental
- **Government buildings** - May have Roman-style arches
- **Modern buildings** - Might have parabolic or flat arches
- **Doorways/Windows** - Various types

Take photos and sketch the arches you find!

EVALUATION

1. Explain the **primary structural function** of an arch. How does it convert forces?
2. What is the "**keystone**," and why is it important to the arch's stability?
3. What is the difference in shape between a **Roman (semicircular) arch** and a **Segmental arch**?
4. Define "**Span**" and "**Rise**." What is the ratio of rise to span in a semicircular arch?
5. Sketch and label: **Voussoir**, **Keystone**, **Intrados**, **Extrados**, **Abutment**.
6. Why did Gothic architects prefer **pointed arches** over Roman arches for cathedrals?
7. What type of arch would you use for a **low bridge** where height is limited?

ASSIGNMENT

Part 1: Drawing Practice

1. On a drawing sheet (A4), using **compass and ruler**:
 - Draw a **Semicircular Arch** with:
 - Span = 100mm
 - Show construction (center point, radius)
 - Label: Span, Rise, Keystone position
2. On the same sheet, draw a **Pointed (Gothic) Arch** with:
 - Same span = 100mm
 - Two curve centers (show construction)
 - Label: Span, Rise, Arc centers

Part 2: Labeling Exercise

3. Sketch a **complete arch** showing:

- Keystone
- Voussoirs (at least 7)
- Springer (both sides)
- Springing Line
- Intrados
- Extrados
- Abutments
- Span dimension
- Rise dimension

Part 3: Research Task

4. Find and photograph (or sketch) **THREE different arches** in your local area:

- Identify the type of each arch
- Note the building/structure name
- Estimate the span (approximate)
- State the probable date of construction
- Explain why that particular arch type was chosen

Example format:

ARCH SURVEY REPORT

Arch 1: St. Mary's Church Entrance

Type: Pointed (Gothic) Arch

Span: Approximately 2.5m

Date: Estimated 1920s

Reason: Creates tall, grand entrance;
traditional church architecture

WEEK 3: ROOFS

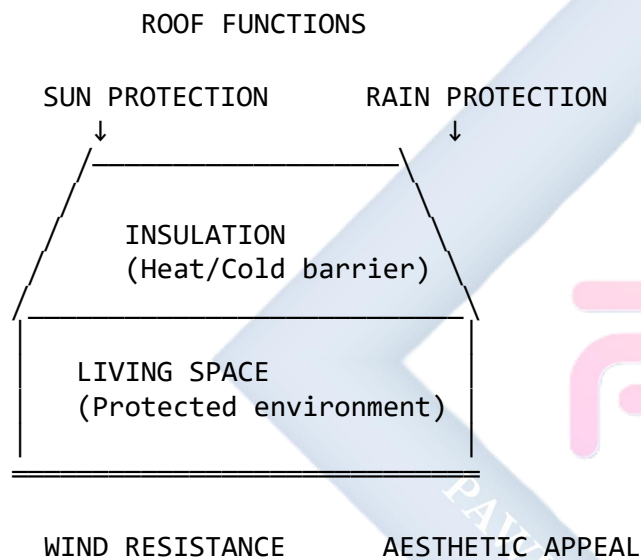
3.1 MEANING OF ROOF

A **roof** is the uppermost structural covering of a building that serves multiple critical functions:

Primary Functions:

1. **Protection** - From rain, sun, wind, snow
2. **Insulation** - Temperature and sound control
3. **Aesthetics** - Defines building appearance
4. **Structural** - Completes building envelope

Diagram 82: Roof Functions

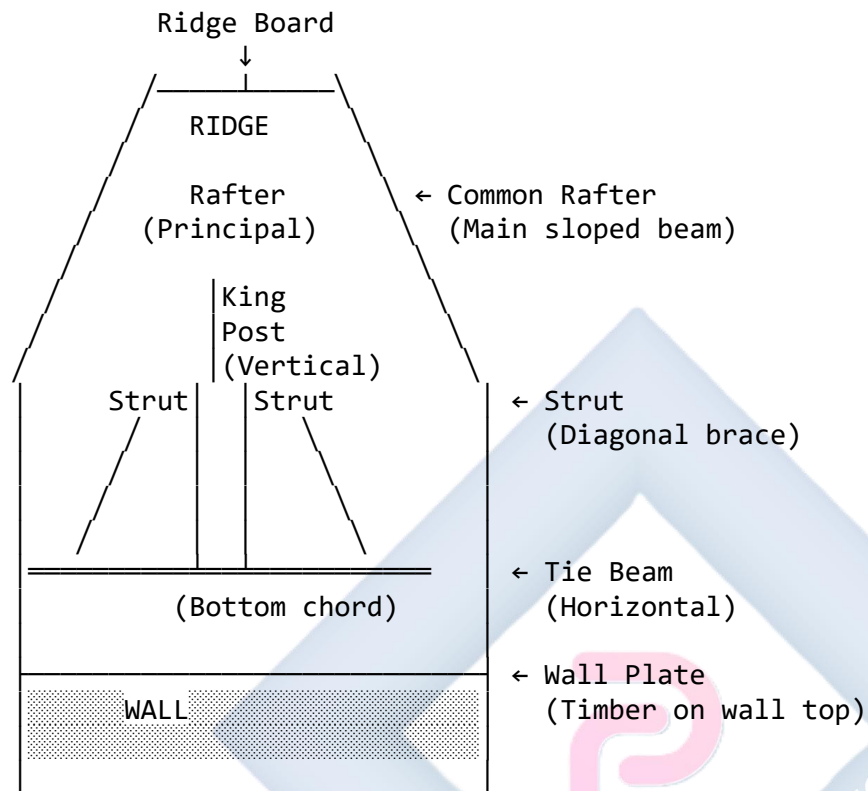


3.2 ROOF TERMINOLOGY (ROOF MEMBERS)

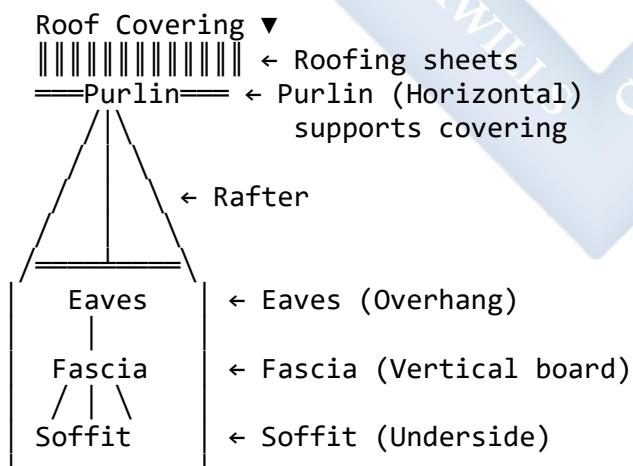
Understanding roof components is essential for reading building sections and constructing roofs properly.

Diagram 83: Complete Roof Structure - King Post Truss

ROOF STRUCTURE COMPONENTS



OUTSIDE VIEW

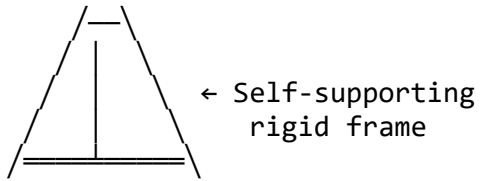


Detailed Component Definitions:

Diagram 84: Roof Members - Individual Components

1. TRUSS

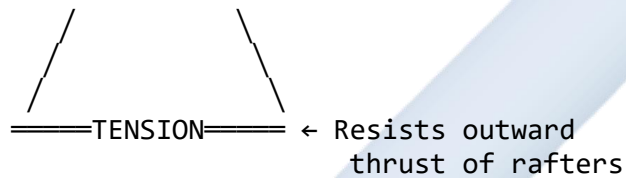
Complete triangular framework



Types: King Post, Queen Post, Fink, Howe, Scissors

2. TIE BEAM

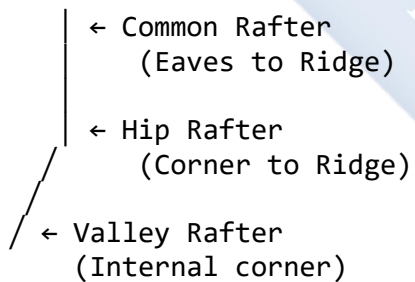
Bottom horizontal member



Function: Prevents walls from spreading

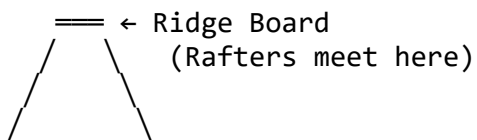
3. RAFTER

Main sloped beam



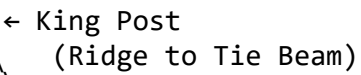
4. RIDGE BOARD/RIDGE

Peak of roof



5. KING POST

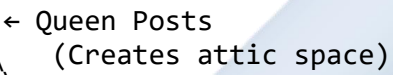
Central vertical member



Function: Supports ridge
Prevents sag

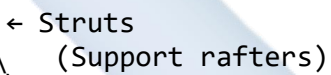
6. QUEEN POSTS

Two vertical members



7. STRUT

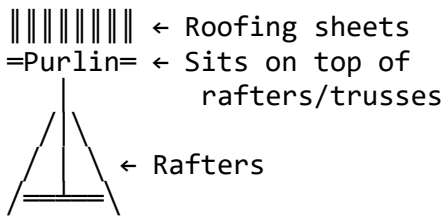
Diagonal brace



Function: Prevent rafter sag
Transfer loads

8. PURLIN

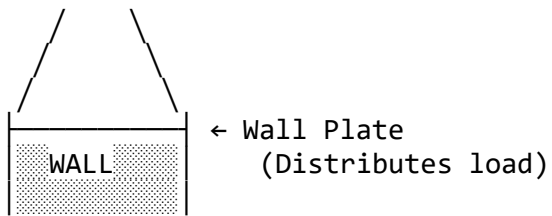
Horizontal beam on rafters



Function: Support roofing
Reduce rafter span

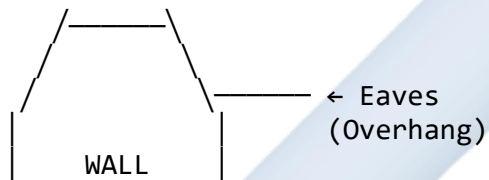
9. WALL PLATE

Timber on wall top



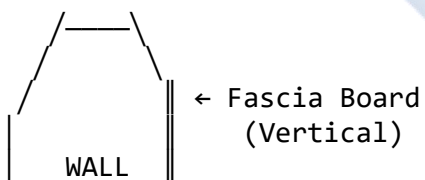
- Function:
- Transfer roof load
 - Base for rafters
 - Level bearing

10. EAVES Roof overhang



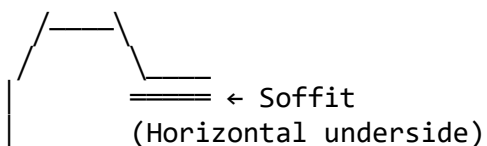
- Purpose:
- Protect walls
 - Direct water away
 - Shade windows

11. FASCIA Vertical board at eaves



- Function:
- Finish eaves
 - Gutter attachment
 - Weather protection

12. SOFFIT Underside of eaves

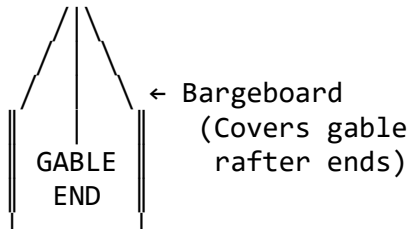


| WALL

- Function:
- Close eaves
 - Ventilation
 - Pest protection

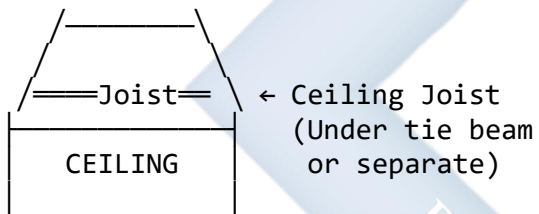
13. BARGEBOARD

Gable end trim



14. CEILING JOIST

Supports ceiling



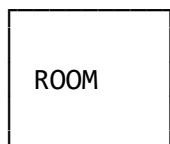
3.3 TYPES OF ROOFS (DESIGN)

Diagram 85: Roof Types - Comprehensive Guide

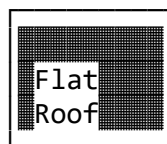
1. FLAT ROOF

Horizontal or slight slope

ELEVATION



PLAN VIEW



Slope: 1-5° (for drainage)

CHARACTERISTICS:

- Modern appearance

- Usable roof space
- Simple construction
- Lower initial cost

USES:

- Commercial buildings
- Modern homes
- Low rainfall areas
- Urban architecture

MATERIALS:

- Bitumen membrane
- PVC membrane
- Concrete

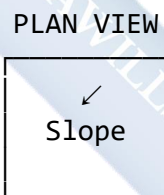
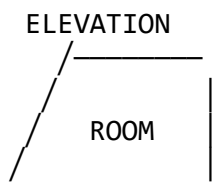
ADVANTAGES:

- Additional space (terrace)
- Easy maintenance access
- HVAC equipment platform

DISADVANTAGES:

- Poor drainage
- Requires maintenance
- Not ideal for heavy rain

2. SHED ROOF (Lean-to) Single slope



One sloped surface

CHARACTERISTICS:

- Simple design
- Economical
- Single direction slope
- Easy construction

USES:

- Additions/extensions
- Sheds/outbuildings
- Verandas/porches
- Modern architecture

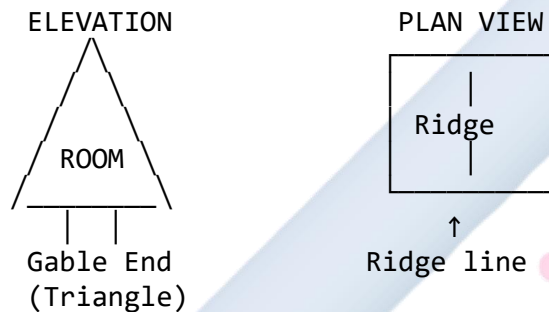
ADVANTAGES:

- Low cost
- Quick construction
- Good water runoff
- Maximizes one wall height

DISADVANTAGES:

- Limited attic space
- Asymmetrical appearance

3. GABLE ROOF (Pitched)
Classic triangular



CHARACTERISTICS:

- Two sloping sides
- Triangular ends (gables)
- Ridge at center
- Most common type

USES:

- Residential houses
- Traditional buildings
- Worldwide standard
- All climates

ADVANTAGES:

- Excellent water drainage
- Attic/storage space
- Natural ventilation
- Simple construction
- Cost-effective

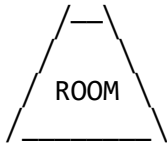
DISADVANTAGES:

- Vulnerable in high winds
- Gable ends need bracing

4. HIP ROOF

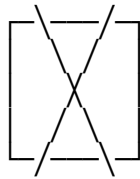
Four sloped sides

ELEVATION



No gable
(All slopes)

PLAN VIEW



← Hip lines
(Diagonal)

↑
Ridge line

CHARACTERISTICS:

- Four sloping sides
- No vertical gable ends
- Hips at corners
- Self-bracing structure

USES:

- Residential houses
- High wind areas
- Coastal regions
- Traditional style

ADVANTAGES:

- Excellent wind resistance
- Even load distribution
- Self-supporting
- Aesthetic appeal
- All-round eaves

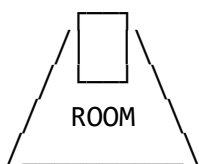
DISADVANTAGES:

- More complex framing
- Higher cost
- Less attic space
- Difficult construction

5. GAMBREL ROOF

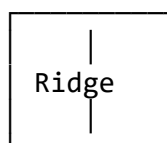
Double slope (Barn roof)

ELEVATION



Two slopes each side:

PLAN VIEW



- Steep lower
- Shallow upper

CHARACTERISTICS:

- Four slope surfaces
- Two angles per side
- Barn-style appearance
- Maximum attic space

USES:

- Barns
- Farm buildings
- Dutch Colonial homes
- Storage buildings

ADVANTAGES:

- Maximum usable attic
- Extra headroom
- Efficient space
- Traditional charm

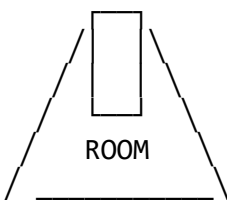
DISADVANTAGES:

- Complex framing
- Higher maintenance
- Snow load on lower slope

6. MANSARD ROOF

French style (Four-sided gambrel)

ELEVATION



PLAN VIEW



Double slope on all 4 sides

CHARACTERISTICS:

- Four-sided gambrel
- Steep lower slopes
- Flat or low-pitch top
- French architecture

USES:

- French buildings
- Victorian houses

- Urban townhouses
- Historic restoration

ADVANTAGES:

- Full extra floor
- Dormer windows
- Maximum space
- Elegant appearance

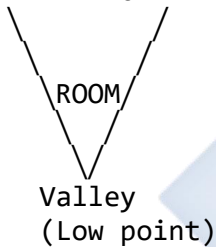
DISADVANTAGES:

- Very complex
- Expensive
- Local codes may restrict

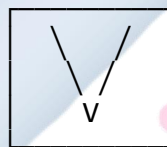
7. BUTTERFLY ROOF

Inverted gable (V-shape)

ELEVATION



PLAN VIEW



Valley
(Center)

CHARACTERISTICS:

- Two slopes inward
- Center valley
- Modern design
- Opposite of gable

USES:

- Modern architecture
- Eco-friendly homes
- Desert climates
- Artistic designs

ADVANTAGES:

- Natural lighting
- Rainwater collection
- Unique appearance
- High ceilings

DISADVANTAGES:

- Water pooling risk
- Complex drainage
- Not for heavy rain

- Higher maintenance

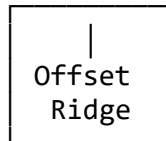
8. SALTBOX ROOF

Asymmetrical gable

ELEVATION



PLAN VIEW



Long back slope

CHARACTERISTICS:

- Asymmetrical
- Long rear slope
- Short front slope
- Colonial American

USES:

- Colonial houses
- Traditional New England
- Historic buildings

ADVANTAGES:

- Extra rear space
- Snow sheds easily
- Unique character
- Weather protection

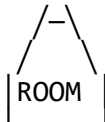
DISADVANTAGES:

- Uneven interior space
- Complex framing
- Less common

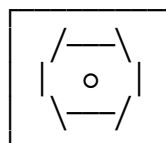
9. DOME ROOF

Hemispherical

ELEVATION



PLAN VIEW



Circular structure

Circular plan

CHARACTERISTICS:

- Hemispherical shape
- No angles/ridges
- Strong structure
- Iconic appearance

USES:

- Mosques/churches
- Planetariums
- Sports arenas
- Observatory

ADVANTAGES:

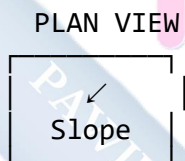
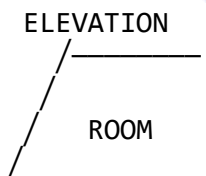
- Very strong
- Minimal materials
- Energy efficient
- Unique aesthetics

DISADVANTAGES:

- Complex construction
- Specialized skills
- Interior space loss

10. SKILLION ROOF

Modern single slope



Similar to shed but steeper

CHARACTERISTICS:

- Single steep slope
- Modern aesthetic
- Simple construction
- Efficient drainage

USES:

- Contemporary homes
- Additions
- Modern commercial

ADVANTAGES:

- Cost-effective
- Quick build
- Good drainage

- Solar panel friendly

DISADVANTAGES:

- Limited ceiling height
- Less attic space

Diagram 86: Roof Type Selection Guide

ROOF TYPE	CLIMATE	COST	COMPLEXITY
Flat	Dry	Low	Simple
Shed	Any	Very Low	Very Simple
Gable	Any	Low	Simple
Hip	Windy	Medium	Medium
Gambrel	Moderate	Medium	Medium
Mansard	Urban	High	Complex
Butterfly	Dry/Mod	Medium	Medium
Saltbox	Cold	Medium	Medium
Dome	Any	High	Very Complex
Skillion	Any	Very Low	Very Simple

RECOMMENDED FOR NIGERIA:

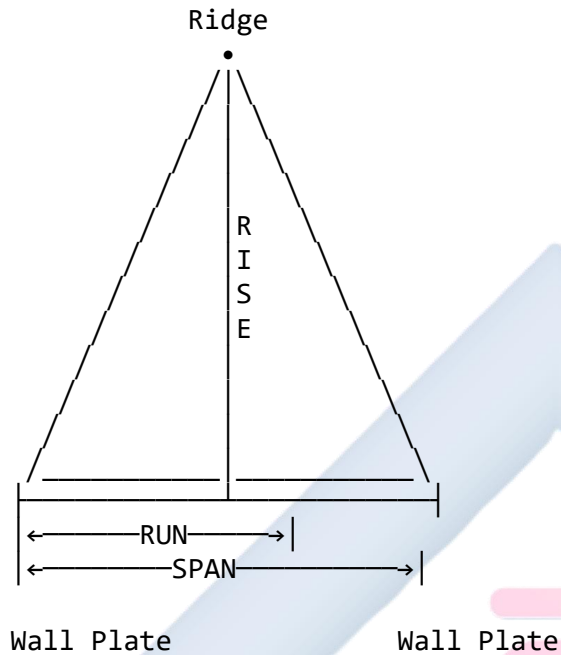
- Gable Roof - Most common, good drainage
- Hip Roof - Wind resistance (coastal areas)
- Shed Roof - Budget construction

3.4 DETERMINING ROOF PITCH

Pitch = Steepness/slope of the roof

Diagram 87: Roof Pitch Calculation

PITCH COMPONENTS



DEFINITIONS:

SPAN: Total width of building
(Wall to wall)

RUN: Half of span
(Wall to center/ridge)
 $\text{Run} = \text{Span} \div 2$

RISE: Vertical height
(Wall plate to ridge)

PITCH FORMULA:

$$\text{Pitch} = \text{Rise} \div \text{Run}$$

WORKED EXAMPLES:

Example 1:
Building span = 10m

Rise = 2.5m

Step 1: Calculate Run

Run = Span $\div$ 2

Run = 10m $\div$ 2 = 5m

Step 2: Calculate Pitch

Pitch = Rise $\div$ Run

Pitch = 2.5m $\div$ 5m

Pitch = 0.5 (or 1:2 or 50%)

Example 2:

Span = 8m

Rise = 3m

Run = 8m $\div$ 2 = 4m

Pitch = 3m $\div$ 4m = 0.75

(or 3:4 or 75%)

Example 3:

Span = 12m

Rise = 4m

Run = 12m $\div$ 2 = 6m

Pitch = 4m $\div$ 6m = 0.67

(or 2:3 or 67%)

Diagram 88: Pitch Expression Methods

EXPRESSING PITCH

1. AS RATIO

Rise : Run

Examples: 1:2, 1:3, 2:5, 3:4

2. AS FRACTION

Rise/Run

Examples: 1/2, 1/3, 2/5, 3/4

3. AS DECIMAL

Rise $\div$ Run

Examples: 0.5, 0.33, 0.4, 0.75

4. AS PERCENTAGE

$$(\text{Rise} \div \text{Run}) \times 100$$

Examples: 50%, 33%, 40%, 75%

5. AS ANGLE (Degrees)

$$\theta = \arctan(\text{Rise} \div \text{Run})$$

Pitch → Angle

1:2 → 26.6°

1:3 → 18.4°

2:5 → 21.8°

3:4 → 36.9°

1:1 → 45°

6. AS "X-IN-12" (Common in USA)

X units rise per 12 units run

Pitch → X-in-12

1:2 → 6:12

1:3 → 4:12

2:5 → 4.8:12

3:4 → 9:12

VISUAL PITCH GUIDE

Very Flat (1:12 = 8.3%)

Flat (2:12 = 16.7%)

/—

Low (3:12 = 25%)

/—

Moderate (4:12 = 33%)

/—

Standard (6:12 = 50%)

/—

Steep (9:12 = 75%)

/—

Very Steep (12:12 = 100% = 45°)

Diagram 89: Pitch Selection Guide

RECOMMENDED PITCH BY ROOFING MATERIAL

MATERIAL	MIN PITCH	IDEAL PITCH
Corrugated Aluminum/Iron	10° (1:6)	15-30° (1:4-1:2)
Asphalt Shingles	15° (1:4)	20-35° (1:3-1:1.5)
Clay/Concrete Tiles	20° (1:3)	30-40° (1:2-1:1.2)
Metal Roofing (Standing seam)	5° (1:12)	15-30° (1:4-1:2)
Thatch	45° (1:1)	50-60° (1.2:1-1.7:1)
Flat Roof (Membrane)	2° (1:30)	3-5° (1:20-1:12)

PITCH BY CLIMATE

Heavy Rainfall: Steep (>30°)
Fast water runoff

Moderate Rain: Standard (20-30°)
Good drainage

Light Rain: Low (10-20°)
Adequate drainage

Snow Areas: Very Steep (>45°)
Snow slides off

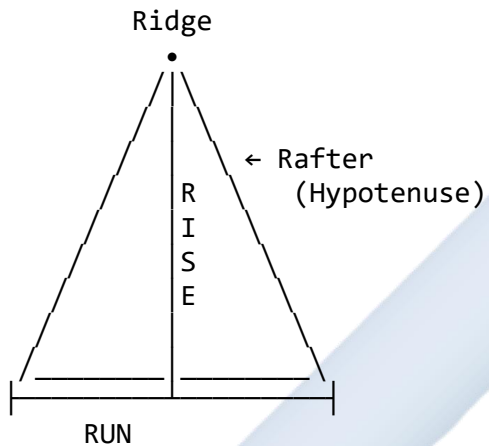
High Winds: Low to Moderate
Less wind resistance

Hot Climate: Steeper better
Attic ventilation

Diagram 90: Calculating Rafter Length

RAFTER LENGTH CALCULATION

Using Pythagoras Theorem



Formula:

$$\text{Rafter Length}^2 = \text{Rise}^2 + \text{Run}^2$$

$$\text{Rafter Length} = \sqrt{(\text{Rise}^2 + \text{Run}^2)}$$

WORKED EXAMPLE:

Given:

Span = 10m

Rise = 3m

Step 1: Find Run

$$\text{Run} = \text{Span} \div 2 = 10\text{m} \div 2 = 5\text{m}$$

Step 2: Calculate Rafter Length

$$\text{Rafter}^2 = \text{Rise}^2 + \text{Run}^2$$

$$\text{Rafter}^2 = 3^2 + 5^2$$

$$\text{Rafter}^2 = 9 + 25$$

$$\text{Rafter}^2 = 34$$

$$\text{Rafter} = \sqrt{34}$$

$$\text{Rafter} = 5.83\text{m}$$

Step 3: Add for overhang
 If eaves overhang = 0.5m
 Total Rafter = 5.83m + 0.5m
 = 6.33m

QUICK REFERENCE TABLE

SPAN	RISE	RUN	RAFTER LENGTH
8m	2m	4m	4.47m
8m	3m	4m	5.00m
10m	2.5m	5m	5.59m
10m	3m	5m	5.83m
12m	3m	6m	6.71m
12m	4m	6m	7.21m

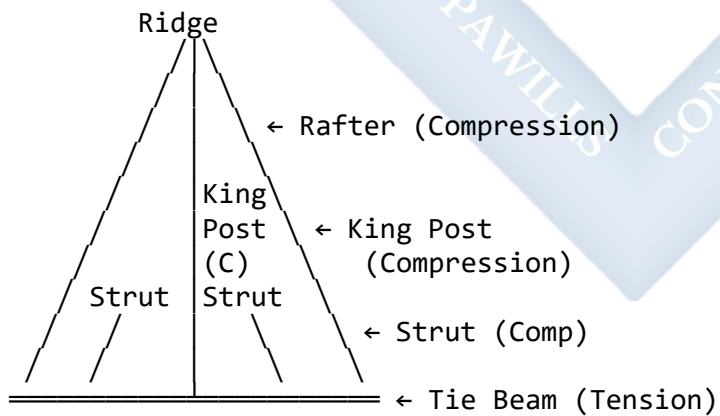
(Add overhang length)

3.5 ROOF TRUSS TYPES (DETAILED)

Diagram 91: Truss Types Comparison

1. KING POST TRUSS

Span: up to 8m



Components: 7 members

C = Compression

T = Tension

Advantages:

- Simple design
- Easy construction
- Cost-effective

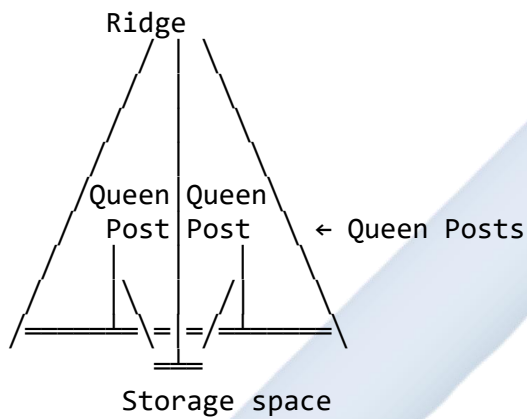
- Proven strength

Uses:

- Small residential
- Garages
- Simple structures

2. QUEEN POST TRUSS

Span: 8-12m



Components: 9 members

Advantages:

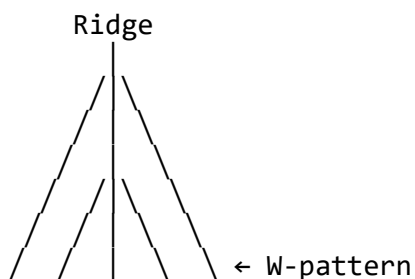
- Attic storage space
- Wider span
- Better for ceiling
- Strong structure

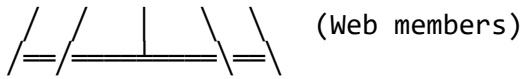
Uses:

- Medium residential
- Storage buildings
- Churches

3. FINK TRUSS (W-Truss)

Span: 8-15m





Components: 11+ members
Modern truss design

Advantages:

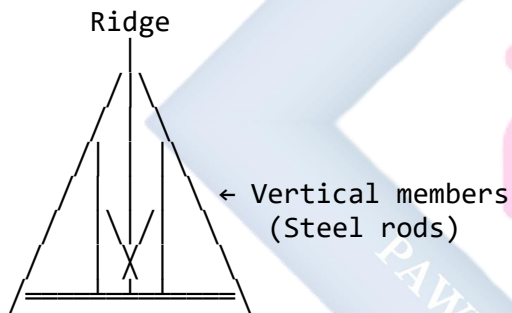
- Efficient material use
- Wide span capability
- Prefabricated
- Easy to install

Uses:

- Modern residential
- Commercial buildings
- Standardized construction

4. HOWE TRUSS

Span: 15-30m



Components: 13+ members
Heavy-duty design

Advantages:

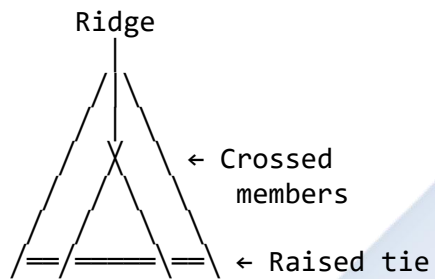
- Very strong
- Long span
- Heavy loads
- Industrial strength

Uses:

- Bridges
- Warehouses
- Large halls
- Industrial buildings

5. SCISSORS TRUSS

Span: 8-15m



Vaulted ceiling space

Advantages:

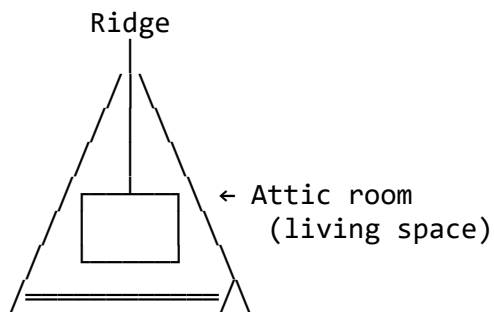
- Cathedral ceiling
- Open interior
- Aesthetic appeal
- More headroom

Uses:

- Churches
- Modern homes
- Living rooms
- Great rooms

6. ATTIC TRUSS

Span: 8-12m



Room within truss

Advantages:

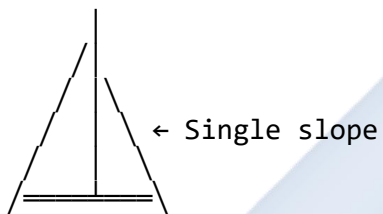
- Additional living space
- No separate ceiling needed
- Efficient design
- Cost-effective expansion

Uses:

- Residential attic rooms
- Guest rooms
- Storage + living

7. MONO TRUSS

Span: 4-10m



Half of a standard truss

Advantages:

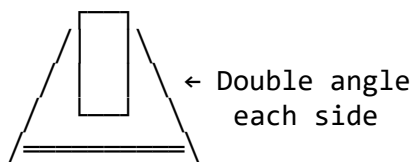
- Simple
- Modern look
- Quick install
- Cost-effective

Uses:

- Additions
- Sheds
- Porches
- Modern architecture

8. GAMBREL TRUSS

Span: 8-15m



Maximum attic space

Advantages:

- Barn-style
- Maximum storage
- Traditional look
- Efficient space use

Uses:

- Barns
- Farm buildings
- Storage buildings
- Dutch Colonial homes

Diagram 92: Truss Selection Guide

TRUSS TYPE	MAX SPAN	USE CASE	TYPICAL SPACING
King Post	8m	Small Res	0.6-1.0m
Queen Post	12m	Medium Res	0.6-1.2m
Fink (W)	15m	Std Res	0.6-1.2m
Howe	30m	Commercial	1.2-2.0m
Scissors	15m	Cathedral	0.6-1.2m
Attic	12m	Extra Room	0.6-1.0m
Mono	10m	Addition	0.6-1.0m
Gambrel	15m	Barn/Store	1.0-1.5m

Res = Residential

Spacing = Distance between trusses

EVALUATION

1. What are the **four primary functions** of a roof?
2. Define "**truss**" and explain why triangular shapes are used.
3. What is the main visual difference between a "**Gable Roof**" and a "**Hip Roof**"? Which is better for high winds?
4. A building has a span of **8 metres** and a roof rise of **2 metres**. Calculate: a) The "**run**" of the roof b) The "**pitch**" (as a ratio, decimal, and percentage) c) The approximate "**rafter length**" (without overhang)
5. List and sketch **three** different types of roof trusses. Label all major members.
6. What is the difference between a **purlin** and a **rafter**?
7. Explain the functions of: **Tie Beam, King Post, Wall Plate, Fascia, Soffit**.

8. What roof pitch would you recommend for: a) Heavy rainfall area? b) High wind coastal area? c) Flat terrain with moderate rain?

ASSIGNMENT

Part 1: Drawing Practice

- On an A4 sheet, draw a **Simple King Post Truss** showing:
 - Span = 8m (scale 1:100, so 80mm on paper)
 - Rise = 2.5m (25mm on paper)
 - Label ALL members:
 - Ridge
 - Rafter (both sides)
 - King Post
 - Struts (both sides)
 - Tie Beam
 - Wall Plate
 - Add dimensions for span and rise
 - Show forces: C (Compression) or T (Tension) on each member

Part 2: Roof Type Identification

- Sketch the following roof types in BOTH elevation and plan view: a) **Gable Roof** (show ridge line in plan) b) **Hip Roof** (show hip lines in plan) c) **Shed Roof** (show slope direction in plan)

Part 3: Pitch Calculations

- Complete this table with calculations:

SPAN (m)	RISE (m)	RUN (m)	PITCH (Ratio)	PITCH (%)	RAFTER LENGTH
10	3	?	?	?	?
12	4	?	?	?	?
8	2	?	?	?	?
14	3.5	?	?	?	?

Show all calculations!

Part 4: Field Survey

- Observe **THREE buildings** in your area:
 - Identify the roof type
 - Estimate the pitch (steep, moderate, flat)

- Note the roofing material used
- Sketch the roof shape
- Explain why that roof type was likely chosen

Example format:

BUILDING 1: Family House, Plot 23

Roof Type: Gable Roof

Pitch: Moderate (approximately 30°)

Material: Corrugated aluminum sheets

Reason: Good drainage for rainfall,
cost-effective, common design



WEEK 4: CONVENTIONAL REPRESENTATION OF MATERIALS AND SYMBOLS

4.1 WHY USE SYMBOLS?

Technical drawing symbols are the universal language of engineering and architecture. They provide:

1. **Speed** - Faster than writing descriptions
2. **Clarity** - Instantly recognizable
3. **Space Efficiency** - Compact representation
4. **Standardization** - Universal understanding
5. **Professional Communication** - Industry standard

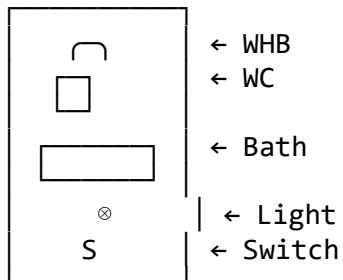
Diagram 93: Symbol vs Description

WITHOUT SYMBOLS (Slow, cluttered)

"Water closet with cistern,
wash hand basin with
mixing tap, bathtub 1700mm
long, ceiling light with
one-way switch"

Time: 2 minutes to write
Space: Large text block

WITH SYMBOLS (Fast, clear)



Time: 30 seconds to draw
Space: Compact, clear

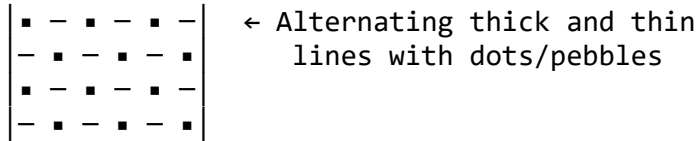
EFFICIENCY: 4x faster!

4.2 CONVENTIONAL REPRESENTATION OF MATERIALS (IN SECTION)

When objects are shown in section, different **hatching patterns** indicate different materials.

Diagram 94: Material Section Symbols - Complete Guide

1. EARTH (SOIL/GROUND)

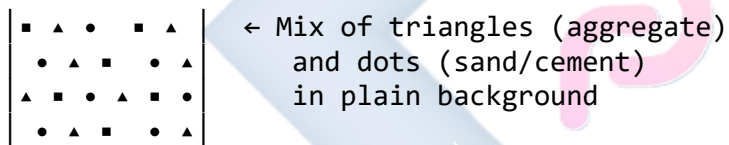


CHARACTERISTICS:

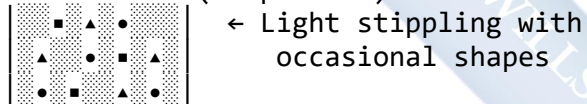
- Random dots (pebbles/soil particles)
- Alternating line thickness
- Irregular spacing
- Natural appearance

USE: Natural ground, excavation, earth fill

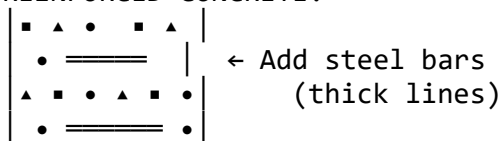
2. CONCRETE (PLAIN/REINFORCED)



ALTERNATIVE (Simplified):



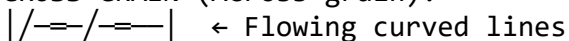
REINFORCED CONCRETE:

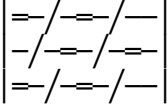


USE: Foundations, slabs, columns, beams

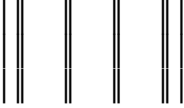
3. WOOD (TIMBER)

CROSS-GRAIN (Across grain):

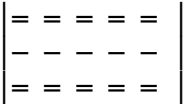



 representing wood grain
 (natural, organic)

ALONG GRAIN (With grain):

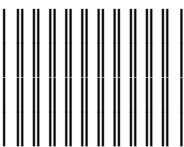

 ← Parallel lines
 (longitudinal section)

PLYWOOD:


 ← Multiple thin layers
 with alternating grain

USE: Structural timber, doors, frames, trusses

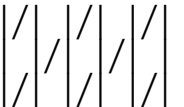
4. STEEL/IRON (METALS)


 ← Evenly spaced thin
 double lines at 45°
 (uniform, mechanical)

ALTERNATIVE (Single lines):


 ← Evenly spaced single
 lines at 45°

CAST IRON:


 ← Heavier, irregular
 spacing

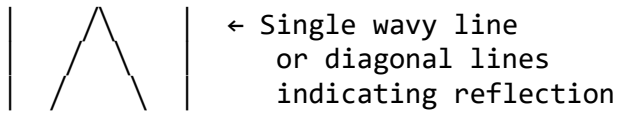
USE: Beams, columns, reinforcement, structural steel

5. GLASS

IN SECTION:

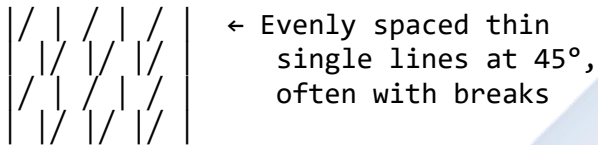

 ← Very thin, widely
 spaced lines at 45°
 (transparent quality)

IN ELEVATION:

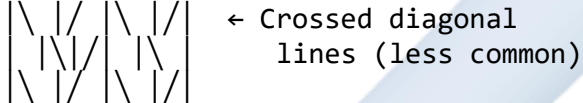


USE: Windows, doors, partitions

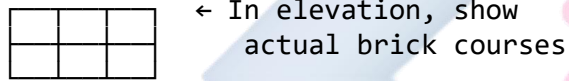
6. BRICKWORK/BLOCKWORK



ALTERNATIVE:

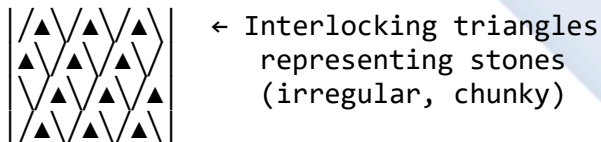


SPECIFIC BRICK PATTERN:

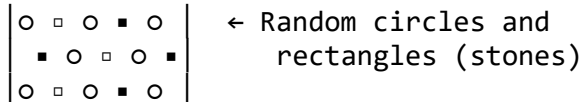


USE: Walls, hollow blocks, brick walls

7. HARDCORE (GRAVEL/STONE FILL)

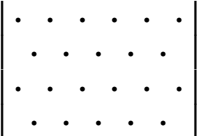


ALTERNATIVE:



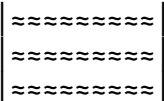
USE: Foundation fill, sub-base, drainage

8. SAND

	← Fine dots (small stippling) evenly distributed
---	--

USE: Sand blinding, bedding, fill

9. INSULATION

	← Wavy lines or zigzag pattern (fibrous quality)
---	--

ALTERNATIVE:

	← Zigzag pattern
---	------------------

USE: Thermal/acoustic insulation

10. WATER

	← Usually blank/white with label
---	-------------------------------------

ALTERNATIVE:

	← Wavy lines (indicating liquid)
---	-------------------------------------

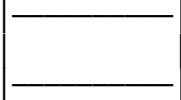
USE: Water tanks, pools, drainage

11. PLASTER/RENDER

	← Light stippling or light gray tone
---	---

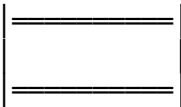
 (thin finish layer)

OR:

 ← Thin solid black line on surface edge

USE: Wall finish, ceiling finish

12. MEMBRANE (DPC/WATERPROOFING)

 ← Thick solid black line or band

ALTERNATIVE:

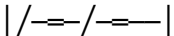
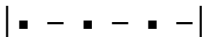
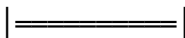
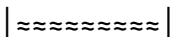
 ← Dense hatching or solid black

USE: Damp proof course, waterproofing

Diagram 95: Material Symbols Practice Sheet

MATERIAL IDENTIFICATION EXERCISE

Match the material to its section symbol:

- | | |
|--|---------------|
| A.  | 1. Concrete |
| B.  | 2. Steel/Iron |
| C.  | 3. Brickwork |
| D.  | 4. Wood |
| E.  | 5. Hardcore |
| F.  | 6. Glass |
| G.  | 7. Earth |
| H.  | 8. Membrane |
| I.  | 9. Sand |

J. | ~ ~ ~ | 10. Insulation

11. Water

Answers: A-2, B-1, C-3, D-4, E-5,
F-6, G-7, H-9, I-10, J-11

4.3 ARCHITECTURAL SYMBOLS (FOR FLOOR PLANS)

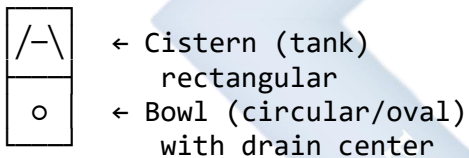
Floor plans use standardized symbols to show fixtures, fittings, and services.

4.3.1 PLUMBING SYMBOLS

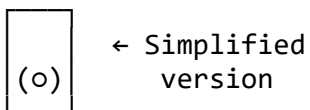
Diagram 96: Plumbing Fixtures - Complete Guide

1. WATER CLOSET (WC) / TOILET

PLAN VIEW:



ALTERNATIVE:



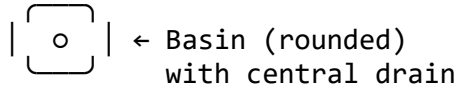
DIMENSIONS (Typical):

- Width: 400-500mm
- Depth: 600-750mm
- Cistern height: 400mm

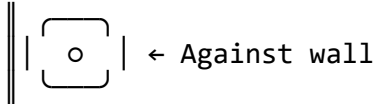
LABEL: "WC" or "W.C."

2. WASH HAND BASIN (WHB) / SINK

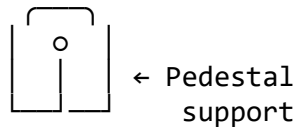
PLAN VIEW:



WALL-MOUNTED:



PEDESTAL BASIN:



CORNER BASIN:



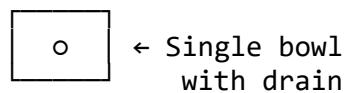
DIMENSIONS (Typical):

- Width: 400-600mm
- Depth: 400-500mm

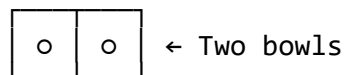
LABEL: "WHB" or "Basin"

3. KITCHEN SINK

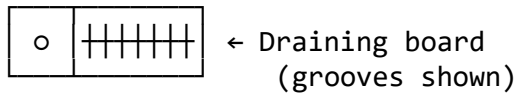
SINGLE BOWL:



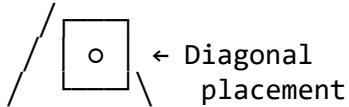
DOUBLE BOWL:



WITH DRAINER:



CORNER SINK:



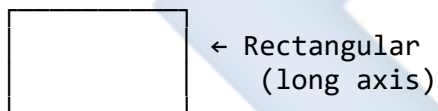
DIMENSIONS (Typical):

- Single: 500×500mm
- Double: 1000×500mm
- With drainer: 1200×600mm

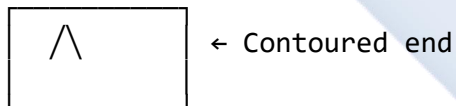
LABEL: "Kitchen Sink" or "KS"

4. BATH TUB

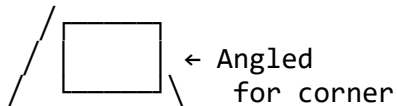
STANDARD:



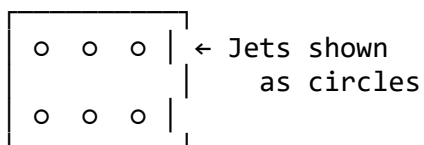
WITH HEAD REST:



CORNER BATH:



JACUZZI/SPA:



DIMENSIONS (Typical):

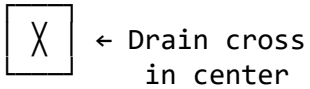
- Standard: 1700×700mm

- Short: 1500x700mm
- Corner: 1400x1400mm

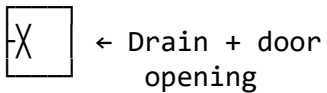
LABEL: "Bath" or "Tub"

5. SHOWER

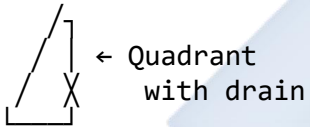
SHOWER TRAY:



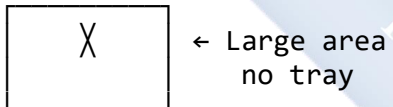
SHOWER CUBICLE:



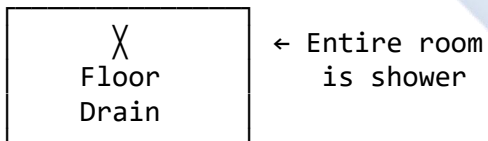
CORNER SHOWER:



WALK-IN SHOWER:



WET ROOM:

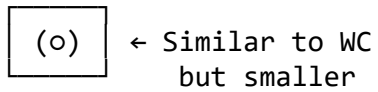


DIMENSIONS (Typical):

- Standard tray: 800x800mm
- Large: 1000x1000mm
- Walk-in: 1200x900mm

LABEL: "Shower" or "S"

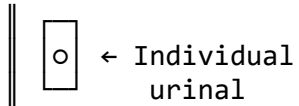
6. BIDET



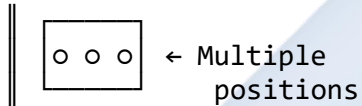
DIMENSIONS: 400×600mm
LABEL: "Bidet"

7. URINAL

WALL-MOUNTED:

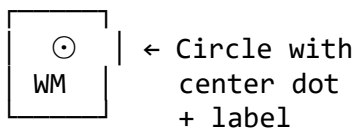


TROUGH:



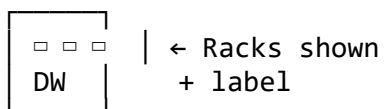
LABEL: "Urinal"

8. WASHING MACHINE



DIMENSIONS: 600×600mm
LABEL: "WM" or "Washing Machine"

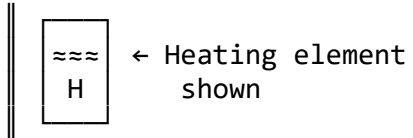
9. DISHWASHER



DIMENSIONS: 600×600mm
LABEL: "DW" or "Dishwasher"

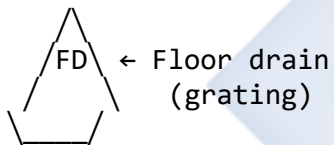
10. WATER HEATER/GEYSER

WALL-MOUNTED:



LABEL: "WH" or "Geyser"

11. FLOOR DRAIN/GULLY



ALTERNATIVE:

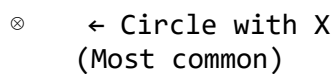


LABEL: "FD" or "Gully"

4.3.2 ELECTRICAL SYMBOLS

Diagram 97: Electrical Symbols - Complete Guide

1. CEILING LIGHT OUTLET



ALTERNATIVE:



PENDANT LIGHT:

 ← Vertical line
shows hanging

RECESSED LIGHT:

• ← Double circle
(recessed/spotlight)

LABEL: Usually unlabeled,
or "L" for light

2. WALL LIGHT OUTLET

 ← Circle with X
attached to wall

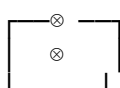
 ← With arm/bracket

SCONCE:

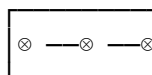
 ← Angled bracket

LABEL: "WL" if needed

3. FLUORESCENT LIGHT

 ← Tube outline
with symbol

LINEAR:

 ← Multiple tubes

LABEL: "FL" or "Fluor"

4. SWITCH (ONE-WAY)

S ← Letter S
(Most common)

—S— ← S on line

o ← Circle with
| vertical line
S

LABEL: "S" or "S1"

5. SWITCH (TWO-WAY)

S₂ ← S with subscript 2

—S₂— ← S2 on line

o
/ | \ ← Y-shaped
S₂

USE: Control one light
from two locations

LABEL: "S2" or "2-Way"

6. DIMMER SWITCH

D ← Letter D

—D— ← D on line

⊙ ← Circle with
D rotating dial

LABEL: "DIM" or "D"

7. SOCKET OUTLET (Power Plug)

|| □ ← D-shape on wall
(Most common)

|| □ ← Facing opposite

|| ● ← Half-filled circle

DOUBLE SOCKET:

|| □ ← Two symbols
|| □ stacked

FLOOR SOCKET:

□ ← In floor
└─┘

DIMENSIONS:

- Height: 300mm (normal)
- Height: 1100mm (kitchen)

LABEL: Usually unlabeled,
or "SO" for Socket Outlet

8. CEILING FAN

⊕ ← Circle with +
/|\ and propeller
blades

ALTERNATIVE:

◎ ← Concentric circles
/|\ with blades

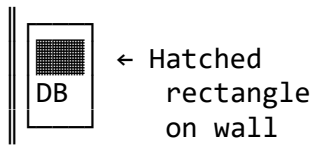
⊗
/|\ ← With X center

WITH LIGHT:

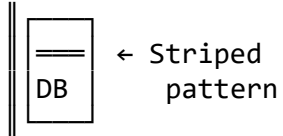
⊗
/|\ ← Combined
fan + light
⊗

LABEL: "CF" or "Fan"

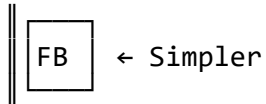
9. DISTRIBUTION BOARD (DB)



ALTERNATIVE:



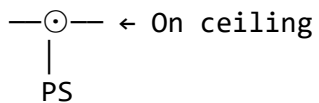
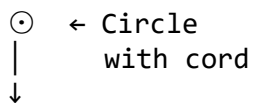
FUSE BOX:



DIMENSIONS: 400x600mm typical

LABEL: "DB" or "Distribution Board"

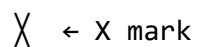
10. PULL SWITCH (Bathroom)



USE: Bathroom lights
(cord operated)

LABEL: "PS"

11. JUNCTION BOX



—+— ← Cross with lines

- ← Small square
(less common)

LABEL: "JB" if needed

12. TELEPHONE OUTLET

|| □ ← Similar to socket
T but with T

|| □ ← Square with T
T

LABEL: "TEL" or "T"

13. TV AERIAL OUTLET

|| □ ← D-shape with
TV TV label

|| ◁ ← Triangle
TV (antenna shape)

LABEL: "TV" or "Aerial"

14. DATA/NETWORK OUTLET

|| □ ← D-shape with
D D label

|| □ ← Or with NET
NET

LABEL: "DATA" or "NET"

15. SMOKE DETECTOR

- ⊙ ← Concentric circles
⊙ (alarm shape)
- ⊗ ← Circle with X
SD and label

LABEL: "SD" or "Smoke Det"

16. BELL/BUZZER

- ◐ ← Half circle
B (bell shape)
- ◑ ← D-shape with
B B label

LABEL: "Bell" or "Buzzer"

17. SECURITY SENSOR

- ◁ ← Triangle
PIR (sensor)
- ← Solid dot
S with label

LABEL: "PIR" or "Sensor"

WIRING CONNECTIONS

Switch to Light:

S> ⊗
(Dotted line shows control)

Multiple Lights on Circuit:

S> ⊗ —⊗ —⊗
 (Solid line = power cable)

Two-way Switching:

S₂> ⊗ <..... S₂
 (Light controlled from 2 places)

LEGEND:

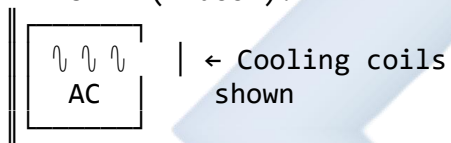
—— Solid line = Power cable
 Dotted line = Control wire

4.3.3 MECHANICAL/HVAC SYMBOLS

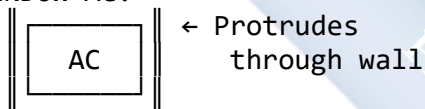
Diagram 98: Mechanical Symbols

1. AIR CONDITIONING UNIT (AC)

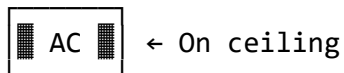
SPLIT UNIT (Indoor):



WINDOW AC:



CEILING CASSETTE:

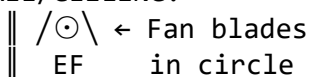


DIMENSIONS: 800x300mm typical

LABEL: "AC" or "A/C Unit"

2. EXTRACTOR FAN

WALL/CEILING:



INLINE DUCT FAN:

==⊙== ← In ductwork

LABEL: "EF" or "Extract Fan"

3. DUCTWORK

SUPPLY AIR:

====> ← Solid line with arrow

RETURN AIR:

<- - - - ← Dashed line with arrow

FLEXIBLE DUCT:

~~~~~> ← Wavy line

DIMENSIONS NOTED:

=300x200=> ← Duct size in mm

---

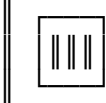
### 4. AIR DIFFUSER/GRILLE

---

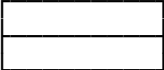
CEILING DIFFUSER:

 ← Cross hatch  
(air outlet)

WALL GRILLE:

 ← Vertical lines  
(grille pattern)

LINEAR SLOT:

 ← Slot diffuser

LABEL: "Diffuser" or "Grille"

---

### 5. THERMOSTAT

---

---

---

|| ⊙ ← Circle with  
T T label

|| □ ← Small square  
T on wall

LABEL: "TSTAT" or "Thermostat"

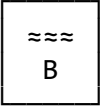
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## 6. BOILER/WATER HEATER

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---

||  ← Heating element

GAS BOILER:

||  ← Flame symbol

LABEL: "Boiler" or "WH"

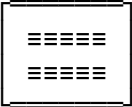
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---

## 7. RADIATOR

---

---

||  ← Horizontal lines  
(heating fins)

DIMENSIONS:

- Length: 600-1800mm
- Height: 600mm typical

LABEL: "RAD" or "Radiator"

---

---

## 8. FIRE EXTINGUISHER

---

---

⊕ ← Circle with cross  
FE (safety symbol)

← Cylinder shape

LABEL: "FE" or "Fire Ext"

---

## 9. FIRE HOSE REEL

---

⊙  
FHR ← Coiled hose  
in cabinet

LABEL: "FHR" or "Hose Reel"

---

## 10. SPRINKLER HEAD

---

⊥ ← T-shape  
○ (sprinkler)

⊙ ← Circle with  
drop line

LABEL: "SPK" if needed

---

## 4.4 DIFFERENTIATION AND APPLICATION

### Common Mistakes to Avoid:

### Diagram 99: Symbol Confusion

#### COMMONLY CONFUSED SYMBOLS

#### X WRONG INTERPRETATION

Ceiling Light ⊗ Shower Drain X

"They both have X!"

✓ CORRECT IDENTIFICATION

Ceiling Light: ⊗ ← Circle with X  
• On ceiling (center of room)  
• Controls lighting  
• Connected to switch

Shower Drain: X ← X in square  
• In shower tray (floor level)  
• Drainage only  
• No electrical connection

X WRONG INTERPRETATION

Socket Outlet □ WHB (Basin) ∩

"Both are on walls!"

✓ CORRECT IDENTIFICATION

Socket Outlet: ||□ ← Electrical  
• Power supply (lower on wall)  
• 300mm height  
• Connected to DB

Wash Basin: ||∩ ← Plumbing  
• Water fixture (counter height)  
• 800mm height  
• Connected to pipes

X WRONG INTERPRETATION

Switch S South S

"Both use letter S!"

✓ CORRECT IDENTIFICATION

Switch: S ← Control device  
• Small symbol (near door)  
• Dotted line to light  
• Electrical symbol

Compass: ⊕ ← With N, S, E, W

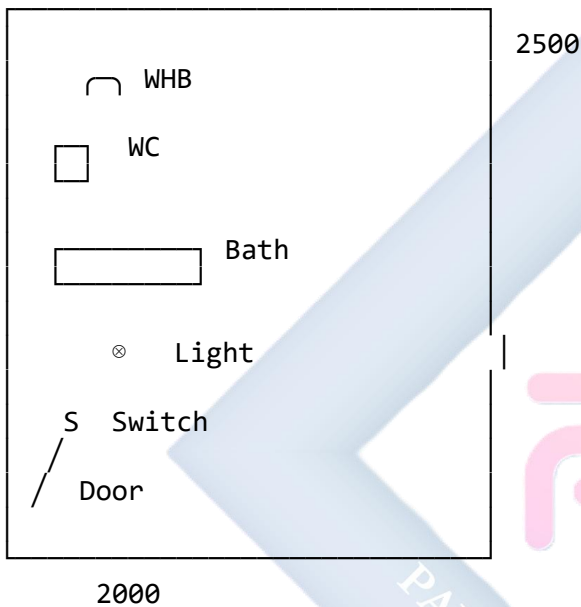
N  
W+E  
S

(on drawings)  
Shows orientation  
Not a switch!

## Diagram 100: Correct Symbol Application

BATHROOM FLOOR PLAN (Correct symbols)

SCALE 1:50

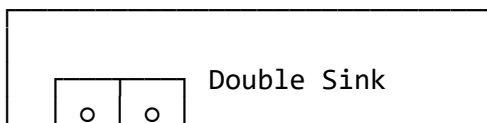


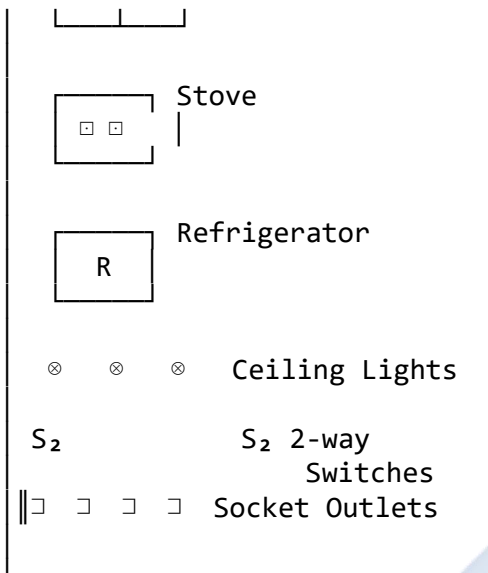
### LEGEND:

- ⊗ = Ceiling Light
- S = One-way Switch
- ┌ = Wash Hand Basin (WHB)
- = Water Closet (WC)
- ▭ = Bath Tub
- .....> = Switch control wire

KITCHEN FLOOR PLAN (Correct symbols)

SCALE 1:50



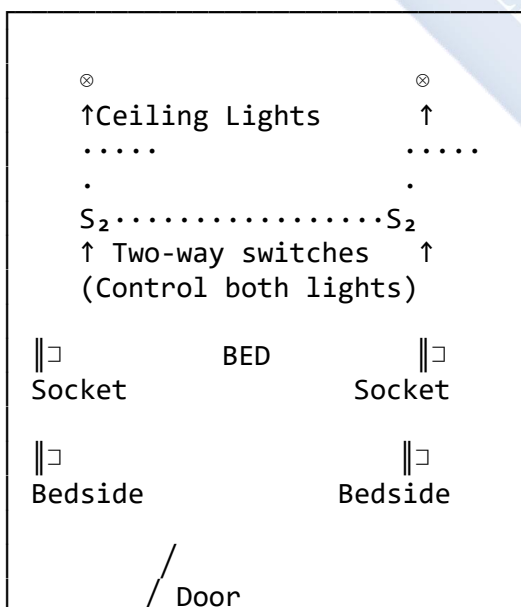


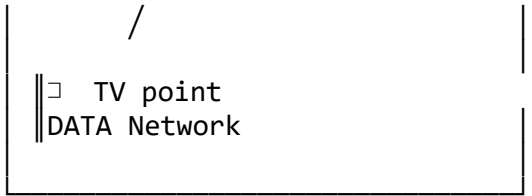
**LEGEND:**

- ⊗ = Ceiling Light
- $S_2$  = Two-way Switch
- ||□ = Socket Outlet
- = Kitchen Sink

**Diagram 101: Wiring Diagram Example**

**ELECTRICAL LAYOUT - BEDROOM**





#### KEY FEATURES:

- Two-way switching from both doors
- Sockets at bedside tables (charging)
- TV and data points on far wall
- Ceiling lights positioned centrally

#### WIRING LEGEND:

- Power cable (live circuit)
- .... Control wire (switch to light)

---

## EVALUATION

1. Why do we use **standard symbols** in technical drawing instead of writing descriptions?
2. Draw the **conventional sectional symbol** for: a) Concrete b) Wood (cross-grain) c) Steel/Iron d) Brickwork e) Hardcore
3. Draw the **floor plan symbol** for: a) Water Closet (WC) b) Wash Hand Basin c) Ceiling Light d) Socket Outlet e) One-way Switch
4. Where on a drawing would you apply **plumbing** and **electrical** symbols?
5. What is the difference between a **one-way switch** (S) and a **two-way switch** (S<sub>2</sub>)?
6. How do you show the **connection** between a switch and the light it controls on a floor plan?
7. In a building section, how would you differentiate between **concrete foundation**, **hardcore fill**, and **earth**?

---

## ASSIGNMENT

### Part 1: Material Symbols Practice

1. On an A4 sheet, create a **Material Symbols Reference Sheet**:
  - Divide the sheet into 10 boxes
  - In each box, draw and label one material symbol:
    - Concrete
    - Wood
    - Steel
    - Brickwork

- Earth
- Hardcore
- Glass
- Sand
- Membrane (DPC)
- Insulation
- Make each symbol at least 50mm × 50mm
- Add a note explaining where each material is used

## Part 2: Bathroom Floor Plan

2. Draw a **Bathroom Floor Plan** (2.5m × 2.0m) at scale 1:50:

- Include and correctly draw symbols for:
  - Water Closet (WC)
  - Wash Hand Basin (WHB)
  - Bath Tub or Shower
  - One ceiling light with one-way switch
  - One socket outlet
  - Door (900mm wide, show swing)
- Add dimensions for:
  - Overall room size
  - Fixture positions
- Include:
  - Title: "BATHROOM FLOOR PLAN"
  - Scale: 1:50
  - North arrow
  - Legend/key for all symbols used
  - 10mm border
  - Title block

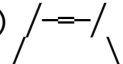
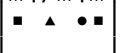
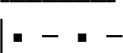
## Part 3: Building Section Detail

3. Draw a **simple wall section** showing materials:

- Draw a vertical section through an external wall (scale 1:20)
- Show from foundation to roof level
- Include and correctly hatch:
  - Foundation (concrete)
  - Hardcore under floor (hardcore symbol)
  - Floor slab (concrete)
  - Wall (brickwork/blockwork)
  - Roof structure (wood)
- Add labels for each material
- Dimensions not required, focus on correct symbols



**Example format:**

|             |                                                                                   |            |
|-------------|-----------------------------------------------------------------------------------|------------|
| Roof (Wood) |  |            |
| Wall        |  | (Brick)    |
| Floor       |  | (Concrete) |
| Hardcore    |  |            |
| Foundation  |  | (Concrete) |
| Earth       |  |            |

**Part 4: Symbol Identification Quiz**

4. Create a **Symbol Quiz** for a classmate:
  - Draw 15 different symbols (mix of plumbing, electrical, mechanical)
  - Number them 1-15
  - Create an answer key on a separate page
  - Exchange with a classmate and test each other

## WEEK 5: MIDTERM TEST

### MIDTERM TEST - SECOND TERM

Time Allowed: 2 Hours Total Marks: 100

---

#### SECTION A: THEORY (40 MARKS)

Answer ALL questions in this section.

---

##### Question 1 (8 marks)

- a) Define the term "**Computer**" and list its four main operations. (4 marks)
  - b) Differentiate between **hardware** and **software**. Give two examples of each. (4 marks)
- 

##### Question 2 (10 marks)

- a) What does **CAD** stand for? (1 mark)
  - b) List FIVE advantages of Computer-Aided Design (CAD) over manual drawing. (5 marks)
  - c) Name TWO popular CAD software programs used for:
    - Architectural design (1 mark)
    - Mechanical engineering design (1 mark)
  - d) What is the specific name for a large-format printer used for printing technical drawings? (1 mark)
  - e) Explain the function of **RAM (Random Access Memory)** in a computer system. (1 mark)
- 

##### Question 3 (8 marks)

- a) Define the following arch-related terms:
  - i) Keystone (1 mark)
  - ii) Voussoir (1 mark)
  - iii) Intrados (1 mark)
  - iv) Springing Line (1 mark)
- b) What is the primary structural function of an arch? (2 marks)
- c) Name TWO types of arches commonly used in:

- i) Roman architecture (1 mark)
  - ii) Gothic architecture (1 mark)
- 

**Question 4** (6 marks)

- a) Define the term "**Pitch**" as it relates to roofs. (2 marks)
- b) A building has a span of 10 metres and a rise of 3 metres. Calculate:
- i) The **run** of the roof (1 mark)
  - ii) The **pitch** (express as a ratio) (2 marks)
- c) What is the difference between a **Gable Roof** and a **Hip Roof**? (1 mark)
- 

**Question 5** (8 marks)

- a) Why do we use conventional symbols in technical drawings? Give TWO reasons. (2 marks)
- b) Describe the sectional representation (hatching pattern) for:
- i) Concrete (1 mark)
  - ii) Wood (1 mark)
  - iii) Steel (1 mark)
  - iv) Brickwork (1 mark)
- c) In a floor plan, what symbol is used to show the connection between a switch and the light it controls? (1 mark)
- d) Where on a drawing would you typically apply electrical and plumbing symbols? (1 mark)
- 
- 

**SECTION B: DRAWING/PRACTICAL (60 MARKS)**

**Answer ALL questions in this section. Use proper drawing instruments and techniques.**

---

**Question 6: Hardware Components (10 marks)**

Draw and label FIVE computer hardware components used in CAD work. Include:

- Monitor
- CPU (System Unit)
- Mouse
- Keyboard

- Plotter

Each component should be clearly sketched and labeled. (2 marks each)

---

### Question 7: Arch Drawing (15 marks)

a) Using a compass and ruler, construct a **Semicircular (Roman) Arch** with:

- Span = 100mm
- Show construction lines (center point, radius)
- (6 marks)

b) On your arch drawing, label the following:

- Keystone
- Voussoir (at least 3)
- Springer (both sides)
- Intrados
- Extrados
- Span dimension
- Rise dimension
- (7 marks)

c) Add a simple title block with: Title, Your Name, Date, Scale 1:1 (2 marks)

---

### Question 8: Roof Truss Drawing (15 marks)

Draw a **King Post Truss** with the following specifications:

- Span = 8m (scale 1:100, so 80mm on paper)
- Rise = 2.5m (25mm on paper)

Your drawing must include and label:

- Ridge/Ridge Board
- Rafters (both sides)
- King Post
- Struts (both sides)
- Tie Beam
- Wall Plates (both ends)

Add dimensions for:

- Total span

- Rise height

Include forces notation:

- C = Compression
- T = Tension (Label which members are in compression and which are in tension)

Include a simple title block. (All components = 15 marks)

### Question 9: Material Symbols (10 marks)

Draw a **simple building section** (sketch, not to scale) showing a wall from foundation to roof. Include the correct sectional (hatching) symbols for:

- Foundation (Concrete) - 2 marks
- Hardcore fill (Gravel) - 2 marks
- Floor slab (Concrete) - 2 marks
- Wall (Brickwork or Blockwork) - 2 marks
- Roof structure (Wood/Timber) - 2 marks

Each material must be correctly hatched and labeled.

**Example format:**

ROOF (Label → Wood)  
WALL (Label → Brick)  
FLOOR (Label → Concrete)  
HARDCORE (Label → Gravel)  
FOUNDATION (Label → Concrete)

### Question 10: Floor Plan Symbols (10 marks)

Draw the standard floor plan symbols for the following (each symbol should be approximately 30mm in size):

- a) Water Closet (WC) - 2 marks b) Wash Hand Basin (WHB) - 2 marks c) Ceiling Light - 1 mark  
d) One-way Switch (S) - 1 mark e) Socket Outlet - 2 marks f) Kitchen Sink (double bowl) - 2 marks

Each symbol must be clearly drawn and labeled.

## MARKING SCHEME

### SECTION A: THEORY (40 Marks)

**Question 1 (8 marks)**

- Definition of computer (1 mark)
- Four operations: Input, Processing, Output, Storage (4 marks: 1 each)
- Hardware vs Software distinction (1 mark)
- Examples: 2 hardware, 2 software (2 marks)

**Question 2 (10 marks)**

- CAD = Computer-Aided Design (1 mark)
- Five advantages (5 marks: 1 each - speed, accuracy, editing, storage, sharing, visualization, collaboration, etc.)
- Architectural software: Revit, ArchiCAD, AutoCAD Architecture (1 mark for 2 correct)
- Mechanical software: SolidWorks, Inventor, CATIA (1 mark for 2 correct)
- Plotter (1 mark)
- RAM function: temporary workspace/memory for active programs (1 mark)

**Question 3 (8 marks)**

- Keystone: topmost central voussoir that locks arch (1 mark)
- Voussoir: wedge-shaped block forming arch (1 mark)
- Intrados: inner concave curve/underside (1 mark)
- Springing line: horizontal line where arch begins to curve (1 mark)
- Arch function: converts vertical load to diagonal/horizontal thrust (2 marks)
- Roman: Semicircular arch (1 mark)
- Gothic: Pointed arch (1 mark)

**Question 4 (6 marks)**

- Pitch: steepness/slope of roof, ratio of rise to run (2 marks)
- Run = Span  $\div$  2 = 10m  $\div$  2 = 5m (1 mark)
- Pitch = Rise  $\div$  Run = 3  $\div$  5 = 3:5 or 0.6 (2 marks)
- Gable vs Hip: Gable has triangular ends; Hip has four slopes, no gable ends (1 mark)

**Question 5 (8 marks)**

- Reasons for symbols: Speed, clarity, standardization, universal understanding (2 marks for any 2)
- Concrete: triangles and dots mixed (1 mark)
- Wood: flowing grain lines (1 mark)
- Steel: evenly spaced thin double lines at 45° (1 mark)
- Brickwork: evenly spaced single lines at 45° (1 mark)
- Connection symbol: dotted line (1 mark)
- Location: floor plans (1 mark)

## SECTION B: DRAWING/PRACTICAL (60 Marks)

### Question 6 (10 marks)

- Each component clearly drawn: 1 mark  $\times$  5
- Each component correctly labeled: 1 mark  $\times$  5
- Total: 10 marks

### Question 7 (15 marks)

- Construction: Accurate semicircle (3 marks)
- Construction lines shown (center, radius) (3 marks)
- All labels correct and clear (7 marks: 1 each for 7 items)
- Title block (2 marks)
- Total: 15 marks

### Question 8 (15 marks)

- Correct proportions (scale) (3 marks)
- All members drawn correctly (6 marks: 1 per member type)
- All labels present and correct (4 marks)
- Dimensions shown (1 mark)
- Forces notation (C/T) (1 mark)
- Total: 15 marks

### Question 9 (10 marks)

- Each material correctly hatched: 5  $\times$  2 marks = 10 marks
- (Deduct 1 mark per material if hatching incorrect)

### Question 10 (10 marks)

- Each symbol correctly drawn and labeled:
  - WC: 2 marks
  - WHB: 2 marks
  - Ceiling Light: 1 mark
  - Switch: 1 mark
  - Socket: 2 marks
  - Kitchen Sink: 2 marks
- Total: 10 marks

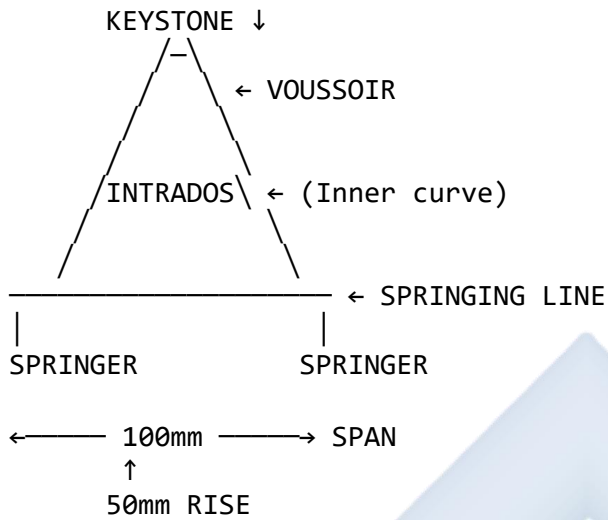
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## SAMPLE STUDENT ANSWER GUIDE

### For Question 7 - Arch Drawing:

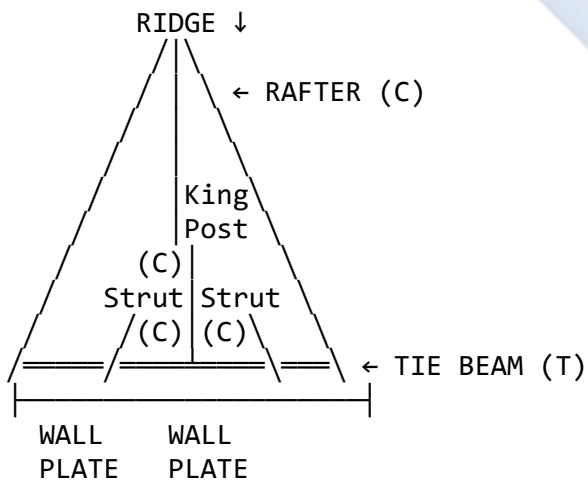
SEMICIRCULAR ARCH  
SCALE: 1:1



TITLE: Roman Arch  
NAME: [Student Name]  
DATE: [Date]  
SCALE: 1:1

**For Question 8 - King Post Truss:**

KING POST TRUSS  
SCALE: 1:100





← 80mm →  
(8m)  
↑  
25mm RISE  
(2.5m)

KEY:

C = Compression

T = Tension

TITLE: King Post Truss

SCALE: 1:100

DATE: [Date]

### For Question 9 - Material Symbols:

#### BUILDING SECTION - MATERIALS

ROOF  
/—\ ← WOOD  
(Flowing grain)

WALL  
| / |  
| / | ← BRICKWORK  
| / | (45° lines)

FLOOR SLAB  
| ■ ▲ ● | ← CONCRETE  
| ● ▲ ■ | (Triangles + dots)

HARDCORE  
| /▲\ /▲\ | ← GRAVEL  
| ▲\▲\ ▲\ | (Interlocking triangles)

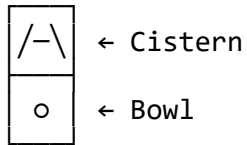
FOUNDATION  
| ■ ▲ ● | ← CONCRETE  
| ● ▲ ■ | (Triangles + dots)

Labels must be clear!

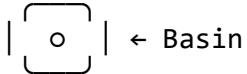
### For Question 10 - Floor Plan Symbols:

## FLOOR PLAN SYMBOLS

### 1. WATER CLOSET (WC)



### 2. WASH HAND BASIN



### 3. CEILING LIGHT



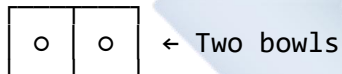
### 4. ONE-WAY SWITCH



### 5. SOCKET OUTLET



### 6. KITCHEN SINK (Double)

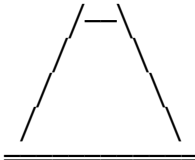


## COMMON MISTAKES TO AVOID

### Diagram 102: Midterm Test - Common Errors

#### ERROR 1: ARCH CONSTRUCTION

✗ WRONG



Flat sides  
(Not circular!)

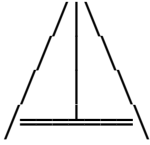
✓ CORRECT



Smooth curve  
(True semicircle)

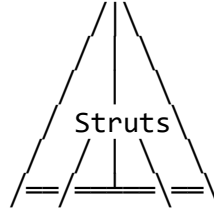
#### ERROR 2: KING POST TRUSS

X WRONG



Missing struts!

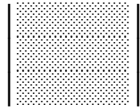
✓ CORRECT



Complete structure

#### ERROR 3: MATERIAL SYMBOLS

X WRONG (Concrete)



Solid fill  
(Incorrect!)

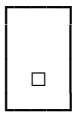
✓ CORRECT



Triangles + dots  
(Shows aggregate)

#### ERROR 4: FLOOR PLAN SYMBOLS

X WRONG (WC)



Just rectangles!

✓ CORRECT



Cistern + bowl  
(Proper symbol)

#### ERROR 5: DIMENSIONS

X WRONG

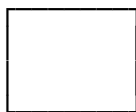
50



No arrows  
(Incomplete!)

✓ CORRECT

← 50 →



↑  
30

Proper dimension lines

#### ERROR 6: SCALE APPLICATION

X WRONG

Scale 1:100

✓ CORRECT

Scale 1:100

Drawing 8m span  
as 8mm!

$$8\text{m} \div 100 = 0.08\text{m} \\ = 8\text{mm} \quad \times$$

Too small!

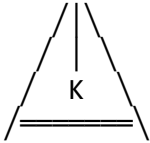
Drawing 8m span  
as 80mm!

$$8\text{m} \div 100 = 0.08\text{m} \\ = 80\text{mm} = 8\text{cm} \quad \checkmark$$

Correct size!

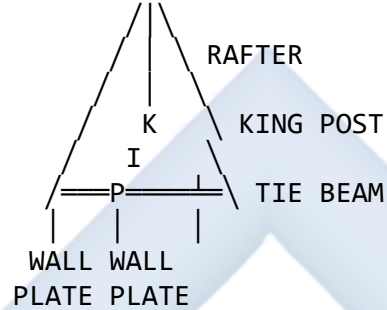
#### ERROR 7: LABELING

**✗ WRONG**



No labels!  
(What is K?)

**✓ CORRECT**



All labeled clearly!

#### ERROR 8: TITLE BLOCK

**✗ WRONG**

(Missing entirely!)

**✓ CORRECT**

TITLE: Arch Dwg  
NAME: J. Ade  
DATE: 10/11/25  
SCALE: 1:1

Always include!

Complete info!

---

## EXAMINATION TIPS

### Time Management:

#### RECOMMENDED TIME ALLOCATION

|                       |            |
|-----------------------|------------|
| Section A (Theory):   | 45 minutes |
| • Read all questions: | 5 min      |
| • Answer Q1-Q5:       | 40 min     |

|                      |                              |
|----------------------|------------------------------|
| Section B (Drawing): | 75 minutes                   |
| • Q6 (Hardware):     | 10 min                       |
| • Q7 (Arch):         | 20 min                       |
| • Q8 (Truss):        | 20 min                       |
| • Q9 (Materials):    | 15 min                       |
| • Q10 (Symbols):     | 10 min                       |
| <b>TOTAL:</b>        | <b>120 minutes (2 hours)</b> |

### Drawing Tips:

1. **Use 2H pencil** for construction lines (light)
2. **Use HB pencil** for final lines (dark)
3. **Draw faintly first**, then darken
4. **Use ruler** for all straight lines
5. **Use compass** for all circles/arcs
6. **Label everything** clearly
7. **Add title block** to every drawing
8. **Check your work** before submission

### Theory Tips:

1. **Read questions carefully** - answer what is asked
2. **Define terms precisely** - use technical language
3. **Show all calculations** - include units (m, mm, etc.)
4. **Use bullet points** for listing
5. **Draw sketches** if helpful for explanations
6. **Write neatly** - examiners must read your work!

---

## REVISION CHECKLIST

### Before the test, can you:

#### Computer Hardware & CAD:

Define computer and list its operations?

List computer hardware components?

Explain RAM vs Storage?

Name popular CAD software?

List 5 advantages of CAD over manual drawing?

Know what a plotter is?

#### Arches:

Define all arch terminology (keystone, voussoir, etc.)?

Explain how arches work structurally?

Draw different arch types?

Construct a semicircular arch?

Label all parts correctly?

**Roofs:**

Define pitch and calculate it?

Identify different roof types?

Draw and label a King Post Truss?

Know all roof members (rafter, ridge, etc.)?

Indicate forces (compression/tension)?

**Material Symbols:**

Draw hatching for concrete?

Draw hatching for wood?

Draw hatching for steel?

Draw hatching for brickwork?

Draw hatching for hardcore?

Draw hatching for earth?

**Floor Plan Symbols:**

Draw WC symbol?

Draw WHB symbol?

Draw ceiling light symbol?

Draw switch symbol?

Draw socket symbol?

Draw kitchen sink symbol?

Show switch-to-light connection?

## PAST QUESTIONS PRACTICE

Students should practice these additional questions:

### Additional Theory Questions:

1. Explain the difference between a **HDD** and an **SSD**. Which is better for CAD work and why?
2. A building has a span of 12m and a pitch of 1:3. Calculate the rise and the approximate rafter length.
3. Why is a **Hip Roof** considered better than a **Gable Roof** in high-wind areas?
4. List **THREE** parts that should be labeled on a King Post Truss.
5. In a building section, how would you differentiate between a **foundation** and **hardcore fill** using hatching patterns?

### Additional Drawing Questions:

1. Draw an **exploded view** showing: Bolt, Plain Washer, and Nut (with part balloons).
2. Draw a **simple floor plan** for a toilet (1.5m      2m) showing: WC, WHB, one light, one switch, and door.
3. Sketch **THREE different arch types** in elevation (Semicircular, Pointed, Segmental).
4. Draw a **Queen Post Truss** and label all members.
5. Draw a **wall section** showing five different materials with correct hatching.

---

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END OF MIDTERM TEST

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## WEEK 6: BLUEPRINT READING, REPRODUCTION SERVICES AND MERCHANDISING

### 6.1 WHAT IS A "BLUEPRINT"?

The term "**blueprint**" has an interesting history and modern application.

#### Diagram 103: Blueprint Evolution

##### BLUEPRINT HISTORY

##### 1840s - INVENTION

Sir John Herschel invented  
cyanotype process

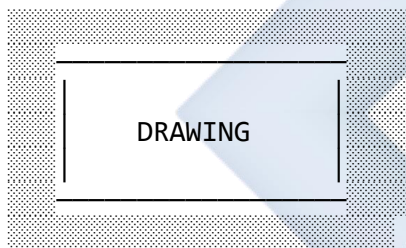
Original Drawing (on tracing paper)



Exposed to UV light + chemicals



Result: WHITE lines on BLUE paper



Blue Background  
White Lines



## 1950s-1980s - DYE-LINE PROCESS

Also called "Ammonia Process"  
or "Whiteprint"

Original Drawing



Light-sensitive paper



Developed with ammonia gas



Result: DARK lines on WHITE paper

DRAWING

White Background  
Dark Lines

Problems:

- Slow process
- Ammonia smell (toxic!)
- Paper yellows over time

## 1990s-PRESENT - DIGITAL PLOTTING

Computer-Aided Design (CAD)



Digital file (DWG, PDF, etc.)



Sent to plotter/printer



Result: BLACK lines on WHITE paper

CAD DRAWING

White Paper  
Black/Color Lines

Advantages:

- Fast (minutes, not hours)
- Clean process

- High quality
- Color possible
- Multiple copies easy

#### TODAY'S MEANING

"Blueprint" now means:

- Any working drawing/plan
- Technical drawing
- Architectural plans
- Construction documents

Even though modern prints are BLACK on WHITE, we still call them "blueprints"!

Like calling all tissues "Kleenex" or all copiers "Xerox"

---

## 6.2 READING AND INTERPRETING DRAWINGS

Reading blueprints is a critical skill. Different professionals read drawings differently based on their needs.

### Diagram 104: Building Drawing Reading Strategy

#### ARCHITECTURAL DRAWING SET

#### SET CONSISTS OF:

1. SITE PLAN (Context)
2. FLOOR PLANS (Layout)
3. ELEVATIONS (Exterior)
4. SECTIONS (Construction)
5. DETAILS (Specific areas)
6. SCHEDULES (Specifications)

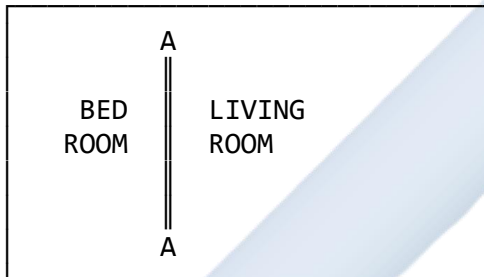
## READING SEQUENCE:

### STEP 1: TITLE BLOCK

- Project name & address
- Sheet title (e.g., "Floor Plan")
- Sheet number (e.g., "A-02")
- Scale (e.g., "1:100")
- Date & revisions
- Architect/Engineer name

Read this FIRST - gives context!

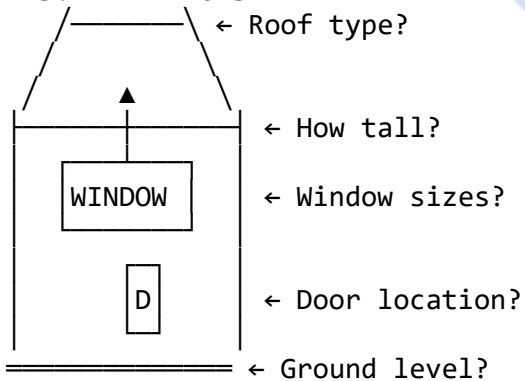
### STEP 2: FLOOR PLAN



Questions to answer:

- What is the building layout?
- Where are the rooms?
- What are room dimensions?
- Where are doors/windows?
- Where is SECTION A-A cut?

### STEP 3: ELEVATIONS

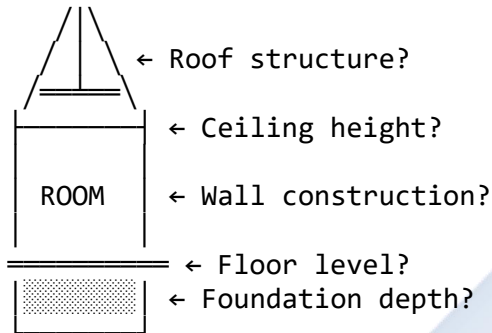


Questions to answer:

- What does building look like?

- How high is it?
- What materials are used?
- Roof shape/pitch?

#### STEP 4: SECTION A-A



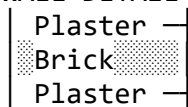
#### Questions to answer:

- How is it built?
- Foundation depth?
- Wall thickness/material?
- Floor-to-ceiling height?
- Roof structure type?

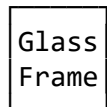
#### STEP 5: DETAILS

Zoom in on specific areas:

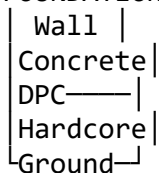
##### WALL DETAIL (1:20)



##### WINDOW DETAIL (1:5)



##### FOUNDATION DETAIL (1:20)



#### STEP 6: CROSS-REFERENCE

Always relate views to each other:

Floor Plan width = Elevation width  
Floor Plan length = Section length  
Section height = Elevation height

All views must ALIGN!

---

## Diagram 105: Engineering Drawing Reading Strategy

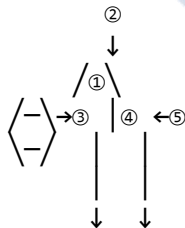
### MECHANICAL/ENGINEERING DRAWING SET

SET CONSISTS OF:

1. ASSEMBLY DRAWING (Overview)
2. DETAIL DRAWINGS (Each part)
3. PARTS LIST/BOM (Inventory)
4. SPECIFICATIONS (Materials/etc)

READING SEQUENCE:

STEP 1: ASSEMBLY DRAWING



Questions:

- How many parts total?
- How do they fit together?
- What is assembly sequence?

STEP 2: PARTS LIST (BOM)

| ITEM | PART NAME | QTY | PART NO |
|------|-----------|-----|---------|
| ①    | Housing   | 1   | HS-001  |
| ②    | Cover     | 1   | CV-002  |
| ③    | Shaft     | 1   | SH-003  |
| ④    | Bearing   | 2   | BR-004  |
| ⑤    | Bolt M10  | 4   | BT-005  |

Questions:

- What materials needed?
- How many of each part?
- Part numbers for ordering?

### STEP 3: DETAIL DRAWINGS

For each part, find its detail drawing:

#### PART ③: SHAFT

TOP VIEW



Ø20

FRONT VIEW



←100→

Questions:

- What are exact dimensions?
- What material?
- What tolerances?
- What surface finish?
- Any special processes?

### STEP 4: MANUFACTURING INFO

From detail drawing, extract:

Material: Mild Steel

Diameter: Ø20mm ±0.05mm

Length: 100mm ±0.2mm

Surface: Turned smooth

Tolerance: H7

Heat Treatment: None

Quantity: 1 per assembly

---

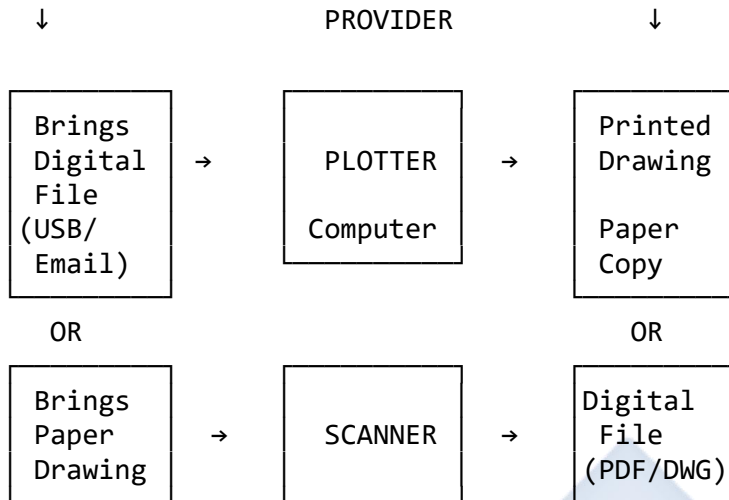
## 6.3 BLUEPRINT REPRODUCTION SERVICES

The business of copying and printing large-format technical drawings.

### Diagram 106: Blueprint Reproduction Business

#### REPRODUCTION SERVICE WORKFLOW

| CUSTOMER | SERVICE | OUTPUT |
|----------|---------|--------|
|----------|---------|--------|



#### DETAILED PROCESS FLOW

##### 1. RECEIVE ORDER

Customer arrives with:

- USB drive (DWG, PDF files)
- Email (send files)
- Paper drawings (to scan)
- Cloud link (Dropbox, etc.)



##### 2. CHECK FILE

Open file on computer  
Verify:

- File opens correctly
- Drawing is complete
- No corrupted elements
- Correct orientation
- Paper size (A0,A1,A2,etc.)



##### 3. GIVE QUOTATION

Calculate cost:

Paper Size × Quantity × Rate

Example:

A1 size = ₦1,000 per copy  
5 copies = ₦5,000

+ Binding (if requested)  
+ Folding (if requested)

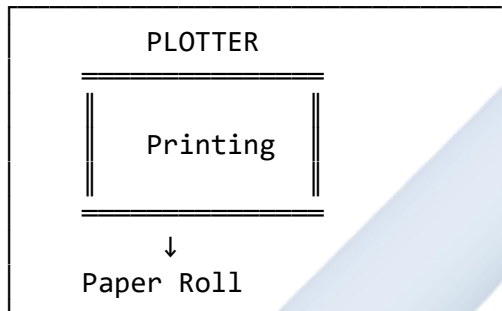
↓

#### 4. PREPARE FILE

- Set correct paper size
- Set scale (1:100, etc.)
- Choose color/monochrome
- Set line weights
- Preview print

↓

#### 5. PRINT



↓

#### 6. QUALITY CHECK

- Inspect print for:
- Clarity (sharp lines)
  - Complete (nothing cut off)
  - Correct size
  - No smudges/defects
  - Proper colors (if color)

↓

#### 7. FINISHING

- Options:
- Trim edges (neat borders)
  - Fold to A4 size
  - Roll and tube
  - Bind multiple sheets
  - Laminate (protection)

↓

#### 8. DELIVERY

- Hand to customer
- Provide receipt
- Store digital backup (for reprints)

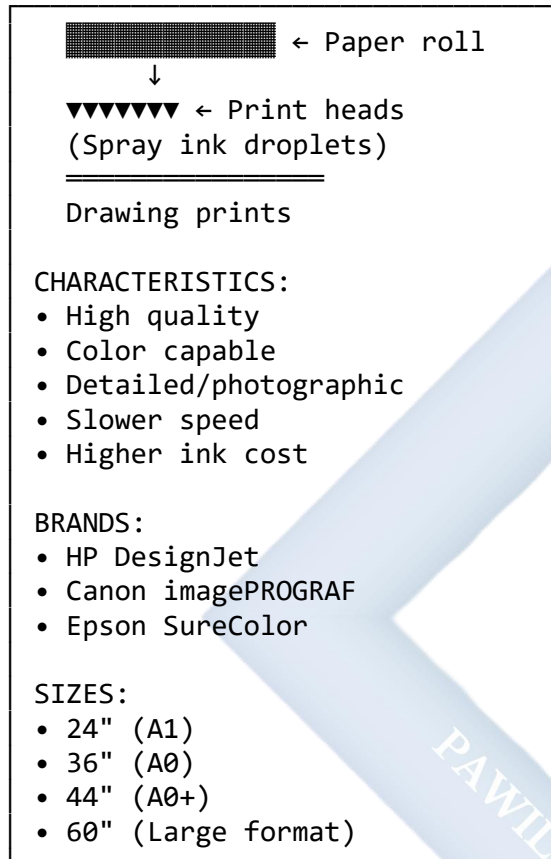


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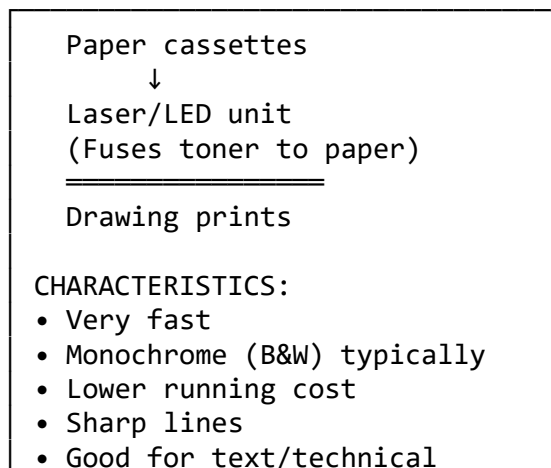
## Diagram 107: Plotter Types

### PLOTTER CATEGORIES

#### 1. INKJET PLOTTER



#### 2. LASER/LED PLOTTER



**BRANDS:**

- Xerox
- Ricoh
- KIP

**SIZES:**

- A1, A0 typical
- Up to 36" width

### 3. PEN PLOTTER (Legacy)

Mechanical arm

 ← Physical pens  
Moves across paper

Drawing traced

**CHARACTERISTICS:**

- Very slow
- Precise lines
- Multiple pen colors
- Nostalgic/artistic
- Mostly obsolete now

**HISTORICAL SIGNIFICANCE:**

- First computer plotters
- Used 1970s-1990s
- Replaced by inkjet/laser

### COMPARISON TABLE

| TYPE   | SPEED  | COST   | QUALITY   |
|--------|--------|--------|-----------|
| Inkjet | Medium | High   | Excellent |
| Laser  | Fast   | Medium | Good      |
| Pen    | Slow   | Low    | Good      |

## 6.4 MERCHANDISING

Additional products/services to increase revenue.

## Diagram 108: Product Range

### BLUEPRINT SERVICE CENTER PRODUCT CATALOG

| CORE SERVICES                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>✓ Plotting (A4 - A0+)</li><li>✓ Scanning (A4 - A0)</li><li>✓ PDF to CAD conversion</li><li>✓ Drawing digitization</li><li>✓ Color printing</li><li>✓ Binding &amp; finishing</li></ul> |

| DRAWING SUPPLIES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>PAPER:</p> <ul style="list-style-type: none"><li>• Drawing sheets (A4-A0)</li><li>• Tracing paper rolls</li><li>• Graph paper</li><li>• Cartridge paper</li></ul> <p>INSTRUMENTS:</p> <ul style="list-style-type: none"><li>• Set squares (45°, 30°/60°)</li><li>• T-squares (various sizes)</li><li>• Compasses (spring bow, beam)</li><li>• Dividers</li><li>• Protractors</li><li>• French curves</li><li>• Drawing boards (A4, A3, A2)</li><li>• Parallel rulers</li></ul> <p>DRAWING TOOLS:</p> <ul style="list-style-type: none"><li>• Technical pens (0.1mm-0.8mm)</li><li>• Pencils (2H, H, HB, B, 2B)</li><li>• Lead holders &amp; leads</li><li>• Erasers (plastic, kneaded)</li><li>• Eraser shields</li><li>• Sharpeners</li><li>• Masking tape</li><li>• Drawing clips</li></ul> <p>SCALES:</p> <ul style="list-style-type: none"><li>• Architects scale</li><li>• Engineers scale</li><li>• Metric scales (1:100, etc.)</li><li>• Triangular scales</li></ul> |

TEMPLATES:

- Circle templates
- Ellipse templates
- Architectural symbols
- Furniture templates
- Lettering guides

OFFICE SUPPLIES

- Folders & binders
- Drawing tubes (storage)
- Plan filing systems
- Staplers & punches
- Calculators (scientific)
- USB drives
- External hard drives

ADDITIONAL SERVICES

- Lamination (A4-A0)
- Mounting on foam board
- CAD training courses
- Drawing digitization service
- Cloud storage solutions
- Delivery service

PRICING STRATEGY

PLOTTING RATES (Sample - Nigeria)

| SIZE | BLACK & WHITE | COLOR  |
|------|---------------|--------|
| A4   | ₦200          | ₦400   |
| A3   | ₦400          | ₦800   |
| A2   | ₦600          | ₦1,200 |
| A1   | ₦1,000        | ₦2,000 |
| A0   | ₦1,500        | ₦3,000 |

VOLUME DISCOUNTS:

- 10+ copies: 10% discount
- 50+ copies: 15% discount

- 100+ copies: 20% discount

**STUDENT DISCOUNTS:**

- 10% off with valid student ID

---

## 6.5 SETTING UP A BLUEPRINT REPRODUCTION SERVICE CENTER

Starting this business requires careful planning.

### Diagram 109: Business Setup Guide

#### BLUEPRINT BUSINESS STARTUP PLAN

---

##### PHASE 1: PLANNING (Months 1-2)

###### 1. MARKET RESEARCH

Identify potential customers:

- Architects
- Engineers
- Builders/Contractors
- Real estate developers
- Government agencies
- Educational institutions

Survey competition:

- How many services in area?
- Their pricing?
- Their service quality?
- Gaps you can fill?

###### 2. BUSINESS PLAN

Executive Summary

- Business name & concept
- Mission statement
- Goals (1yr, 3yr, 5yr)

Financial Projections

- Startup costs
- Monthly operating costs
- Revenue projections
- Break-even analysis

Marketing Strategy

- Target customers
- Pricing strategy
- Promotional plans

## PHASE 2: FUNDING (Months 2-3)

### STARTUP COST BREAKDOWN (Nigeria - 2025)

#### EQUIPMENT:

|                                                                                      |                   |
|--------------------------------------------------------------------------------------|-------------------|
| Large Format Plotter<br>(HP DesignJet T520 36")                                      | ₦2,500,000        |
| Computer (High-spec)<br>- i7 processor<br>- 16GB RAM<br>- 512GB SSD<br>- 27" monitor | ₦400,000          |
| Scanner (A0 size)                                                                    | ₦1,000,000        |
| UPS/Inverter System<br>(Power backup)                                                | ₦300,000          |
| Generator (5KVA)                                                                     | ₦250,000          |
| Furniture<br>(Desk, chairs, shelves)                                                 | ₦200,000          |
| Software Licenses<br>(AutoCAD, PDF, etc.)                                            | ₦150,000          |
| <b>EQUIPMENT TOTAL:</b>                                                              | <b>₦4,800,000</b> |

#### INITIAL STOCK:

|                       |                 |
|-----------------------|-----------------|
| Paper rolls (various) | ₦200,000        |
| Ink cartridges        | ₦100,000        |
| Drawing supplies      | ₦150,000        |
| Office supplies       | ₦50,000         |
| <b>STOCK TOTAL:</b>   | <b>₦500,000</b> |

#### OTHER COSTS:

|                       |            |
|-----------------------|------------|
| Shop rent (3 months)  | ₦300,000   |
| Renovation/Setup      | ₦200,000   |
| Business registration | ₦50,000    |
| Marketing materials   | ₦100,000   |
| Contingency (10%)     | ₦595,000   |
| OTHER TOTAL:          | ₦1,245,000 |

|                      |            |
|----------------------|------------|
| GRAND TOTAL STARTUP: | ₦6,545,000 |
|----------------------|------------|

#### FUNDING SOURCES:

1. Personal Savings
2. Family Loans
3. Bank Loan (SME)
4. Microfinance Institution
5. Government Grants:
  - SMEDAN (Small & Medium Enterprises Development Agency of Nigeria)
  - BOI (Bank of Industry)
  - State youth empowerment programs

---

#### PHASE 3: LOCATION (Month 3)

##### IDEAL LOCATION CRITERIA:

- ✓ Near architects/engineers offices
- ✓ Near construction companies
- ✓ Near technical colleges
- ✓ High visibility (main road)
- ✓ Adequate parking
- ✓ Reliable electricity
- ✓ Good internet connection
- ✓ Security
- ✓ Accessible by vehicle  
(customers bring large drawings)

##### SHOP SIZE:

- Minimum: 20sqm (for equipment)
- Ideal: 30-40sqm (with stock room)

---

## PHASE 4: SETUP (Months 4-5)

### 1. SHOP PREPARATION

- Paint & renovate
- Install air conditioning
- Setup workstations
- Install shelving
- Setup display area
- Install signage

### 2. EQUIPMENT INSTALLATION

- Plotter setup & calibration
- Computer network setup
- Scanner installation
- Software installation
- Test all equipment
- Train on equipment

### 3. STOCK PROCUREMENT

- Order paper rolls
- Order ink/toner
- Stock drawing supplies
- Organize inventory system

### 4. BUSINESS REGISTRATION

- Register business name
  - Obtain necessary permits
  - Open business bank account
  - Get insurance
- 

## PHASE 5: LAUNCH (Month 6)

### MARKETING ACTIVITIES:

#### 1. Grand Opening Event

- Invite potential customers
- Offer opening discounts
- Demonstrate equipment
- Distribute flyers/cards

#### 2. Online Presence

- Create social media accounts  
(Facebook, Instagram, Twitter)
- Post sample work
- Share tips on CAD/drawing
- Engage with customers



### 3. Direct Marketing

- Visit architects' offices
- Visit engineering firms
- Visit construction sites
- Leave business cards
- Offer introductory discounts

### 4. Advertising

- Local newspaper ads
- Radio spots (if budget allows)
- Roadside banners
- Vehicle branding
- Online ads (Google, social media)

---

## ONGOING OPERATIONS

### MONTHLY EXPENSES (Estimate):

|                          |                 |
|--------------------------|-----------------|
| Rent                     | ₦100,000        |
| Electricity              | ₦50,000         |
| Staff salary (1 person)  | ₦50,000         |
| Internet                 | ₦10,000         |
| Consumables (ink, paper) | ₦150,000        |
| Maintenance              | ₦20,000         |
| Marketing                | ₦30,000         |
| Miscellaneous            | ₦20,000         |
| <b>TOTAL MONTHLY:</b>    | <b>₦430,000</b> |

### REVENUE TARGETS:

To break even, need:

₦430,000 ÷ ₦1,000 (avg A1 price)

= 430 prints per month

= 15 prints per day (30-day month)

Realistic with good location!

---

## SUCCESS FACTORS

### ✓ QUALITY SERVICE

- Fast turnaround
- High-quality prints
- Friendly service

- Technical expertise
  - ✓ RELIABILITY
    - Consistent quality
    - Meet deadlines
    - Backup equipment
    - Power backup
  - ✓ CUSTOMER SERVICE
    - Professional attitude
    - Technical advice
    - Follow-up
    - Customer feedback
  - ✓ CONTINUOUS IMPROVEMENT
    - Upgrade equipment
    - Learn new software
    - Expand services
    - Stay current with technology
- 

## EVALUATION

1. Why are working drawings traditionally called "**blueprints**" even though modern prints are black on white?
  2. What is the modern device used to print **large-format** technical drawings?
  3. Describe the **workflow** for a typical blueprint reproduction service from when a customer brings a file to delivery.
  4. What is the difference between reading an **Assembly Drawing** and a **Detail Drawing** in mechanical engineering?
  5. List **five items** that a blueprint reproduction service center might sell as **merchandise** (besides plotting services).
  6. What are **three key pieces of equipment** needed to set up a blueprint reproduction business?
  7. Explain how a **Quantity Surveyor** uses building drawings differently from how a **Contractor** uses them.
  8. What are **two challenges** of running a blueprint reproduction business in Nigeria?
- 

## ASSIGNMENT

### Part 1: Service Center Visit

1. Visit a local **blueprint service**, **cyber cafe**, or **business center** in your area.

2. Interview the owner/manager and gather the following information:
  - Do they offer plotting/printing services?
  - What size paper can they print? (A4, A3, A2, A1, A0?)
  - What type of plotter/printer do they use? (Brand/Model?)
  - How much do they charge per sheet for different sizes?
  - What other services do they offer?
  - What challenges do they face in the business?
  - How many customers do they serve per day/week?
3. Write a **report** (1-2 pages) summarizing your findings.

## Part 2: Business Plan Outline

4. Create a simple **business plan outline** for starting your own blueprint reproduction service. Include:
  - a) **Business Name** - Choose a creative name
  - b) **Location** - Where would you set up? Why?
  - c) **Equipment Needed** - List main items
  - d) **Estimated Costs** - Research and estimate:
    - Equipment costs
    - Rent (3 months)
    - Initial stock
    - Total startup capital needed
  - e) **Target Customers** - Who will you serve?
  - f) **Services Offered** - What will you provide?
  - g) **Pricing Strategy** - How much will you charge?
  - h) **Marketing Plan** - How will you attract customers?

## Part 3: Drawing Reading Exercise

5. Obtain a **simple building drawing** (floor plan or section) - you can use one from your textbook or that your teacher provides.
6. Study the drawing and answer:
  - What is the title of the drawing?
  - What is the scale?
  - What are the overall dimensions?
  - List 5 specific pieces of information you can extract from the drawing
  - How would an electrician use this drawing?
  - How would a plumber use this drawing?

- How would a builder use this drawing?

#### **Part 4: Create a Service Price List**

7. Design a **professional price list** for a blueprint service including:

- Company name and logo (design your own)
- Services offered (plotting, scanning, binding, etc.)
- Prices for different paper sizes (A4 to A0)
- Black & white vs Color pricing
- Volume discounts
- Additional services pricing
- Contact information
- Opening hours

Make it visually appealing - you can draw it by hand or use a computer if available.



## WEEK 7: MIDTERM BREAK

*This week is for the midterm break. Students should use this time to:*

### MIDTERM BREAK ACTIVITIES

#### 1. Review & Consolidation

- Review all work from Weeks 1-6
- Organize notes and drawings
- Complete any unfinished assignments
- Practice drawing symbols from memory

#### 2. Practical Skills Practice

- Practice drawing material symbols (hatching)
- Practice drawing floor plan symbols
- Sketch different arch types
- Draw roof trusses with proper labels

#### 3. Field Observation

- Visit construction sites (with permission)
- Observe buildings and identify:
  - Arch types used
  - Roof types and structures
  - Building materials
- Take photographs for your portfolio

#### 4. Portfolio Preparation

- Gather your best drawings from the term
- Organize them in a folder/portfolio
- Label each drawing clearly
- Create a table of contents

#### 5. Reading & Research

- Read ahead: Blueprint Reading topics
- Watch CAD tutorial videos online
- Research local architectural firms
- Learn about building materials used in Nigeria

#### 6. Equipment Maintenance

- Clean your drawing instruments
- Sharpen pencils

- Organize your drawing supplies
  - Prepare materials for Week 8
- 



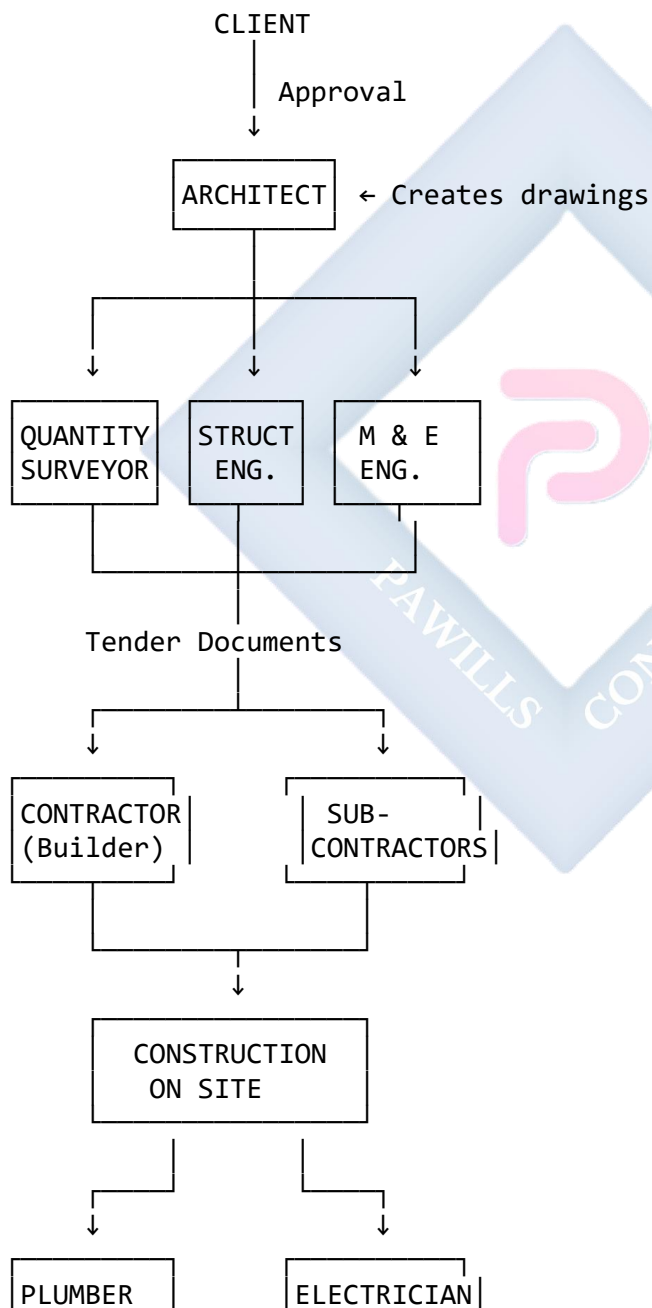
## WEEK 8: BLUEPRINT READING (CONTINUED) - CONSTRUCTION APPLICATIONS

### 8.1 CONSTRUCTION/PRODUCTION APPLICATIONS

Different professionals use the same blueprint for different purposes. Understanding these uses is critical for effective communication in construction.

**Diagram 110: Who Uses Blueprints and How**

BUILDING CONSTRUCTION TEAM



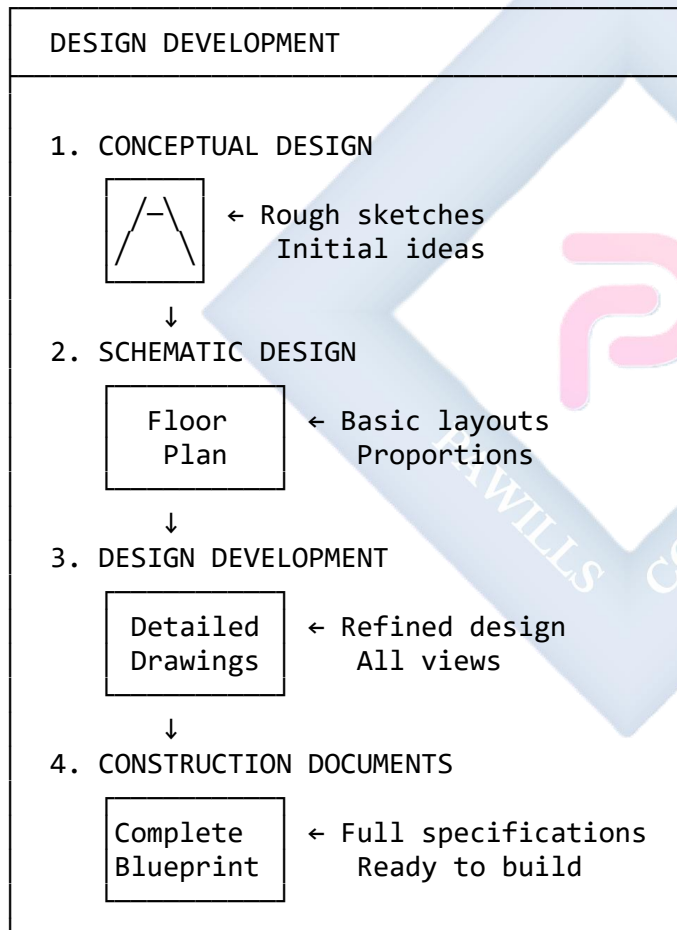
Each professional reads the SAME drawings but extracts DIFFERENT information based on their role!

## 8.1.1 HOW THE ARCHITECT USES THE DRAWING

### Diagram 111: Architect's Perspective

#### ARCHITECT'S USE OF DRAWINGS

PURPOSE: Design Communication & Coordination



#### WHAT ARCHITECT CHECKS:

- ✓ SPATIAL RELATIONSHIPS
  - Do rooms flow logically?
  - Are spaces properly sized?



- Good circulation/movement?
- ✓ PROPORTIONS & AESTHETICS
  - Visual balance
  - Harmonious design
  - Pleasing appearance
- ✓ FUNCTIONALITY
  - Does it meet client needs?
  - Practical room layouts?
  - Adequate natural light?
- ✓ BUILDING CODES
  - Fire safety requirements
  - Accessibility standards
  - Minimum room sizes
  - Ventilation requirements
- ✓ COORDINATION
  - Do all drawings align?
  - Plan matches elevations?
  - Sections match plans?
  - No conflicts between trades?
- ✓ CONSTRUCTABILITY
  - Can this be built?
  - Are details practical?
  - Reasonable costs?
  - Available materials?

DRAWING SET CREATED BY ARCHITECT:

| ARCHITECTURAL DRAWINGS (A-Series) |
|-----------------------------------|
| A-01: Site Plan                   |
| A-02: Ground Floor Plan           |
| A-03: First Floor Plan            |
| A-04: Roof Plan                   |
| A-05: North Elevation             |
| A-06: South Elevation             |
| A-07: East Elevation              |
| A-08: West Elevation              |
| A-09: Section A-A                 |
| A-10: Section B-B                 |
| A-11: Details (Wall sections)     |
| A-12: Details (Windows/Doors)     |
| A-13: Door/Window Schedule        |
| A-14: Finish Schedule             |

## 8.1.2 HOW THE QUANTITY SURVEYOR (QS) USES THE DRAWING

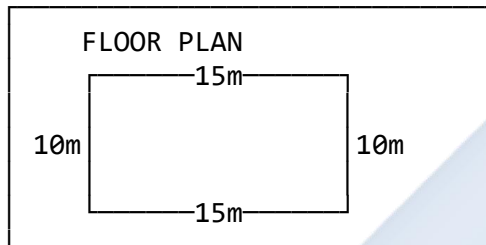
### Diagram 112: Quantity Surveyor's Perspective

#### QUANTITY SURVEYOR'S USE OF DRAWINGS

PURPOSE: Cost Estimation & Budgeting

PROCESS: "TAKING OFF" QUANTITIES

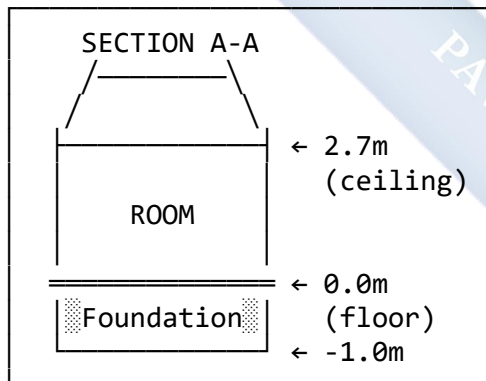
#### STEP 1: MEASURE WALLS



#### CALCULATION:

$$\begin{aligned}\text{Perimeter} &= (15\text{m} + 10\text{m}) \times 2 \\ &= 50\text{m total length}\end{aligned}$$

#### STEP 2: FIND WALL HEIGHT



$$\begin{aligned}\text{Wall height} &= 2.7\text{m} + 0.3\text{m (to wall plate)} \\ &= 3.0\text{m}\end{aligned}$$

#### STEP 3: CALCULATE WALL AREA

$$\begin{aligned}\text{Area} &= \text{Length} \times \text{Height} \\ &= 50\text{m} \times 3.0\text{m} \\ &= 150\text{m}^2\end{aligned}$$

#### STEP 4: DEDUCT OPENINGS

From Floor Plan, count:

- Doors: 4 (each  $2\text{m} \times 1\text{m} = 2\text{m}^2$ )
- Windows: 6 (each  $1.5\text{m} \times 1.2\text{m} = 1.8\text{m}^2$ )

Deductions:

Doors:  $4 \times 2\text{m}^2 = 8\text{m}^2$

Windows:  $6 \times 1.8\text{m}^2 = 10.8\text{m}^2$

Total deduct =  $18.8\text{m}^2$

Net wall area =  $150\text{m}^2 - 18.8\text{m}^2$   
=  $131.2\text{m}^2$

#### STEP 5: CALCULATE BLOCKS NEEDED

Wall thickness: 225mm (9-inch block)

Block size: 450mm  $\times$  225mm  $\times$  225mm

Blocks per  $\text{m}^2$ :

$1\text{m}^2 \div (0.45\text{m} \times 0.225\text{m}) = 9.88 \text{ blocks/m}^2$   
 $\approx 10 \text{ blocks per m}^2$

Total blocks =  $131.2\text{m}^2 \times 10$   
= 1,312 blocks

Add 5% wastage:

$1,312 \times 1.05 = 1,378 \text{ blocks}$

#### STEP 6: CALCULATE OTHER QUANTITIES

FOUNDATION:

From Section:

Width = 600mm

Depth = 500mm

Length = 50m (perimeter)

Volume =  $0.6\text{m} \times 0.5\text{m} \times 50\text{m}$   
=  $15\text{m}^3$  concrete

HARDCORE:

From Section:

Area = Floor area =  $15\text{m} \times 10\text{m} = 150\text{m}^2$

Thickness = 150mm = 0.15m

Volume =  $150\text{m}^2 \times 0.15\text{m}$   
=  $22.5\text{m}^3$

ROOFING:

From Roof Plan & Section:

Roof area calculation (including slopes)  
 $\approx 200\text{m}^2$  (for pitched roof)

Aluminum sheets needed:

Sheet size =  $3\text{m} \times 0.75\text{m} = 2.25\text{m}^2$

Sheets =  $200\text{m}^2 \div 2.25\text{m}^2$

= 89 sheets

Add 10% overlap/waste = 98 sheets

**BILL OF QUANTITIES (BOQ) EXTRACT:**

| ITEM | DESCRIPTION       | UNIT           | QTY   | RATE    |
|------|-------------------|----------------|-------|---------|
| 1.0  | SUBSTRUCTURE      |                |       |         |
| 1.1  | Excavation        | m <sup>3</sup> | 20    | ¥5,000  |
| 1.2  | Hardcore fill     | m <sup>3</sup> | 22.5  | ¥8,000  |
| 1.3  | Concrete 1:3:6    | m <sup>3</sup> | 15    | ¥45,000 |
| 1.4  | DPC membrane      | m <sup>2</sup> | 50    | ¥2,500  |
| 2.0  | SUPERSTRUCTURE    |                |       |         |
| 2.1  | 9" Hollow blocks  | No             | 1,378 | ¥250    |
| 2.2  | Mortar for blocks | m <sup>3</sup> | 3     | ¥25,000 |
| 2.3  | Lintels concrete  | m              | 20    | ¥8,000  |
| 3.0  | ROOFING           |                |       |         |
| 3.1  | Roof trusses      | No             | 15    | ¥35,000 |
| 3.2  | Purlins 2x3"      | m              | 100   | ¥1,200  |
| 3.3  | Aluminum sheets   | No             | 98    | ¥4,500  |
| 4.0  | FINISHES          |                |       |         |
| 4.1  | Plastering        | m <sup>2</sup> | 262   | ¥2,500  |
| 4.2  | Floor tiles       | m <sup>2</sup> | 150   | ¥5,000  |
| 4.3  | Painting          | m <sup>2</sup> | 262   | ¥1,500  |
| 5.0  | DOORS & WINDOWS   |                |       |         |
| 5.1  | Hardwood doors    | No             | 4     | ¥85,000 |
| 5.2  | Aluminum windows  | m <sup>2</sup> | 10.8  | ¥25,000 |

**QS DELIVERABLES:**

- Bill of Quantities (BOQ)
- Cost estimate
- Budget breakdown
- Material schedules
- Tender documents

## 8.1.3 HOW THE CONTRACTOR/BUILDER USES THE DRAWING

### Diagram 113: Contractor's Perspective

#### CONTRACTOR'S USE OF DRAWINGS

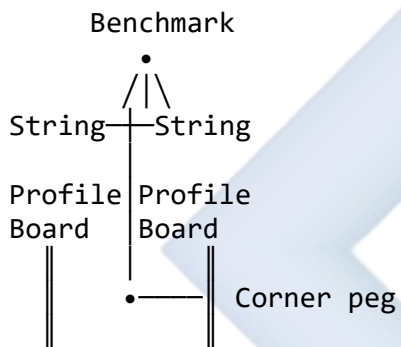
PURPOSE: Construction Execution

#### PHASE 1: SITE SET-OUT

From SITE PLAN & FLOOR PLAN

1. Locate property boundaries
2. Mark building corners
3. Set up profile boards
4. String lines for walls

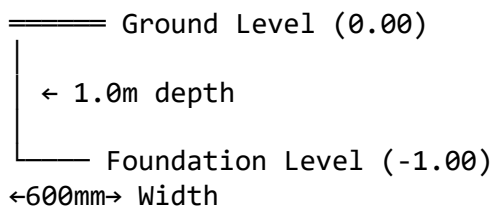
SITE SETUP:



Mark all corners accurately!  
Check diagonal measurements!

#### PHASE 2: EXCAVATION

From SECTION DRAWING



TRENCH DIMENSIONS:

- Depth: 1.0m below ground
- Width: 600mm
- Follow wall centerlines

CHECK:

- ✓ Depth with measuring rod
- ✓ Level with spirit level
- ✓ Width with tape measure

### PHASE 3: FOUNDATION LAYING

From SECTION & SPECIFICATIONS

Concrete  
1:3:6 ← 500mm deep  
600mm wide

#### PROCESS:

1. Place hardcore (if needed)
2. Compact hardcore
3. Pour concrete mix 1:3:6
4. Level and cure

#### MATERIALS NEEDED:

(From BOQ by QS)

- Cement: X bags
- Sand: Y m<sup>3</sup>
- Granite: Z m<sup>3</sup>
- Water: W liters

### PHASE 4: BLOCKWORK/WALLING

From FLOOR PLAN

Wall positions:

External 225mm  
Internal  
150mm blocks

#### LEAVE OPENINGS FOR:

- Doors (dimensions from plan)
- Windows (dimensions from plan)

#### BUILD TO HEIGHT:

From SECTION: 3.0m (to wall plate)

#### CHECK:

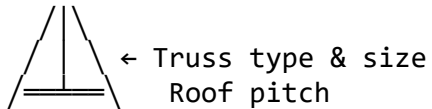
- ✓ Plumb (vertical) with plumb bob
- ✓ Level with spirit level

- ✓ Square corners (3-4-5 rule)
- ✓ Wall thickness correct
- ✓ Door/window openings accurate

## PHASE 5: ROOFING

From SECTION, ROOF PLAN, DETAILS

SECTION shows:



ROOF PLAN shows:

- Truss spacing (e.g., 1.0m c/c)
- Ridge position
- Overhang distances
- Total roof area

CONSTRUCTION SEQUENCE:

1. Fix wall plates
2. Lift trusses into position
3. Brace trusses
4. Fix purlins
5. Fix roofing sheets
6. Install fascia & soffit
7. Fix gutters

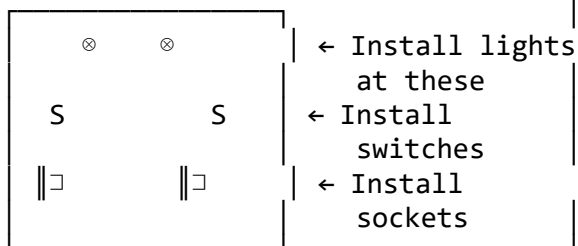
MATERIALS:

From BOQ:

- Trusses: 15 No
- Purlins: 100m
- Roofing sheets: 98 No
- Roofing nails: 5 kg

## PHASE 6: SERVICES INSTALLATION

ELECTRICAL (From Floor Plan)



PLUMBING (From Floor Plan)



WHB  
WC

- ← Run pipes to basin
- ← Run pipes to toilet

Coordinate with:

- Electrician
- Plumber
- Other subcontractors

## PHASE 7: FINISHING

From FINISH SCHEDULE

### WALLS:

- Plaster both sides
- Paint (color from schedule)

### FLOORS:

- Screed
- Tiles (type from schedule)

### CEILING:

- Install joists
- Fix ceiling boards/POP
- Paint

### DOORS & WINDOWS:

- Install as per schedule
- Hardware as specified

## CONTRACTOR'S DAILY TASKS:

### Morning:

- Check drawings
- Brief workers
- Check material stock
- Measure/set out work

### During Day:

- Supervise workers
- Check quality
- Verify dimensions



- Solve problems
- Coordinate trades

Evening:

- Check progress vs drawings
- Plan next day's work
- Order materials needed
- Report to client/architect

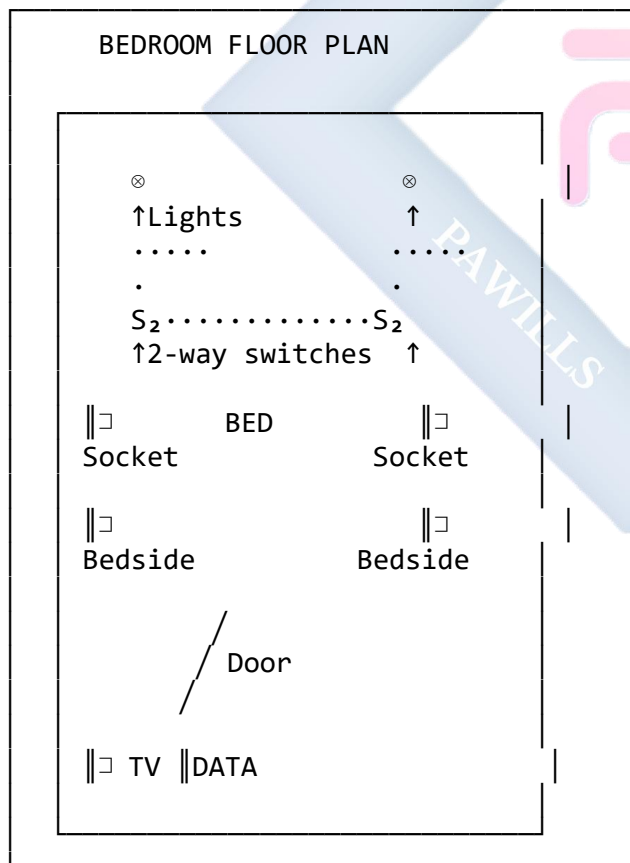
## 8.1.4 HOW THE ELECTRICIAN USES THE DRAWING

### Diagram 114: Electrician's Perspective

ELECTRICIAN'S USE OF DRAWINGS

PURPOSE: Electrical Installation

FROM FLOOR PLAN - EXTRACT ELECTRICAL INFO

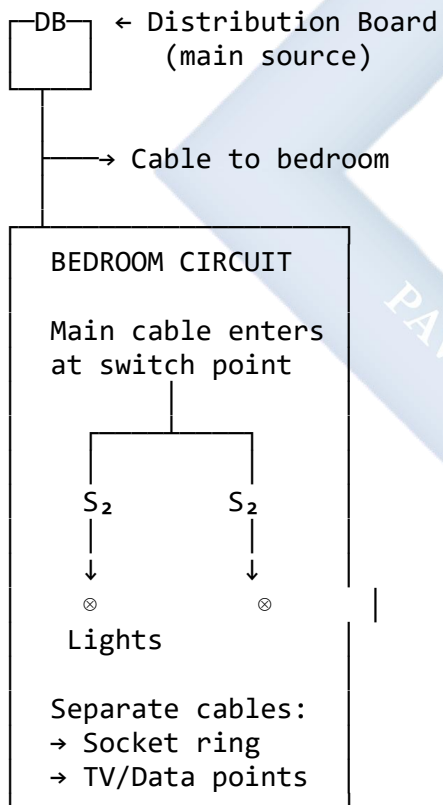


## ELECTRICIAN'S CHECKLIST:

### 1. COUNT ALL FIXTURES:

| ITEM             | QTY |
|------------------|-----|
| Ceiling Lights   | 2   |
| Switches (2-way) | 2   |
| Socket Outlets   | 4   |
| TV Point         | 1   |
| Data Point       | 1   |

### 2. PLAN CABLE ROUTES:



### 3. DETERMINE CABLE SIZES:

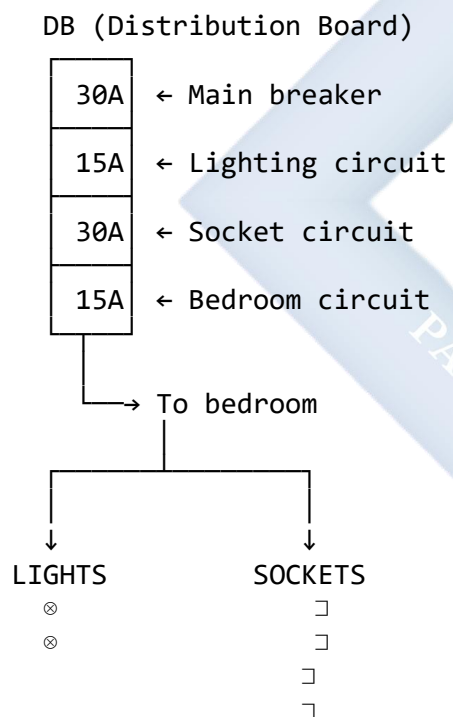
| CIRCUIT  | CABLE              |
|----------|--------------------|
| Lighting | 1.5mm <sup>2</sup> |

|                |                    |
|----------------|--------------------|
| Socket outlets | 2.5mm <sup>2</sup> |
| TV/Data        | Coaxial/<br>Cat6   |

#### 4. INSTALLATION SEQUENCE:

- a) Chase/conduit walls
- b) Run cables from DB
- c) Install switch boxes
- d) Install socket boxes
- e) Pull cables through
- f) Connect at DB
- g) Test circuits
- h) Install fixtures
- i) Final testing

#### WIRING DIAGRAM (Created by Electrician):



#### TESTING CHECKLIST:

- ✓ Continuity test (all cables)
- ✓ Insulation resistance test
- ✓ Polarity check (Live/Neutral/Earth)
- ✓ RCD/MCB operation test
- ✓ Load test (all circuits)
- ✓ Final inspection

MATERIALS NEEDED (For this bedroom):

| ITEM                     | QTY | UNIT  |
|--------------------------|-----|-------|
| 1.5mm <sup>2</sup> cable | 30m | metre |
| 2.5mm <sup>2</sup> cable | 40m | metre |
| PVC conduit 20mm         | 50m | metre |
| Switch boxes             | 2   | No    |
| Socket boxes             | 4   | No    |
| 2-way switches           | 2   | No    |
| Socket outlets           | 4   | No    |
| Ceiling roses            | 2   | No    |
| Light bulbs              | 2   | No    |
| TV outlet                | 1   | No    |
| Data outlet              | 1   | No    |
| Cable ties               | 50  | No    |
| Conduit clips            | 100 | No    |

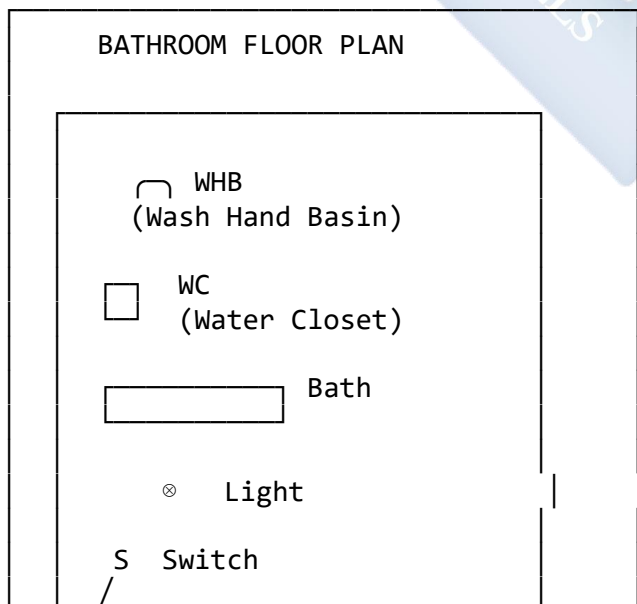
## 8.1.5 HOW THE PLUMBER USES THE DRAWING

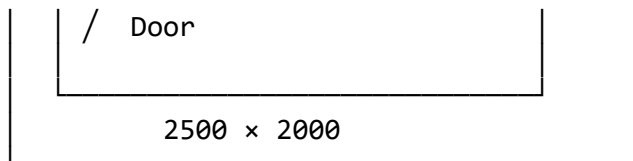
### Diagram 115: Plumber's Perspective

PLUMBER'S USE OF DRAWINGS

PURPOSE: Plumbing Installation

FROM FLOOR PLAN - EXTRACT PLUMBING INFO



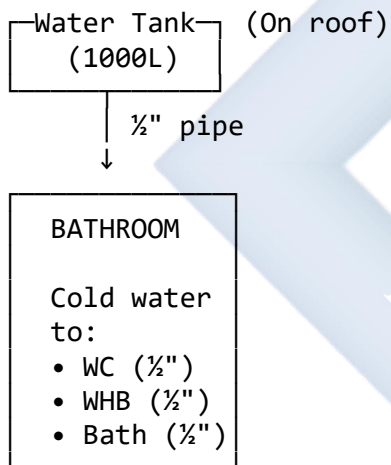


## PLUMBER'S CHECKLIST:

### 1. IDENTIFY ALL FIXTURES:

| FIXTURE           | QTY | SIZE      |
|-------------------|-----|-----------|
| Water Closet (WC) | 1   | P-trap    |
| Wash Hand Basin   | 1   | 400mm dia |
| Bath Tub          | 1   | 1700mm    |
| Floor Drain       | 1   | 100mm     |

### 2. WATER SUPPLY SYSTEM:

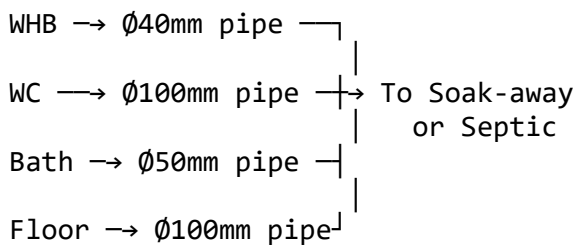


If hot water needed:

Water Tank → Heater → Fixtures

### 3. DRAINAGE SYSTEM:

All fixtures drain to:



#### 4. PIPE SIZING:

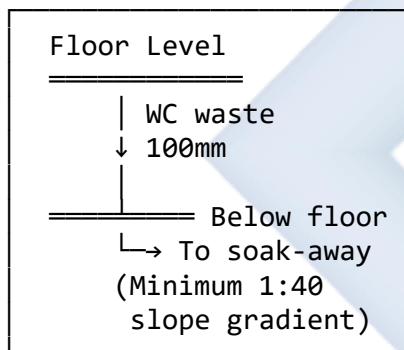
| FIXTURE     | SUPPLY    | WASTE      |
|-------------|-----------|------------|
| WC          | ½" (15mm) | 4" (100mm) |
| WHB         | ½" (15mm) | 1½" (40mm) |
| Bath        | ½" (15mm) | 2" (50mm)  |
| Floor drain | ---       | 4" (100mm) |

#### INSTALLATION SEQUENCE:

##### PHASE 1: SOIL & WASTE PIPES

First install because:

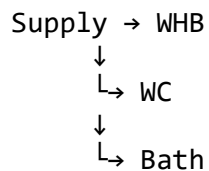
- Largest pipes (100mm)
- Fixed positions
- Must slope correctly



##### PHASE 2: WATER SUPPLY PIPES

Can be:

- Surface mounted
- Concealed in walls
- Under floor



Include:

- Shut-off valves
- Air vents

- Access points

### PHASE 3: FIXTURE INSTALLATION

#### WC Installation:

1. Set in position
2. Connect soil pipe
3. Connect water supply
4. Seal base with silicone
5. Bolt down
6. Test flush

#### WHB Installation:

1. Fix brackets/pedestal
2. Set basin level
3. Connect waste pipe
4. Connect taps
5. Connect supply
6. Seal edges
7. Test for leaks

#### Bath Installation:

1. Set on bath feet/support
2. Check level
3. Connect waste/overflow
4. Connect supply to mixer
5. Test for leaks
6. Seal to wall

### PHASE 4: TESTING

#### Water Supply Test:

- ✓ Turn on water
- ✓ Check all joints
- ✓ Look for leaks
- ✓ Check flow rates
- ✓ Test all taps

#### Drainage Test:

- ✓ Run water in each fixture
- ✓ Check waste flows freely

- ✓ No leaks at joints
- ✓ Proper drainage speed
- ✓ No gurgling sounds

#### MATERIALS NEEDED:

| ITEM                | QTY | UNIT   |
|---------------------|-----|--------|
| WC suite (complete) | 1   | set    |
| WHB with pedestal   | 1   | set    |
| Bath tub (acrylic)  | 1   | No     |
| Taps (basin)        | 1   | pair   |
| Taps (bath)         | 1   | set    |
| ½" UPVC pipe        | 15m | metre  |
| 40mm PVC pipe       | 3m  | metre  |
| 50mm PVC pipe       | 4m  | metre  |
| 100mm PVC pipe      | 5m  | metre  |
| P-trap (WC)         | 1   | No     |
| Trap (WHB)          | 1   | No     |
| Floor drain         | 1   | No     |
| Stop valves         | 3   | No     |
| Pipe fittings       | As  | needed |
| Silicone sealant    | 2   | tubes  |
| PTFE tape           | 3   | rolls  |
| Pipe clips          | 30  | No     |

#### COMMON PROBLEMS & SOLUTIONS:

| PROBLEM            | CAUSE                      | SOLUTION         |
|--------------------|----------------------------|------------------|
| Low water pressure | Pipe too small/<br>blocked | Use larger pipe  |
| Slow drainage      | Wrong slope                | Check gradient   |
| Gurgling sounds    | No vent                    | Install air vent |
| Leaking joints     | Poor sealing               | Re-seal properly |



## 8.2 MANAGING BLUEPRINT SERVICES (THE BUSINESS)

Running a successful blueprint reproduction service requires good management.

### Diagram 116: Business Management

#### BLUEPRINT SERVICE BUSINESS MANAGEMENT

##### DAILY OPERATIONS

###### OPENING (8:00 AM)

- ✓ Turn on equipment
- ✓ Boot up computer
- ✓ Check plotter status
- ✓ Review day's orders
- ✓ Check paper/ink levels

###### MORNING SHIFT (8:00 AM - 1:00 PM)

- Receive customer orders
- Process urgent jobs first
- Quality check all prints
- Maintain clean workspace

###### AFTERNOON SHIFT (2:00 PM - 6:00 PM)

- Continue processing orders
- Deliver completed jobs
- Restock materials
- Clean and maintain equipment

###### CLOSING (6:00 PM)

- ✓ Complete pending jobs
- ✓ Shut down equipment properly
- ✓ Secure shop
- ✓ Record day's transactions
- ✓ Plan next day

##### CUSTOMER SERVICE EXCELLENCE

###### GREETING

"Good morning! Welcome to [Business Name]. How may I help you today?"

###### RECEIVING ORDER

1. "What size do you need?"
2. "How many copies?"

3. "Color or black and white?"
4. "When do you need it?"
5. Check file compatibility
6. Give clear quotation
7. Get customer contact

#### QUALITY ASSURANCE

- Preview before printing
- Show sample to customer
- Reprint if unsatisfactory
- Deliver on time

#### FOLLOW-UP

- Call when ready
- Thank customer
- Ask for feedback
- Offer future discounts

### RECORD KEEPING

#### DAILY SALES RECORD

Date: 10/11/2025

| No  | Customer | Size | Qty | Amount |
|-----|----------|------|-----|--------|
| 001 | Arc.Ade  | A1   | 5   | ₦5,000 |
| 002 | Eng.Ltd  | A2   | 10  | ₦6,000 |
| 003 | Builder  | A3   | 3   | ₦1,200 |

Total Sales: ₦12,200

Cost of materials: ₦3,500

Profit: ₦8,700

### CHALLENGES & SOLUTIONS

#### CHALLENGE: Power Failure

##### PROBLEM:

- Plotter stops mid-print
- Computer shuts down

- Lose unsaved work

**SOLUTIONS:**

- ✓ UPS/Inverter (2-3 hours)
- ✓ Generator backup
- ✓ Auto-save enabled
- ✓ Print queue manager
- ✓ Inform customer of delay

**CHALLENGE: File Incompatibility**

**PROBLEM:**

- Corrupt files
- Wrong file format
- Missing fonts

**SOLUTIONS:**

- ✓ Multiple software installed (AutoCAD, Revit, PDF, etc.)
- ✓ Font library
- ✓ File conversion tools
- ✓ Work with customer to fix

**CHALLENGE: Equipment Breakdown**

**PROBLEM:**

- Plotter malfunction
- Print head clog
- Paper jam

**SOLUTIONS:**

- ✓ Regular maintenance
- ✓ Keep spare parts
- ✓ Technician on call
- ✓ Service contract
- ✓ Backup plotter (if possible)

**CHALLENGE: Competition**

**SOLUTIONS:**

- ✓ Better quality
- ✓ Faster service
- ✓ Lower prices
- ✓ Better customer service
- ✓ Additional services
- ✓ Loyalty programs

---

## EVALUATION

1. Which professional uses building drawings to **calculate the number of blocks** needed? What is this calculation process called?
2. Which drawing view (Plan, Elevation, or Section) does the **contractor** primarily use to know the **foundation depth**?
3. Explain how an **electrician** and a **plumber** use the same floor plan differently. What specific information does each extract?
4. What are **three major challenges** of running a blueprint reproduction business in Nigeria? Suggest one solution for each.
5. Describe the **site set-out process** that a contractor performs using the floor plan.
6. If a room on a floor plan shows dimensions "4000      3000", and the section shows "floor to ceiling height 2700", what are the complete dimensions needed for: a) Plastering the walls? b) Tiling the floor? c) Installing the ceiling?
7. Why is **power supply** such a critical issue for blueprint reproduction services?

---

## ASSIGNMENT

### Part 1: Professional Role Analysis

1. Choose **ONE** professional role:
  - Architect
  - Quantity Surveyor
  - Contractor/Builder
  - Electrician
  - Plumber
2. Create a **detailed guide** (2-3 pages) explaining:
  - How they use building drawings
  - What specific information they extract
  - What tools/equipment they need
  - Step-by-step work process
  - Common problems they face
  - Include diagrams/sketches

### Part 2: Quantity Take-Off Exercise

3. You are given a simple rectangular building:

- Length: 12m
- Width: 8m
- Wall height: 3.0m
- Wall thickness: 225mm (9-inch blocks)
- Doors: 3 (each  $2\text{m} \times 1\text{m}$ )
- Windows: 4 (each  $1.5\text{m} \times 1.2\text{m}$ )

Calculate: a) Perimeter of walls (in metres) b) Total wall area (in  $\text{m}^2$ ) c) Area of openings (doors + windows) d) Net wall area (wall area minus openings) e) Number of 9-inch blocks needed (10 blocks per  $\text{m}^2$ ) f) Add 5% for wastage

Show all calculations clearly!

### Part 3: Service Business Plan

4. You want to start a blueprint reproduction service. Create a simple **one-page business plan** including:

- Business name
- Location (where and why)
- Target customers
- Equipment needed
- Estimated startup costs
- Services offered
- Pricing strategy
- How you will handle power challenges
- Marketing plan

### Part 4: Drawing Reading Exercise

5. Obtain a building drawing (floor plan with section).

Study it and answer as different professionals:

#### As Contractor:

- Where will you dig foundations?
- How deep?
- What dimensions for walls?

#### As Electrician:

- How many lights needed?
- How many switches?
- How many socket outlets?
- Where is the distribution board?

#### As Plumber:

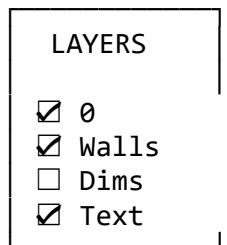
- What plumbing fixtures are shown?
- Where will supply pipes run?
- Where will waste pipes go?

Present your findings in a table format.

---



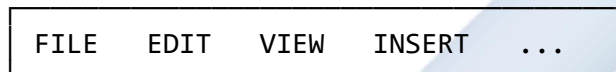




## Key Interface Components:

### Diagram 118: Interface Elements Explained

#### 1. MENU BAR



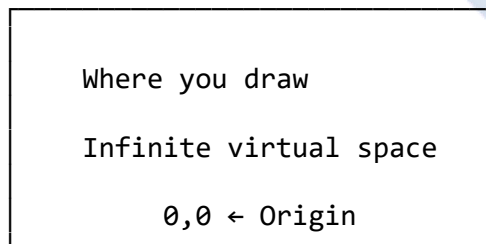
- Traditional drop-down menus
- Access all commands
- File operations (New, Open, Save)

#### 2. RIBBON/TOOLBAR



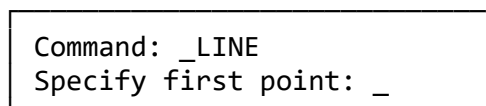
- Quick access to drawing tools
- Icons for common commands
- Organized by function (Draw, Modify)

#### 3. DRAWING AREA/CANVAS



- Main workspace
- Coordinate system based
- Pan and zoom freely

#### 4. COMMAND LINE





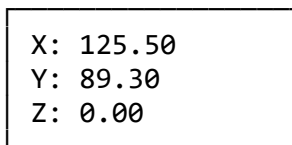
- Shows active command
- Displays prompts
- Enter commands via keyboard
- Shows coordinates

#### 5. STATUS BAR



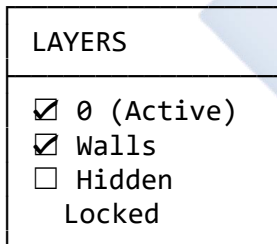
- Toggle drawing aids
- Show/hide features
- Quick settings access

#### 6. COORDINATE DISPLAY



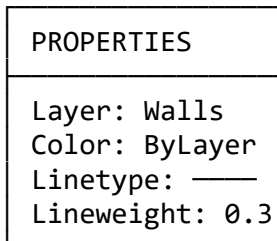
- Shows cursor position
- Real-time coordinates
- Useful for precision

#### 7. LAYERS PANEL



- Organize drawing elements
- Control visibility
- Lock/unlock layers

#### 8. PROPERTIES PANEL



- Object properties
- Modify attributes
- Current settings

## 9.2 FUNDAMENTAL CAD COMMANDS

CAD commands fall into three categories: **Drawing**, **Modifying**, and **Management**.

---

### 9.2.1 DRAWING COMMANDS

#### Diagram 119: Basic Drawing Commands

---

---

##### 1. LINE COMMAND

---

Command: LINE (or L)

Method:

1. Specify first point: Click or type coordinates
2. Specify next point: Click or type
3. Continue or press Enter to finish

Example:

Point 1 (0,0) —→ Point 2 (100,0)  
100 units long

Shortcut: L [Enter]

Icon: 

---

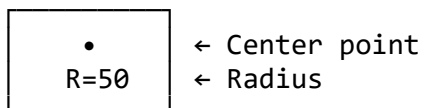
##### 2. CIRCLE COMMAND

---

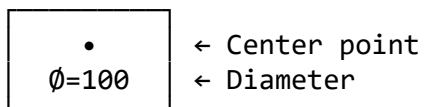
Command: CIRCLE (or C)

Options:

a) Center, Radius



b) Center, Diameter

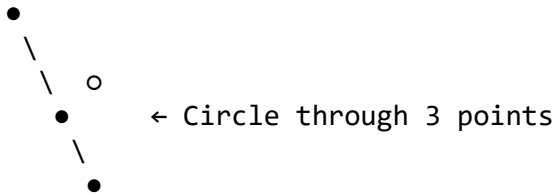


c) 2-Point (diameter)

Point 1 • ————— • Point 2

(Diameter)

d) 3-Point (on circumference)



Shortcut: C [Enter]

Icon: ○

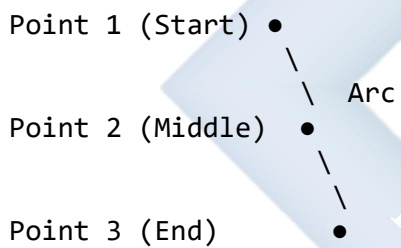
---

### 3. ARC COMMAND

---

Command: ARC (or A)

Common method: 3-Point



Other options:

- Start, Center, End
- Start, Center, Angle
- Start, End, Radius

Shortcut: A [Enter]

Icon: ⤿

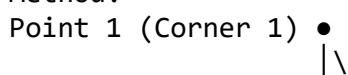
---

### 4. RECTANGLE COMMAND

---

Command: RECTANGLE (or REC)

Method:



Point 2 (Corner 2)



• (Opposite corner)

Can specify:

- Dimensions (Length, Width)
- Area
- Chamfer/Fillet corners

Shortcut: REC [Enter]

Icon: 

---

## 5. POLYGON COMMAND

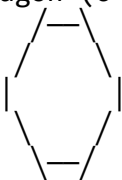
---

Command: POLYGON (or POL)

Options:

- Number of sides (3-1024)
- Inscribed in circle
- Circumscribed about circle

Hexagon (6 sides):



Shortcut: POL [Enter]

Icon: 

---

## 6. ELLIPSE COMMAND

---

Command: ELLIPSE (or EL)

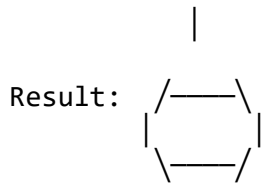
Method: Axis endpoints

Point 1 • ————— • Point 2

(Major axis)



o ← Half minor axis



Shortcut: EL [Enter]  
Icon: ○

---

---

## 7. POLYLINE COMMAND

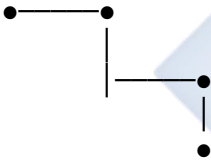
---

---

Command: PLINE (or PL)

Like LINE but:

- Connected segments
- Single object
- Can have width
- Can include arcs



All one object!

Shortcut: PL [Enter]

Icon:

---

---

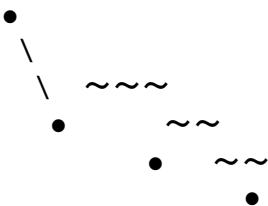
## 8. SPLINE COMMAND

---

---

Command: SPLINE (or SPL)

Smooth curve through points:



Used for: Organic shapes,  
Contours, etc.

Shortcut: SPL [Enter]

Icon: ~

---

## 9. HATCH COMMAND

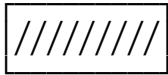
---

Command: HATCH (or H)

Fill area with pattern:



← Solid



← Lines (ANSI31)



← Custom pattern

Used for: Material indication,  
Emphasis, sections

Shortcut: H [Enter]

Icon: 

---

## 10. TEXT COMMAND

---

Command: TEXT or MTEXT

TEXT (Single line):

SINGLE LINE OF TEXT

MTEXT (Multi-line):

Multiple lines  
of text in a  
bounded area

Settings:

- Font (Arial, Romans, etc.)
- Height
- Justification
- Rotation

Shortcut: T [Enter] or MT [Enter]

Icon: A

---

## 9.2.2 MODIFYING COMMANDS

**The most powerful CAD commands!**

### Diagram 120: Modify Commands

---

---

#### 1. MOVE COMMAND

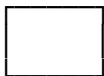
---

Command: MOVE (or M)

Process:

1. Select object(s)
2. Specify base point
3. Specify destination

BEFORE:



AFTER:



Shortcut: M [Enter]

Icon: ⊕→

---

#### 2. COPY COMMAND

---

Command: COPY (or CO)

Process:

1. Select object(s)
2. Specify base point
3. Specify copy location(s)

#### 4. Can create multiple copies

ORIGINAL:      COPIES:



Shortcut: CO [Enter]

Icon: ⊕ ⊕

---

### 3. ROTATE COMMAND

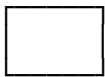
---

Command: ROTATE (or R0)

Process:

1. Select object(s)
2. Specify base point (pivot)
3. Specify rotation angle

BEFORE:      AFTER (45°):



Shortcut: R0 [Enter]

Icon: ↻

---

### 4. SCALE COMMAND

---

Command: SCALE (or SC)

Process:

1. Select object(s)
2. Specify base point
3. Specify scale factor  
(2 = double size)  
(0.5 = half size)

BEFORE:      AFTER (x2):







Shortcut: SC [Enter]

Icon:

---

## 5. TRIM COMMAND ★ Very Important!

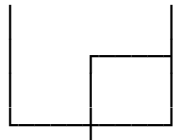
---

Command: TRIM (or TR)

Process:

1. Select cutting edge(s)
2. Press Enter
3. Select part to trim

BEFORE:



Cut here ↑

AFTER:



↑ Trimmed!

Shortcut: TR [Enter]

Icon:

---

## 6. EXTEND COMMAND

---

Command: EXTEND (or EX)

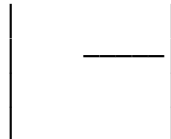
Process:

1. Select boundary edge(s)
2. Press Enter
3. Select line to extend

BEFORE:



AFTER:



Shortcut: EX [Enter]

Icon:

---

## 7. OFFSET COMMAND ★ Very Important!

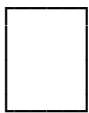
---

Command: OFFSET (or O)

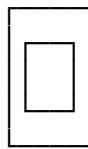
Process:

1. Specify distance
2. Select object
3. Click which side to offset

BEFORE:



AFTER (offset 50):



← 50mm  
inside

Uses:

- Wall thickness
- Parallel lines
- Border creation

Shortcut: O [Enter]

Icon: =

---

## 8. MIRROR COMMAND

---

Command: MIRROR (or MI)

Process:

1. Select object(s)
2. Specify mirror line (2 points)
3. Keep or delete original

BEFORE:



→

AFTER:



Mirror line ↑

Shortcut: MI [Enter]

Icon: |●

---

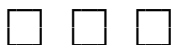
## 9. ARRAY COMMAND

---

Command: ARRAY (or AR)

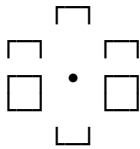
Types:

a) RECTANGULAR ARRAY



← Rows × Columns  
with spacing

b) POLAR ARRAY (Circular)



← Around center  
at equal angles

c) PATH ARRAY



← Along a path

Shortcut: AR [Enter]

Icon: 

---

## 10. FILLET COMMAND

---

Command: FILLET (or F)

Creates rounded corner:

BEFORE:



Sharp corner

AFTER (R=10):



Rounded!

Process:

1. Command: FILLET
2. Set Radius (R)
3. Select first line
4. Select second line

Radius = 0 creates sharp corner

Shortcut: F [Enter]

Icon: 

---

## 11. CHAMFER COMMAND

---

Command: CHAMFER (or CHA)

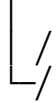
Creates angled corner:

BEFORE:



Sharp corner

AFTER:



Beveled!

Set distances for each side

Shortcut: CHA [Enter]

Icon: 

---

## 12. ERASE COMMAND

---

Command: ERASE (or E)

Simply deletes selected objects

Select → Press Delete key

OR

Command: ERASE → Select → Enter

Shortcut: E [Enter] or Del key

Icon: 

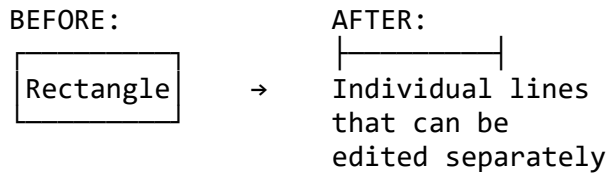
---

## 13. EXPLODE COMMAND

---

Command: EXPLODE (or X)

Breaks complex objects into parts:



Use on: Rectangles, Polylines,  
Blocks, Hatches

Shortcut: X [Enter]

Icon:

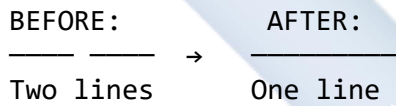
---

## 14. JOIN COMMAND

---

Command: JOIN (or J)

Combines separate lines into one:



Lines must be:

- Collinear (same line)
- Touching or nearly touching

Shortcut: J [Enter]

Icon:

---

### 9.2.3 MANAGEMENT COMMANDS

#### Diagram 121: Layer Management

---

---

LAYERS - THE MOST IMPORTANT CONCEPT!

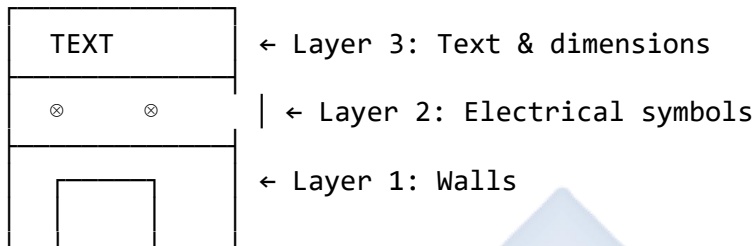
---

WHAT ARE LAYERS?

Think of layers as transparent sheets stacked on

top of each other. Each layer can contain different drawing elements.

#### VISUAL CONCEPT:



All layers combined = Complete drawing

#### LAYER PROPERTIES:

| NAME       | COLOR   | LINETYPE   | ON/OFF                              |         |
|------------|---------|------------|-------------------------------------|---------|
| 0          | White   | Continuous | <input checked="" type="checkbox"/> | Default |
| Walls      | Red     | Continuous | <input checked="" type="checkbox"/> | Active  |
| Doors      | Green   | Continuous | <input checked="" type="checkbox"/> |         |
| Windows    | Blue    | Continuous | <input checked="" type="checkbox"/> |         |
| Dimensions | Cyan    | Continuous | <input type="checkbox"/>            | Hidden  |
| Text       | Yellow  | Continuous | <input checked="" type="checkbox"/> |         |
| Electrical | Magenta | Continuous | <input checked="" type="checkbox"/> |         |
| Plumbing   | Brown   | Continuous | <input checked="" type="checkbox"/> |         |
| Hidden     | Gray    | Dashed     | <input type="checkbox"/>            | Locked  |

#### WHY USE LAYERS?

##### 1. ORGANIZATION

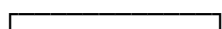
Keep different elements separate:

- Walls on "Walls" layer
- Dimensions on "Dimensions" layer
- Text on "Text" layer

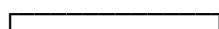
##### 2. VISIBILITY CONTROL

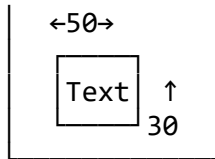
Turn layers ON/OFF:

All layers ON:

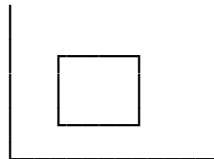


Dimensions OFF:





Cluttered!



Clean for presentation!

### 3. COLOR CODING

Different colors for different elements:

- Red = Walls (thick)
- Blue = Windows
- Green = Doors
- Yellow = Text

### 4. EDITING CONTROL

Lock layers to prevent accidental changes:

Locked layer = Can see but can't edit

### 5. PRINTING CONTROL

Choose which layers to print/plot

LAYER COMMANDS:

LAYER (or LA)

| LAYER PROPERTIES MANAGER                      |                                 |
|-----------------------------------------------|---------------------------------|
| <input checked="" type="checkbox"/> New Layer | <input type="checkbox"/> Delete |
| Set Color                                     | Linetype                        |
| Lock/Unlock                                   | On/Off                          |

LAYMCUR (Make Current)

- Select object
- Its layer becomes active

LAYISO (Layer Isolate)

- Select object
- All other layers turn off

LAYUNISO (Layer Unisolate)

- Restore all layers

LAYER NAMING CONVENTIONS:

Good names:

✓ A-Wall-Exterior

Bad names:

✗ Layer1

- ✓ A-Door
- ✓ A-Window
- ✓ E-Lighting
- ✓ E-Power
- ✓ P-Water-Supply
- ✓ P-Drainage
- ✗ New Layer
- ✗ Stuff
- ✗ Things
- ✗ asdf

Prefix codes:

A- = Architectural

E- = Electrical

P- = Plumbing

S- = Structural

M- = Mechanical

#### TYPICAL ARCHITECTURAL LAYER STRUCTURE:

| LAYER NAME      | COLOR   | USE        |
|-----------------|---------|------------|
| A-Wall-Exterior | Red     | Thick      |
| A-Wall-Interior | Yellow  | Thin       |
| A-Door          | Green   | Doors      |
| A-Window        | Blue    | Windows    |
| A-Furn          | Brown   | Furniture  |
| A-Dim           | Cyan    | Dimensions |
| A-Text          | White   | Labels     |
| A-Hatch         | Gray    | Patterns   |
| E-Light         | Yellow  | Lights     |
| E-Power         | Magenta | Sockets    |
| E-Switch        | Orange  | Switches   |
| P-Fixture       | Blue    | WC, Basin  |
| P-Supply        | Cyan    | Water      |
| P-Waste         | Brown   | Drainage   |

#### LAYER STATES:

##### BEFORE PLOTTING:

- ☒ A-Wall-Exterior
- ☒ A-Door
- ☒ A-Window
- ☒ A-Dim (Show dimensions)
- ☒ A-Text
- ☐ A-Construction (Hide guides)
- ☐ E-Light (Hide electrical)



☐ P-Fixture (Hide plumbing)

Select what to show in final plot!

---

## 9.3 PRODUCING A TECHNICAL DRAWING USING CAD

### Diagram 122: CAD Drawing Process

COMPLETE CAD WORKFLOW

---

---

#### STEP 1: SETUP

---

Command: NEW (or Ctrl+N)

- Start new drawing

Command: UNITS

- Set drawing units:

Length Type:

- Decimal
- Architectural

Precision: 0.00

Angle Type:

- Decimal Degrees

For architecture: Millimeters

For engineering: Millimeters

Command: LIMITS

- Set drawing boundaries
  - Lower left: 0,0
  - Upper right: 420,297 (for A3)

Command: ZOOM ALL (or Z, A)

- View entire drawing area

---

#### STEP 2: CREATE LAYERS

---

Command: LAYER (or LA)

Create layers systematically:

1. Walls - Red - Continuous
2. Doors - Green - Continuous
3. Windows - Blue - Continuous
4. Furniture - Brown - Continuous
5. Dimensions - Cyan - Continuous
6. Text - White - Continuous
7. Centerlines - Yellow - Center
8. Hidden - Gray - Dashed

Set "Walls" as current layer

---

### STEP 3: DRAW BASE GEOMETRY

---

Example: Simple Room Floor Plan

On "Walls" layer:

Command: RECTANGLE  
First corner: 0,0  
Other corner: @6000,4000  
(Creates 6m x 4m rectangle)

Result:



4000mm

Command: OFFSET  
Distance: 225 (wall thickness)  
Select: Outer rectangle  
Side: Inside

Result:





---

#### STEP 4: ADD OPENINGS

---

Switch to "Doors" layer

Door opening (900mm):

Command: LINE

Draw temporary lines for door position

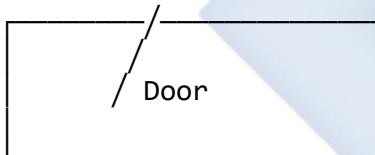
Command: TRIM

Trim wall lines to create opening

Command: ARC

Create door swing (90° arc)

Result:

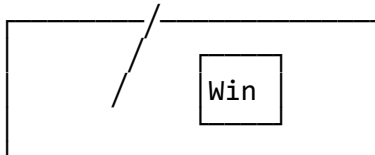


Switch to "Windows" layer

Command: LINE

Create window openings (1200mm each)

Result:



---

#### STEP 5: ADD DETAILS

---

Switch to "Furniture" layer

Add bathroom fixtures:

- WC symbol
- Basin symbol
- Bath tub

Command: CIRCLE (for basin)

Command: RECTANGLE (for bath)

Use previously learned symbols

---

## STEP 6: DIMENSION

---

Switch to "Dimensions" layer

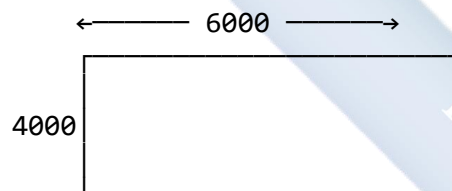
Command: DIMLINEAR (or DLI)

Select first point

Select second point

Place dimension line

Result:



Command: DIMSTYLE (or D)

Set dimension style:

- Text height: 3mm (scaled)
- Arrow size: 2mm
- Units: Decimal, 0 precision

---

## STEP 7: ANNOTATE

---

Switch to "Text" layer

Command: MTEXT (or MT)

Add room labels:

- "BATHROOM"
- "2500 x 2000"
- Material notes

Command: LEADER

Add specific notes with arrows

---

## STEP 8: ADD TITLE BLOCK

---

Command: INSERT

Insert pre-made title block

OR

Command: LINE, MTEXT

Draw title block manually

Fill in:

- Drawing title
- Scale
- Date
- Drawn by
- Project info

---

## STEP 9: PLOT/PRINT

---

Command: PLOT (or Ctrl+P)

Settings:

Printer: DWG To PDF  
 Paper size: A3  
 Plot area: Window  
 Scale: 1:50  
 Plot style: monochrome

- ☒ Center plot
- ☒ Plot with lineweights

Preview before plotting!

Command: PREVIEW

---

---

## STEP 10: SAVE

---

---

Command: SAVE (or Ctrl+S)  
File format: .DWG (native AutoCAD)

Also save as:

Command: SAVEAS

- PDF (for sharing)
- DXF (for compatibility)

Naming convention:

ProjectName\_FloorPlan\_Rev01\_Date.dwg

Example:

NigerianHouse\_GroundFloor\_R01\_20251110.dwg

---

## 9.4 COMPARING MANUAL DRAWING AND CAD

### Diagram 123: Comprehensive Comparison

---

---

#### DETAILED COMPARISON: MANUAL vs CAD

---

---

##### 1. STARTUP COSTS

---

MANUAL:

Initial Cost: ₦50,000 - ₦100,000

- Drawing board: ₦15,000
- T-square: ₦3,000
- Set squares: ₦2,000
- Compass set: ₦5,000
- Scales: ₦3,000
- Pencils/pens: ₦5,000
- Paper: ₦10,000
- Misc tools: ₦10,000

CAD:

Initial Cost: ₦500,000 - ₦1,500,000

- Computer: ₦400,000

- Monitor (large): ₦100,000
- Mouse/keyboard: ₦20,000
- CAD software: ₦500,000/year  
(or free alternatives)
- Plotter (optional): ₦2,000,000

WINNER: Manual (Lower initial investment)

## 2. SPEED

---

### MANUAL:

Simple drawing (A3 floor plan): 4-8 hours

- Layout: 30 min
- Draw walls: 1 hour
- Add details: 2 hours
- Dimension: 1 hour
- Title block: 30 min
- Cleanup: 30 min

### CAD:

Same drawing: 1-2 hours

- Setup layers: 10 min
- Draw walls (offset): 15 min
- Add details (copy/block): 30 min
- Dimension (automatic): 15 min
- Title block (template): 5 min

WINNER: CAD (3-4x faster)

## 3. ACCURACY

---

### MANUAL:

Typical accuracy:  $\pm 0.5\text{mm}$

Limited by:

- Hand steadiness
- Tool quality
- Paper stability
- Human error

### CAD:

Accuracy:  $\pm 0.001\text{mm}$  or better

- Computer precision
- Snap to exact points
- Coordinate input
- No measurement error

Example:

Manual: "approximately 2450mm"

CAD: Exactly 2450.000mm

WINNER: CAD (Perfect precision)

#### 4. EDITING

---

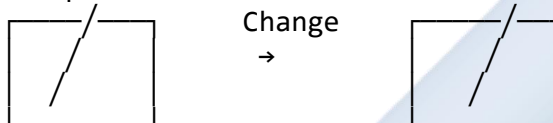
##### MANUAL:

Changes require:

- Erase (damages paper)
- Redraw completely
- Or start new sheet

Time: 30min - 2hours per change

Example: Move door 500mm



##### Process:

1. Erase old door (messy)
2. Fill in wall
3. Draw new opening
4. Redraw door
5. Fix dimensions
6. Redraw title block (date)

##### CAD:

Same change: 30 seconds

Commands:

1. MOVE (door)
2. TRIM (wall)
3. UPDATE (dimension automatically)

WINNER: CAD (100x faster edits)

#### 5. STORAGE

---

##### MANUAL:

- Physical sheets (large)
- Requires flat file cabinet
- Risk of:
  - Fire
  - Water damage
  - Tearing
  - Fading



- Pest damage
- Lost/stolen

Storage for 100 A1 drawings:

- Cabinet: ₦150,000
- Space: 2m x 1m x 0.5m
- Weight: 50kg

CAD:

- Digital files (tiny)
- 100 drawings ≈ 100MB
- Stored on:
  - Computer hard drive
  - USB drive (₦5,000)
  - Cloud (Google Drive, etc.)
- Backups easy
- No degradation
- Searchable

WINNER: CAD (Infinitely better)

## 6. SHARING

---

MANUAL:

Options:

a) Original:

- Hand deliver
- Risk of damage/loss

b) Copy (Blueprint service):

- Cost: ₦1,000 per A1 sheet
- Time: Travel + waiting
- Quality: May degrade
- Send by: Courier (₦2,000+)
- Delivery: 1-3 days

Total for 5 copies to 3 locations:

Cost: ₦15,000 + ₦6,000 = ₦21,000

Time: 2-3 days

CAD:

Options:

a) Email:

- Cost: Free (internet data only)
- Time: Seconds
- Quality: Perfect
- Recipients: Unlimited

b) Cloud sharing:

- Google Drive
- Dropbox
- OneDrive
- Instant access
- Version control

Total: £0, Instant delivery

WINNER: CAD (Free + Instant)

## 7. REPETITION

---

MANUAL:

Drawing same symbol 100 times:

Example: Window symbol

●—● × 100

Time per symbol: 2 minutes

Total: 200 minutes = 3.3 hours

Must draw each one individually!

CAD:

Same task:

1. Draw symbol once: 2 minutes
2. COPY command: 30 seconds
3. Or save as BLOCK
4. INSERT 99 times: 1 minute

Total: 3.5 minutes

WINNER: CAD (60x faster)

## 8. 3D VISUALIZATION

---

MANUAL:

Options:

- Isometric drawing (difficult)
- Oblique drawing (time-consuming)
- Multiple 2D views
- Physical model (expensive)

Time for isometric view: 4-6 hours

Skill level: Very high

CAD:

Options:

- Convert 2D to 3D: Minutes
- Rotate any angle: Real-time
- Realistic rendering: Hours
- Virtual walkthrough: Yes

Time: 10 minutes - 2 hours

Skill level: Moderate

Example:



WINNER: CAD (Easy 3D)

## 9. COLLABORATION

---

MANUAL:

Team work:

- One person draws at a time
- Pass sheet person to person
- Risk of damage
- No simultaneous editing
- Hard to track changes

CAD:

Team work:

- Multiple people
- Simultaneous editing (cloud)
- Layer-based division
- Track changes
- Version control
- Comments/markup
- Real-time updates

Example:

Architect draws walls

Engineer adds structure

Electrician adds electrical

ALL AT SAME TIME!

WINNER: CAD (True collaboration)

## 10. COST OVER TIME

---

MANUAL:

Ongoing costs per year:

- Paper: ₦50,000
  - Pens/pencils: ₦20,000
  - Erasers: ₦5,000
  - Replacement tools: ₦10,000
  - Printing copies: ₦100,000
- Total: ₦185,000/year

5-year total: ₦925,000

CAD:

Ongoing costs per year:

- Software subscription: ₦500,000  
(or free: LibreCAD, DraftSight)
- Electricity: ₦50,000
- Computer upgrade: ₦100,000/5yrs

Total: ₦570,000/year (paid)

Or ₦60,000/year (free software)

5-year total: ₦2,850,000 (paid)

Or ₦300,000 (free)

WINNER: Depends on software choice

## 11. SKILL REQUIREMENTS

---

MANUAL:

Must learn:

- Drawing techniques
- Line weights
- Lettering
- Perspective
- Hand-eye coordination
- Takes: 2-4 years to master

CAD:

Must learn:

- Software interface
- Commands
- Layer management
- Computer basics
- Takes: 3-6 months to be productive

WINNER: CAD (Faster learning)

## 12. QUALITY & CONSISTENCY

---

#### MANUAL:

Quality depends on:

- Draughtsman's skill
- Time of day (fatigue)
- Lighting conditions
- Tool condition
- Varies person to person

Consistency: Variable

#### CAD:

Quality depends on:

- Command knowledge
- Settings
- Always perfect lines
- Always consistent
- Same quality everyone

Consistency: Perfect

WINNER: CAD (Consistent quality)

---

#### FINAL VERDICT:

| CATEGORY       | MANUAL | CAD  | WINNER |
|----------------|--------|------|--------|
| Startup Cost   | ★★★    | ★    | Manual |
| Speed          | ★      | ★★★★ | CAD    |
| Accuracy       | ★★     | ★★★★ | CAD    |
| Editing        | ★      | ★★★★ | CAD    |
| Storage        | ★      | ★★★★ | CAD    |
| Sharing        | ★      | ★★★★ | CAD    |
| Repetition     | ★      | ★★★★ | CAD    |
| 3D Visual      | ★      | ★★★★ | CAD    |
| Collaboration  | ★      | ★★★★ | CAD    |
| Long-term Cost | ★★     | ★★   | Tie    |
| Learning Curve | ★★     | ★★★★ | CAD    |
| Quality        | ★★     | ★★★★ | CAD    |

OVERALL WINNER: CAD (10 out of 12 categories)

BUT: Manual drawing skills are still important

for understanding fundamentals!

---

## EVALUATION

1. Name **three main sections** of the CAD interface and explain the function of each.
2. What is a "**Layer**" in CAD, and why is it the most important organizational tool? Give three reasons.
3. Name **four Drawing commands** and **four Modifying commands** in CAD. Briefly state what each does.
4. Explain the difference between the **TRIM** and **EXTEND** commands. When would you use each?
5. What is the **OFFSET** command used for? Give two practical examples in architectural drawing.
6. List **five advantages** of CAD over manual drawing.
7. What is a "**Block**" in CAD, and why is it a time-saver? Give an example.
8. Explain how **layers** can be used to control what appears on a printed drawing.

## ASSIGNMENT

### Part 1: CAD Command Research

1. Watch a **beginner's AutoCAD tutorial** video (15-30 minutes) online.
2. Write down **ten CAD commands** you learned from the video:
  - Command name
  - Keyboard shortcut
  - What it does
  - When to use it

**Example format:**

| COMMAND | SHORTCUT | FUNCTION           | USE            |
|---------|----------|--------------------|----------------|
| LINE    | L        | Draw straight line | Connect points |
| CIRCLE  | C        | Draw circle        | Holes, curves  |

### Part 2: Layer Planning Exercise

3. You are designing a **2-bedroom house**. Plan a complete layer structure:

Create a table showing:

| LAYER NAME | COLOR | LINETYPE   | USE         |
|------------|-------|------------|-------------|
| A-Wall-Ext | Red   | Continuous | Outer walls |

Include layers for:

- External walls
- Internal walls
- Doors
- Windows
- Furniture
- Dimensions
- Text
- Electrical (lights, switches, sockets)
- Plumbing (fixtures)
- Centerlines
- Hidden lines

Minimum: 12 layers with proper names, colors, and purposes.

### Part 3: Command Sequence

4. Write the **step-by-step command sequence** to draw a simple bathroom (2m × 2.5m) in CAD:

#### Example:

STEP 1: Setup  
Command: UNITS  
Set to: Millimeters

STEP 2: Create layers  
Command: LAYER  
New layers: Walls (Red), Doors (Green), etc.

STEP 3: Draw outer walls  
Command: RECTANGLE  
First point: 0,0  
Other corner: @2000,2500

[Continue with all steps...]

Include commands for:

- Room outline
- Wall thickness (using OFFSET)
- Door opening (using TRIM)
- WC symbol
- Basin symbol
- Dimensions

#### **Part 4: Comparison Essay**

5. Write a **short essay** (1-2 pages) comparing manual drawing and CAD based on YOUR experience:

Topics to cover:

- Which do you find easier to learn?
- Which produces better quality?
- Which would you prefer to use professionally?
- Will manual drawing skills still be needed?
- Justify your answers with examples

#### **Part 5: Practical Exercise (If computer available)**

6. If you have access to CAD software (AutoCAD, LibreCAD, DraftSight, etc.):

Draw a **simple floor plan** showing:

- One rectangular room (3m × 4m)
- Wall thickness (225mm using OFFSET)
- One door (900mm with swing)
- One window (1200mm)
- Room label ("BEDROOM")
- Basic dimensions

Save as: YourName\_RoomPlan.dwg (or PDF) Print or screenshot your work.

---



## WEEK 10: COMPUTER-AIDED DESIGN (ADVANCED APPLICATIONS)

### 10.1 ADVANCED CAD TECHNIQUES

Building on basic commands, we now explore powerful productivity features.

---

#### 10.1.1 BLOCKS - THE ULTIMATE TIME-SAVER

##### Diagram 124: Understanding Blocks

---

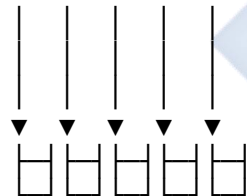
###### WHAT ARE BLOCKS?

A BLOCK is a collection of objects saved as a single named unit that can be inserted repeatedly.

###### CONCEPT:

###### Without Blocks:

Draw chair 50 times



Time: 100 minutes

###### With Blocks:

Draw chair once

Save as block

Insert 50 times



└─ Single unit

Insert, insert, insert...



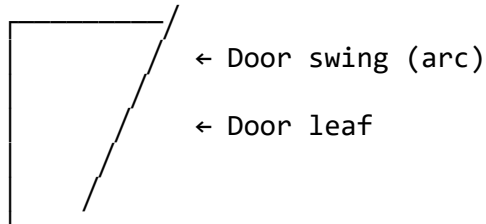
Time: 5 minutes!

---

###### CREATING A BLOCK

Example: Door Symbol

###### STEP 1: Draw the door



STEP 2: Command: BLOCK (or B)

BLOCK DEFINITION DIALOG:

|                                                                                                                            |                         |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Name: DOOR_900                                                                                                             | ← Give descriptive name |
| Base point:<br>[Pick Point] ← Click corner                                                                                 |                         |
| Objects:<br>[Select Objects] ← Select all                                                                                  |                         |
| <input checked="" type="checkbox"/> Retain<br><input type="checkbox"/> Convert to block<br><input type="checkbox"/> Delete | ← Keep original         |

Base point = Insertion point  
(Usually a corner or center)

STEP 3: Click OK  
Block is now saved in drawing!

---

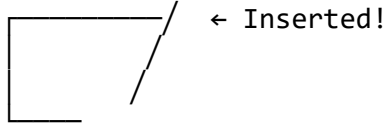
INSERTING A BLOCK

Command: INSERT (or I)

INSERT DIALOG:

|                                                                           |                    |
|---------------------------------------------------------------------------|--------------------|
| Name: [DOOR_900 ▼]                                                        | ← Select block     |
| Insertion point:<br><input checked="" type="checkbox"/> Specify On-screen | ← Click location   |
| Scale:<br>X: 1.0    Y: 1.0                                                | ← Keep 1:1         |
| Rotation:<br>Angle: 0° <input checked="" type="checkbox"/> On-screen      | ← Rotate if needed |

Result:

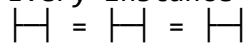


---

## BLOCK ADVANTAGES

### 1. CONSISTENCY

Every instance identical



No variation!

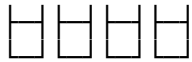
### 2. SPEED

Insert in 2 seconds  
vs Draw in 2 minutes

### 3. EDITING

Edit block definition once:  
ALL instances update automatically!

Before:



After edit:



Edit: BEDIT or REFEDIT

ALL 50 chairs update instantly!

### 4. FILE SIZE

50 separate drawings: LARGE file  
1 block × 50 references: SMALL file  
(Saves disk space)

### 5. ATTRIBUTES

Can add data to blocks:

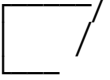
- Door number
- Material
- Cost
- Supplier

Generates automatic schedules!

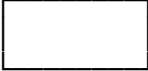
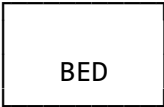
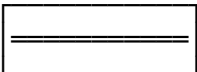
---

## COMMON ARCHITECTURAL BLOCKS

PLUMBING:

|                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DOOR_SINGLE_900<br>                                                                                                                                                                       |
| DOOR_DOUBLE_1800<br>                                                                                                                                                                      |
| WINDOW_SLIDING_1200<br>                                                                                                                                                                   |
| WC_SUITE<br><div><div></div><div>← Cistern</div></div> <div><div></div><div>← Bowl</div></div>           |
| WHB_PEDESTAL<br><div><div></div><div>← Basin</div></div> <div><div></div><div>← Pedestal</div></div> |
| BATH_1700<br>                                                                                                                                                                           |

#### FURNITURE:

|                                                                                                           |
|-----------------------------------------------------------------------------------------------------------|
| CHAIR_PLAN<br>           |
| TABLE_DINING_6SEATER<br> |
| BED_DOUBLE<br>           |
| SOFA_3SEATER<br>         |

#### ELECTRICAL:

|                                                                                                      |
|------------------------------------------------------------------------------------------------------|
| LIGHT_CEILING<br> |
| SWITCH_1WAY<br>S                                                                                     |
| SWITCH_2WAY<br>S <sub>2</sub>                                                                        |
| SOCKET_SINGLE<br> |
| SOCKET_DOUBLE<br> |

---

#### BLOCK LIBRARIES

Professional firms maintain libraries:

## COMPANY BLOCK LIBRARY

### Architectural/

- └ Doors/
  - └ Single\_900.dwg
  - └ Single\_1200.dwg
  - └ Double\_1800.dwg
- └ Windows/
  - └ Sliding\_1200.dwg
  - └ Casement\_900.dwg
- └ Furniture/
  - └ Chairs/
  - └ Tables/
  - └ Beds/
- └ Fixtures/
  - └ WC.dwg
  - └ Basin.dwg
  - └ Bath.dwg

### Electrical/

- └ Lights.dwg
- └ Switches.dwg
- └ Sockets.dwg

### Symbols/

- └ North\_Arrow.dwg
- └ Scale\_Bar.dwg
- └ Title\_Block.dwg

## ONLINE RESOURCES:

- CADBlocksFree.com
- CADdetails.com
- Autodesk Content Library
- Manufacturer websites  
(Kohler, TOTO, etc. provide  
CAD blocks of their products!)

---

## DYNAMIC BLOCKS (Advanced)

Regular Block:



Fixed size

Dynamic Block:



← Stretch!

Adjustable

Dynamic blocks have:

- Grips for stretching
- Flip options
- Multiple configurations
- Visibility states

Example: Door block

- Flip left/right
- Change width (600-1200mm)
- Show/hide frame

ALL FROM ONE BLOCK!

Command: BEDIT (Block Editor)

Add parameters & actions

---

WBLOCK (Write Block)

Save block as separate file:

Command: WBLOCK (or W)

Saves block to:

C:/Blocks/DOOR\_900.dwg

Can now:

- Share with others
- Insert into any drawing
- Build block library

vs BLOCK command:

- Only saved in current drawing
- Not accessible elsewhere

---

### ***10.1.2 ATTRIBUTES - DATA IN BLOCKS***

#### **Diagram 125: Block Attributes**

---

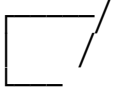
ATTRIBUTES = TEXT DATA ATTACHED TO BLOCKS

Purpose: Extract information automatically

## Example: Door Schedule

### STEP 1: CREATE BLOCK WITH ATTRIBUTES

Drawing door:



Command: ATTDEF (Attribute Definition)

Attribute 1:

|                        |                    |
|------------------------|--------------------|
| Tag: DOOR_NO           | ← Database field   |
| Prompt: "Door number?" | ← Question to user |
| Default: D01           | ← Default value    |

Attribute 2:

|                       |
|-----------------------|
| Tag: DOOR_WIDTH       |
| Prompt: "Width (mm)?" |
| Default: 900          |

Attribute 3:

|                   |
|-------------------|
| Tag: DOOR_TYPE    |
| Prompt: "Type?"   |
| Default: Hardwood |

Now create block with these attributes:

Command: BLOCK

Select: Door + All attributes

### STEP 2: INSERT BLOCK

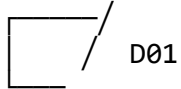
Command: INSERT → DOOR\_900

Dialog appears:

|                     |                   |
|---------------------|-------------------|
| Door number? [D01_] | ← Enter: D01      |
| Width (mm)? [900_]  | ← Enter: 900      |
| Type? [Hardwood_]   | ← Enter: Mahogany |



Result:



Block inserted with data!

### STEP 3: EXTRACT DATA

Command: EATTEXT (Extract Attributes)

OR: DATAEXTRACTION wizard

Creates table/Excel:

#### DOOR SCHEDULE

| MARK | WIDTH | TYPE     | QUANTITY |
|------|-------|----------|----------|
| D01  | 900   | Mahogany | 4        |
| D02  | 1200  | Mahogany | 2        |
| D03  | 900   | Flush    | 6        |
| D04  | 800   | Hardwood | 1        |

Automatic from drawing!  
No manual counting!

---

### PRACTICAL APPLICATIONS

1. DOOR SCHEDULE  
Extract: Mark, Size, Type, Material
2. WINDOW SCHEDULE  
Extract: Mark, Size, Type, Glazing
3. ELECTRICAL SCHEDULE  
Extract: Fixture type, Wattage, Quantity
4. FURNITURE LIST  
Extract: Item, Model, Cost
5. ROOM DATA  
Extract: Name, Area, Finish

## 6. EQUIPMENT TAG

Extract: Equipment ID, Manufacturer, Model

---

### ATTRIBUTE MODES

- ☒ Invisible  
Attribute hidden but data stored
  - ☒ Constant  
Same value for all inserts
  - ☒ Verify  
Confirm value on insertion
  - ☒ Preset  
Use default without prompting
- 

### EDITING ATTRIBUTES

After insertion:

Command: ATTEDIT  
Or double-click block

Change:  
D01 → D02  
900 → 1200  
Mahogany → Teak

Command: BATTMAN (Block Attribute Manager)  
Edit all attributes in drawing at once

---

### ***10.1.3 XREF (EXTERNAL REFERENCES)***

#### **Diagram 126: Working with XREFs**

---

WHAT ARE XREFS?

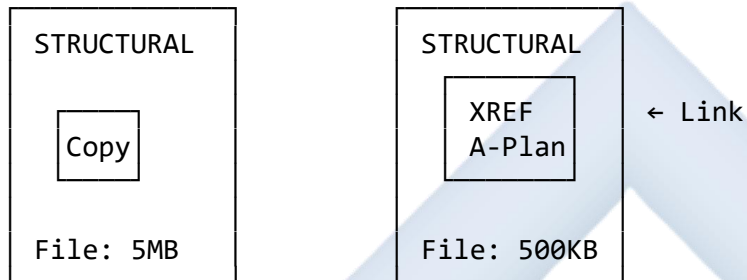
XREF = External Reference  
Link to another drawing file

Like a window to another drawing

#### CONCEPT:

Without XREF:  
Copy architectural  
drawing into  
structural drawing

With XREF:  
Link to  
architectural  
drawing



If architect changes:  
✗ Must copy again  
✗ Out of sync

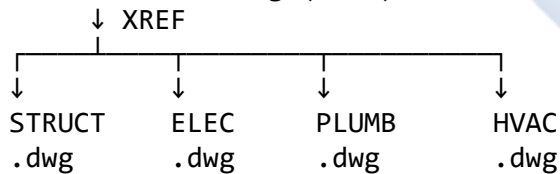
If architect changes:  
✓ Auto-updates!  
✓ Always current

#### WHY USE XREFS?

##### 1. COORDINATION

Multiple disciplines work on same base:

ARCHITECTURAL.dwg (Base)



All reference same base!  
Base updates → All see changes

##### 2. FILE SIZE

Main drawing stays small  
Only reference, not copy

##### 3. TEAM WORK

Architect works on ARCH.dwg  
Engineer works on STRUCT.dwg  
Both see changes in real-time!

#### 4. VERSIONS

Easy to swap versions:

XREF: FloorPlan\_V1.dwg

Change to: FloorPlan\_V2.dwg

---

#### ATTACHING AN XREF

Command: XREF (or XR)

XREF MANAGER:

|                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [Attach] [Detach] [Reload]                                                                                                                                                      |
| Reference File:<br><input checked="" type="checkbox"/> Architectural_Plan.dwg<br><input type="checkbox"/> Electrical_Layout.dwg<br><input type="checkbox"/> Plumbing_Layout.dwg |

Click [Attach]:

1. Browse to file
2. Select insertion point
3. Set scale (usually 1:1)
4. Click OK

XREF appears in current drawing  
But NOT copied - just linked!

---

#### XREF LAYER CONTROL

XREFed drawings have prefix:

XREF File: ARCH.dwg

Layers:

- Walls
- Doors
- Windows

In current drawing appears as:

- ARCH|Walls
- ARCH|Doors
- ARCH|Windows

Can control:

- ☒ Freeze ARCH|Doors (hide doors)
- ☒ Color override (make all gray)
- Lock all XREF layers

---

## XREF STATES

| Status:      |                    |
|--------------|--------------------|
| ✓ Loaded     | - Currently shown  |
| ↻ Reload     | - Refresh changes  |
| ⊗ Unload     | - Hide temporarily |
| ⊙ Detach     | - Remove reference |
| ? Not Found  | - File moved!      |
| ⚠ Unresolved | - Missing file     |

---

## XREF PATH TYPES

### 1. FULL PATH (Absolute)

C:/Projects/Building\_A/Floor\_Plans/Ground.dwg

Problem: If folder moves, XREF breaks!

### 2. RELATIVE PATH

../Floor\_Plans/Ground.dwg

Better: Works if whole project folder moves

### 3. NO PATH

Ground.dwg

Best: Must be in same folder

---

## TYPICAL WORKFLOW

### ARCHITECTURAL FIRM:

1. Architect creates base drawing  
File: A-100\_Ground\_Floor.dwg
2. Sends to consultants:
  - Structural Engineer
  - Electrical Engineer
  - Plumbing Engineer
3. Each consultant:
  - Creates new drawing
  - XREFs A-100\_Ground\_Floor.dwg
  - Adds their discipline work
4. Architect updates base
  - Changes wall location
5. All consultant drawings:
  - Auto-update next opening
  - See new wall location
  - Adjust their work accordingly

---

## XREF vs BLOCK

|           | BLOCK               | XREF            |
|-----------|---------------------|-----------------|
| Copied?   | Yes                 | No              |
| File size | Larger              | Smaller         |
| Updates   | Manual              | Auto            |
| Use for   | Repetitive symbols  | Base drawing    |
| Best for  | Furniture, fixtures | Consult. coord. |

---

## BIND XREF

Convert XREF to permanent part:

Command: XREFBIND

Options:

- Bind - Make permanent copy
- Insert - Merge into drawing

Use when:

- Sending to client
- Final archive
- No more changes expected

After binding:

ARCH|Walls → Walls  
(Becomes regular layer)

---

### 10.1.4 ANNOTATIVE SCALING

#### Diagram 127: Annotative Objects

---

##### THE SCALING PROBLEM

You draw at 1:1 (real size):  
Wall is 5000mm → Draw 5000mm

But printing at different scales:

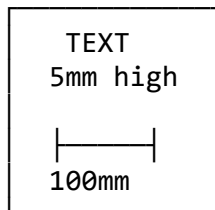
- 1:50 for floor plan
- 1:100 for site plan
- 1:20 for details

Text needs to be:

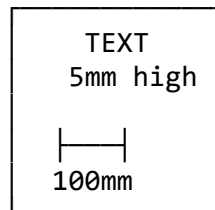
- Readable on paper
- Same printed size
- But different scales!

PROBLEM:

At 1:50 scale:



At 1:100 scale:



Printed text:  
5mm on paper

Printed text:  
5mm on paper

But wall:  
 $100\text{mm} \times 50 = 2\text{mm}$

But wall:  
 $100\text{mm} \times 100 = 1\text{mm}$

Text TOO BIG!

Text HUGE!

## SOLUTION: ANNOTATIVE OBJECTS

---

ANNOTATIVE = Auto-scaling

Text/Dims that adjust to drawing scale

### SETTING UP ANNOTATIVE TEXT

Command: STYLE (Text Style)

Style: Standard  
Font: Arial  
Height: 3mm

☒ Annotative ← CHECK THIS!

Paper text height: 3mm  
(Actual printed size)

Now create text:

Command: TEXT

"BEDROOM"

In Model Space at 1:1:

Text appears HUGE (maybe 300mm!)

But in Viewport at 1:100:

Text appears 3mm on paper ✓

---

## HOW IT WORKS



ANNOTATION SCALE setting:

Status bar: [1:100 ▼]  
↑ Current scale

Change to 1:50:

- Annotative objects resize!
- Still print at 3mm

Change to 1:20:

- Resize again!
- Still print at 3mm

#### MULTI-SCALE SUPPORT

One text, multiple scales:

Object properties:

- ☒ Scale 1:50
- ☒ Scale 1:100
- ☒ Scale 1:20

Text appears correctly at ALL scales!

---

#### ANNOTATIVE DIMENSIONS

Command: DIMSTYLE (or D)

Dimension Style: Standard

Text height: 3mm

Arrow size: 2mm

☒ Annotative

Adjust for: Paper space

Result:

Dimensions always legible  
regardless of drawing scale!

---

## ANNOTATIVE BLOCKS

Symbols (North arrow, etc.)

Command: BLOCK

Name: NORTH\_ARROW

☒ Annotative

Paper size: 20mm

North arrow always 20mm on paper  
whether 1:50 or 1:1000 scale!

---

## SCALE LIST

Set scales used in project:

Command: SCALELISTADD

Add:

- 1:1
- 1:5
- 1:10
- 1:20
- 1:50
- 1:100
- 1:200

Annotative objects use this list

---

## 10.2 SPECIALIZED CAD APPLICATIONS

Different fields use CAD differently.

**Diagram 128: CAD by Discipline**

---

## 1. ARCHITECTURAL CAD (AutoCAD, Revit, ArchiCAD)

### PRIMARY USE:

- Building design
- Space planning
- Documentation

### TYPICAL DRAWINGS:

- Site plans
- Floor plans
- Elevations
- Sections
- Details
- 3D renderings

### SPECIAL FEATURES:

- Wall tools (auto-cleanup)
- Door/window insertion
- Space/area calculation
- Material libraries
- Rendering

### EXAMPLE WORKFLOW:

1. Site analysis
2. Concept sketches
3. Floor plan development
4. 3D massing
5. Detail design
6. Construction docs
7. Rendering/presentation

---

## 2. STRUCTURAL ENGINEERING CAD (AutoCAD, STAAD, SAP)

### PRIMARY USE:

- Structural analysis
- Load calculations
- Foundation design
- Steel/concrete detailing

### TYPICAL DRAWINGS:

- Foundation plans

- Framing plans
- Column schedules
- Beam details
- Reinforcement details
- Connection details

SPECIAL FEATURES:

- Load analysis
- Structural member library
- Rebar detailing
- Connection design
- Bill of materials

INTEGRATION:

XREF architectural plans

Add structural elements on top

---

### 3. MECHANICAL ENGINEERING CAD (SolidWorks, Inventor)

---

PRIMARY USE:

- Product design
- Machine parts
- Assemblies
- Manufacturing drawings

TYPICAL DRAWINGS:

- Part drawings (detailed)
- Assembly drawings
- Exploded views
- BOM (Bill of Materials)
- Machining drawings

SPECIAL FEATURES:

- Parametric modeling
- Constraints
- Motion simulation
- Stress analysis
- GD&T (Geometric tolerancing)

3D FOCUS:

Design in 3D first

2D drawings extracted automatically

---

#### 4. CIVIL ENGINEERING CAD (Civil 3D, 12d)

---

PRIMARY USE:

- Road design
- Land development
- Surveying
- Drainage design

TYPICAL DRAWINGS:

- Survey plans
- Contour maps
- Road profiles
- Drainage layouts
- Earthwork calculations

SPECIAL FEATURES:

- Terrain modeling
- Road alignment
- Pipe networks
- Quantity takeoff
- GPS integration

UNIQUE TOOLS:

- Digital elevation models (DEM)
- Cross-sections
- Volume calculations

---

#### 5. ELECTRICAL CAD (AutoCAD Electrical, EPLAN)

---

PRIMARY USE:

- Circuit design
- Panel layouts
- Wiring diagrams
- Control systems

TYPICAL DRAWINGS:

- Schematic diagrams

- Single-line diagrams
- Panel layouts
- Wiring diagrams
- Cable schedules

SPECIAL FEATURES:

- Symbol libraries (IEEE/IEC)
- Wire numbering
- Component database
- PLC integration
- Automatic reports

---

## 6. PLUMBING/MEP CAD (Revit MEP, AutoCAD MEP)

---

PRIMARY USE:

- Piping layouts
- HVAC design
- Plumbing systems
- Coordination

TYPICAL DRAWINGS:

- Piping isometrics
- HVAC layouts
- Plumbing plans
- Riser diagrams
- Equipment schedules

SPECIAL FEATURES:

- Pipe routing
- Duct sizing
- Clash detection
- Equipment libraries
- Load calculations

---

## 7. LANDSCAPE CAD (AutoCAD, Vectorworks Landmark)

---

PRIMARY USE:

- Site planning

- Garden design
- Irrigation
- Hardscape

#### TYPICAL DRAWINGS:

- Site plans
- Planting plans
- Grading plans
- Irrigation layouts
- Detail drawings

#### SPECIAL FEATURES:

- Plant libraries
- Site analysis
- 3D terrain
- Sunlight analysis
- Material schedules

## 10.3 BUILDING INFORMATION MODELING (BIM)

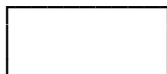
### The Future of CAD

#### Diagram 129: BIM Explained

#### TRADITIONAL CAD vs BIM

#### TRADITIONAL CAD (2D)

Just lines and circles:



← Rectangle (4 lines)

Software sees:

- Line 1: From (0,0) to (5000,0)
- Line 2: From (5000,0) to (5000,3000)
- Line 3: From (5000,3000) to (0,3000)

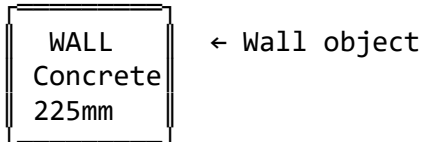
- Line 4: From (0,3000) to (0,0)

NO intelligence!  
Just geometry.

## BIM (Building Information Modeling)

---

Smart objects:



Software knows:

- Type: Exterior wall
- Material: Concrete block
- Thickness: 225mm
- Height: 3000mm
- Thermal properties: R-value 2.5
- Cost: ₦15,000/m<sup>2</sup>
- Fire rating: 2 hours
- Supplier: Dangote
- Weight: 450kg/m<sup>2</sup>

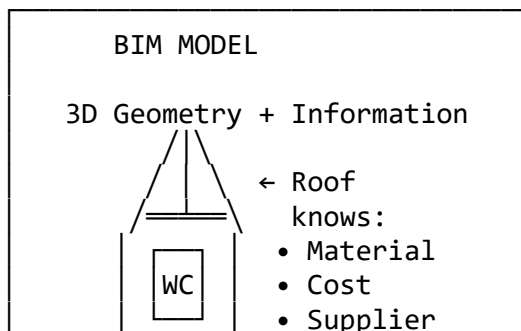
INTELLIGENT object!  
Contains DATA.

---

BIM MODEL = VIRTUAL BUILDING

One 3D model contains:

- Geometry (shape)
- Data (properties)
- Relationships (connections)





|                                      |                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------|
| <div> <div></div> <div></div> </div> | <ul style="list-style-type: none"> <li>• Quantity</li> </ul>                                    |
| ←                                    | Foundation knows:                                                                               |
|                                      | <ul style="list-style-type: none"> <li>• Depth</li> <li>• Concrete</li> <li>• Reinf.</li> </ul> |

## BIM ADVANTAGES

### 1. SINGLE SOURCE OF TRUTH

One model → All drawings

Change wall in 3D:

- Floor plan updates ✓
- Elevation updates ✓
- Section updates ✓
- Schedule updates ✓
- Quantities update ✓

vs CAD:

Change wall:

- Must edit 5+ drawings ✗
- Easy to miss one ✗
- Inconsistencies ✗

### 2. AUTOMATIC SCHEDULES

Count objects automatically:

DOOR SCHEDULE (Auto-generated)

| MARK | SIZE | TYPE  | QTY |
|------|------|-------|-----|
| D01  | 900  | Wood  | 4   |
| D02  | 1200 | Wood  | 2   |
| D03  | 2100 | Steel | 1   |

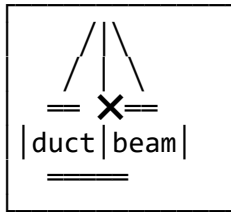
← From model

Add door → Schedule updates!

vs CAD: Manual counting ✗

### 3. CLASH DETECTION

Find conflicts before building:



← BEAM through DUCT!  
Detected!

Software finds:

- Pipe through beam
- Duct through wall
- Light in ceiling void

Fix in model → Avoid site problem!

#### 4. QUANTITY TAKEOFF

Automatic material quantities:

Click wall → Software shows:

- Concrete blocks: 1,245 No
- Mortar: 2.3m<sup>3</sup>
- Plaster: 87m<sup>2</sup>
- Paint: 15 liters
- Cost: ₦428,750

Real-time cost estimation!

#### 5. 4D (TIME)

Add construction schedule:

Week 1: Foundation —■  
Week 2: Walls —■  
Week 3: Roof —■  
Week 4: Finishes —■

Visualize construction sequence  
Before building starts!

#### 6. 5D (COST)

Link to pricing database:

Wall (225mm concrete):

- Blocks: ₦250 each
- Labor: ₦2,000/m<sup>2</sup>
- Total wall cost: ₦487,500

Change material → Cost updates!  
Budget tracking real-time

---

## BIM SOFTWARE

### ARCHITECTURAL :

Autodesk Revit (Most common)

- Parametric
- Families (objects)
- Multi-discipline

ArchiCAD (Graphisoft)

- User-friendly
- Mac compatible
- Good visualization

Vectorworks Architect

- Hybrid 2D/3D
- Good for small firms

### STRUCTURAL :

Revit Structure

Tekla Structures

- Steel detailing
- Connection design

### MEP:

Revit MEP

- Piping
- Ductwork
- Electrical

---

## BIM WORKFLOW

### TRADITIONAL :

Architect draws

### BIM:

Architect builds

- |                     |                    |
|---------------------|--------------------|
| → Paper to engineer | 3D BIM model       |
| → Engineer redraws  | → Share model      |
| → QS counts         | → Engineer opens   |
| → Contractor plans  | → QS extracts      |
|                     | → Contractor views |

|              |               |
|--------------|---------------|
| Sequential ✗ | Concurrent ✓  |
| Slow ✗       | Fast ✓        |
| Errors ✗     | Coordinated ✓ |

---

## BIM CHALLENGES IN NIGERIA

### 1. HIGH SOFTWARE COST

Revit: \$2,700/year (₦4,000,000)

Solution:

- Student versions (free)
- Small firm bundles
- Rent, not buy

### 2. COMPUTER REQUIREMENTS

Need powerful PC:

- i7 processor
- 16GB+ RAM
- Dedicated graphics
- Cost: ₦800,000+

### 3. TRAINING

Steep learning curve  
3-6 months to proficiency

Solution:

- Online courses
- YouTube tutorials
- Practice projects

### 4. INDUSTRY ADOPTION

Many Nigerian firms still use 2D

But changing!

- Large projects require BIM
- Government mandates (coming)
- Competitive advantage

### 5. INTERNET/POWER

Large files need:

- Fast internet (cloud)
- Stable power (UPS)

---

## THE FUTURE

BIM is becoming standard globally

Nigeria following:

- NIOB (Nigerian Institute of Building) promoting BIM
- Major projects requiring BIM
- Universities teaching BIM
- Young professionals adopting

YOUR GENERATION:

Start learning BIM now!  
Will be essential skill

---

## EVALUATION

1. What is a **Block** in CAD? Explain three advantages of using blocks over drawing individual objects.
2. What is the difference between **BLOCK** and **WBLOCK** commands?
3. What are **Attributes** in CAD blocks? Give a practical example of how they are used.
4. Explain what an **XREF** (External Reference) is and why it's useful for team collaboration.
5. What is **Annotative** text in CAD? Solve this problem: If you have annotative text set to 3mm paper height, how tall will it appear in your drawing at scale 1:100?
6. Compare **Traditional CAD** and **BIM**. List four key differences.
7. What does "**Clash Detection**" mean in BIM? Why is it important?
8. Name three BIM software programs and identify which discipline each is best suited for.

---

## ASSIGNMENT

### Part 1: Block Creation Exercise

1. If you have access to CAD software:

Create THREE blocks for architectural use: a) **Door Block** (900mm single door with swing) b) **Window Block** (1200mm sliding window) c) **WC Block** (toilet with cistern and bowl)

For each block:

- Draw accurately
- Set appropriate base point
- Save with descriptive name
- Test by inserting multiple times

Save your work as: YourName\_Blocks.dwg

If no computer access:

- Draw the three symbols on paper
- Indicate the base point (insertion point) for each
- Label clearly

## Part 2: Block Library Planning

2. Plan a complete **Block Library** for a small architectural firm:

Create a folder structure showing:

```
CAD_BLOCKS/  
├── Architectural/  
│   ├── Doors/  
│   ├── Windows/  
│   ├── Fixtures/  
│   └── Furniture/  
├── Electrical/  
└── Plumbing/
```

Under each category, list minimum 5 blocks needed with:

- Block name
- Description
- Typical size
- Where used

### Example:

Block: DOOR\_SINGLE\_900

Description: Single leaf door, 900mm wide

Size: 900mm x 50mm

Use: Interior rooms, bathrooms

## Part 3: BIM Research

3. Research one **BIM software** (Revit, ArchiCAD, or Tekla):

Write a report (1-2 pages) covering:

- Software name and company
- Primary use (architectural/structural/MEP)
- Key features (minimum 5)
- System requirements (computer specs needed)
- Cost (subscription or perpetual license)
- Learning resources available
- Adoption in Nigeria (if you can find info)
- Would you recommend it? Why?

#### Part 4: Comparison Table

4. Create a detailed **comparison table** between:
- Manual Drawing
  - 2D CAD (AutoCAD)
  - BIM (Revit)

Compare on these criteria:

- Initial cost
- Learning time
- Speed of production
- Ease of editing
- Collaboration ability
- File storage
- 3D capability
- Quantity takeoff
- Industry adoption
- Future prospects

Use this format:

| CRITERIA     | MANUAL         | CAD                | BIM           |
|--------------|----------------|--------------------|---------------|
| Initial cost | Low<br>₦50,000 | Medium<br>₦500,000 | High<br>₦1.5M |

#### Part 5: Future Technology Essay

5. Write a short essay (1 page) titled:

**"How will CAD and BIM change the Nigerian construction industry in the next 10 years?"**

Consider:

- Current state of adoption
  - Benefits for Nigerian context
  - Challenges specific to Nigeria
  - Role of young professionals
  - Your predictions
  - How you will prepare yourself
- 





# WEEK 11: REVISION AND EXAMINATION PREPARATION

## 11.1 COMPREHENSIVE TERM REVIEW

This week focuses on consolidating all knowledge from Weeks 1-10.

---

### TOPIC-BY-TOPIC REVISION GUIDE

#### Diagram 130: Term Overview Map

##### SECOND TERM JSS2 TECHNICAL DRAWING

| COMPUTER & CAD FUNDAMENTALS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Week 1: Computer Hardware</p> <ul style="list-style-type: none"><li>• Input devices (keyboard, mouse, scanner)</li><li>• Output devices (monitor, printer, plotter)</li><li>• Storage devices (HDD, SSD, USB, cloud)</li><li>• Processing unit (CPU, RAM, motherboard)</li></ul> <p>Week 1-2: CAD Introduction</p> <ul style="list-style-type: none"><li>• Definition &amp; advantages</li><li>• CAD software types</li><li>• Applications in various fields</li></ul>                                              |
| ARCHITECTURAL ELEMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>Week 2: Arches</p> <ul style="list-style-type: none"><li>• Arch terminology (keystone, voussoir, etc.)</li><li>• Arch types (semicircular, pointed, etc.)</li><li>• Structural function</li><li>• Drawing arches</li></ul> <p>Week 3: Roofs</p> <ul style="list-style-type: none"><li>• Roof types (gable, hip, flat, etc.)</li><li>• Roof terminology (ridge, rafter, purlin)</li><li>• Roof pitch calculation</li><li>• Truss types (King Post, Queen Post, etc.)</li><li>• Drawing trusses with labels</li></ul> |
| SYMBOLS & CONVENTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Week 4: Material & Symbol Representation                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

- Material symbols (concrete, wood, steel, etc.)
- Plumbing symbols (WC, basin, bath, etc.)
- Electrical symbols (lights, switches, etc.)
- Mechanical symbols (AC, fans, etc.)
- When and how to use each

#### PROFESSIONAL PRACTICE

##### Week 6: Blueprint Reading & Services

- Blueprint history and meaning
- Reading drawings (architect, QS, etc.)
- Reproduction services business
- Merchandising opportunities

##### Week 8: Construction Applications

- How different professionals use drawings
- Architect's use
- Quantity Surveyor's use (take-off)
- Contractor's use (site work)
- Electrician's use
- Plumber's use

#### CAD PRACTICAL SKILLS

##### Week 9: Basic CAD

- CAD interface (menu, toolbar, canvas, etc.)
- Drawing commands (line, circle, arc, etc.)
- Modify commands (trim, offset, copy, etc.)
- Layer management
- Drawing workflow

##### Week 10: Advanced CAD

- Blocks and attributes
- External references (XREF)
- Annotative scaling
- BIM introduction
- CAD vs BIM comparison

## 11.2 KEY CONCEPTS SUMMARY

### Diagram 131: Essential Knowledge Checklist

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---

## COMPUTER HARDWARE & CAD

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- ☐ Can name and explain 5 input devices
- ☐ Can name and explain 5 output devices
- ☐ Understand difference between HDD and SSD
- ☐ Know what RAM does
- ☐ Can explain what a plotter is
- ☐ Know what CAD stands for
- ☐ Can list 5 advantages of CAD over manual drawing
- ☐ Can name 3 CAD software programs
- ☐ Understand CAD applications in different fields

---

---

## ARCHES

---

---

- ☐ Can define: Keystone, Voussoir, Springer
- ☐ Can define: Intrados, Extrados, Soffit
- ☐ Can define: Span, Rise, Springing Line
- ☐ Can explain how arches work structurally
- ☐ Can draw and label 3 arch types:
  - Semicircular (Roman)
  - Pointed (Gothic)
  - Segmental
- ☐ Know which cultures used which arch types
- ☐ Can construct semicircular arch with compass
- ☐ Can label all parts on arch drawing

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## ROOFS

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- ☐ Can name 5 roof types:
  - Flat, Gable, Hip, Mansard, Shed
- ☐ Can define: Pitch, Span, Rise, Run
- ☐ Can calculate pitch from span and rise
- ☐ Know pitch formula:  $\text{Rise} \div \text{Run}$
- ☐ Can name all parts of a roof:
  - Ridge, Rafter, Purlin, Tie Beam
  - Wall Plate, King Post, Strut
- ☐ Can draw King Post Truss with labels
- ☐ Can draw Queen Post Truss with labels

- ☐ Know which members are in:
  - Compression (C)
  - Tension (T)
- ☐ Understand advantages of different roof types

---

## SYMBOLS & CONVENTIONS

---

### MATERIAL SYMBOLS (Sections):

- ☐ Can draw hatching for:
  - Concrete (triangles + dots)
  - Wood (flowing grain lines)
  - Steel (evenly spaced 45° lines)
  - Brickwork (45° lines with breaks)
  - Earth (random dots and lines)
  - Hardcore (interlocking triangles)
  - Glass (thin widely-spaced lines)
  - Sand (fine dots)

### PLUMBING SYMBOLS (Plans):

- ☐ Can draw:
  - Water Closet (WC)
  - Wash Hand Basin (WHB)
  - Bath Tub
  - Shower
  - Kitchen Sink
  - Floor Drain

### ELECTRICAL SYMBOLS (Plans):

- ☐ Can draw:
  - Ceiling Light (⊗ )
  - Wall Light
  - Switch (S)
  - Two-way Switch (S<sub>2</sub>)
  - Socket Outlet (□ )
  - Ceiling Fan
  - Distribution Board
- ☐ Know how to show switch-to-light connection

### MECHANICAL SYMBOLS:

- ☐ Can draw:
  - Air Conditioning Unit
  - Extractor Fan
  - Radiator

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## BLUEPRINT READING

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- ☐ Know what "blueprint" means historically
- ☐ Know modern prints are black on white
- ☐ Can explain what plotters do
- ☐ Know how to read a title block
- ☐ Understand drawing scales (1:50, 1:100, etc.)
- ☐ Can identify information different professionals need:
  - Architect: Design, aesthetics, function
  - QS: Quantities, materials, costs
  - Contractor: Construction sequence
  - Electrician: Circuit layouts
  - Plumber: Pipe routes, fixtures

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## QUANTITY SURVEYING BASICS

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- ☐ Know what "taking off" means
- ☐ Can calculate wall area from plan and section
- ☐ Can deduct openings (doors, windows)
- ☐ Can calculate number of blocks needed
- ☐ Understand the concept of Bill of Materials (BOM)
- ☐ Know to add wastage percentage (5-10%)

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## CAD INTERFACE

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- ☐ Can identify CAD interface parts:
  - Menu bar
  - Ribbon/Toolbar
  - Drawing area/Canvas
  - Command line
  - Status bar
  - Layers panel
  - Properties panel
- ☐ Understand coordinate system (0,0 origin)

## CAD DRAWING COMMANDS

---

☐ Know these commands and shortcuts:

- LINE (L)
- CIRCLE (C)
- ARC (A)
- RECTANGLE (REC)
- POLYGON (POL)
- ELLIPSE (EL)
- POLYLINE (PL)
- HATCH (H)
- TEXT (T) / MTEXT (MT)

---

## CAD MODIFY COMMANDS

---

☐ Know these commands and shortcuts:

- MOVE (M)
- COPY (CO)
- ROTATE (RO)
- SCALE (SC)
- TRIM (TR) ★ Very important!
- EXTEND (EX)
- OFFSET (O) ★ Very important!
- MIRROR (MI)
- ARRAY (AR)
- FILLET (F)
- CHAMFER (CHA)
- ERASE (E)
- EXPLODE (X)

---

## CAD LAYERS

---

- ☐ Understand what layers are (transparent sheets)
- ☐ Know why layers are important:
  - Organization
  - Visibility control
  - Color coding
  - Editing control
  - Printing control
- ☐ Can create layers with proper names

- ☐ Know layer properties:
    - Name
    - Color
    - Linetype
    - On/Off
    - Lock/Unlock
  - ☐ Understand layer naming conventions:
    - A- = Architectural
    - E- = Electrical
    - P- = Plumbing
- 
- 

## ADVANCED CAD

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### BLOCKS:

- ☐ Know what blocks are (reusable groups)
- ☐ Understand advantages:
  - Speed
  - Consistency
  - Easy editing (edit once, all update)
  - Smaller file size
- ☐ Know difference between BLOCK and WBLOCK
- ☐ Know common block examples (doors, furniture, etc.)

### ATTRIBUTES:

- ☐ Know attributes add data to blocks
- ☐ Understand use for automatic schedules

### XREF:

- ☐ Know XREF = External Reference
- ☐ Understand it links (not copies) drawings
- ☐ Know advantages for team collaboration
- ☐ Know XREFs auto-update when source changes

### BIM:

- ☐ Know BIM = Building Information Modeling
- ☐ Understand BIM objects are "intelligent"
- ☐ Know key advantages:
  - Single source of truth
  - Automatic schedules
  - Clash detection
  - Quantity takeoff
  - 4D (time) and 5D (cost)
- ☐ Can name BIM software (Revit, ArchiCAD, etc.)
- ☐ Understand difference between CAD and BIM

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## MANUAL vs CAD

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- ☐ Can compare manual drawing and CAD on:
  - Startup cost
  - Speed
  - Accuracy
  - Editing ease
  - Storage
  - Sharing
  - Collaboration
  - Long-term cost
- ☐ Know CAD advantages outweigh disadvantages
- ☐ Understand manual skills still important for fundamentals

---

## 11.3 PRACTICE QUESTIONS

### Diagram 132: Exam-Style Questions

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#### SECTION A: MULTIPLE CHOICE (40 Questions)

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##### COMPUTER HARDWARE & CAD (Questions 1-8)

1. Which of the following is an input device?
  - A. Monitor
  - B. Speaker
  - C. Scanner ✓
  - D. Projector
2. CAD stands for:
  - A. Computer Automatic Design
  - B. Computer-Aided Design ✓
  - C. Computer Architecture Drawing
  - D. Creative Art Design
3. Which storage device has the fastest data access?
  - A. HDD
  - B. SSD ✓
  - C. CD-ROM
  - D. Floppy disk
4. The main advantage of CAD over manual drawing is:



- A. Cheaper startup cost
  - B. Easier to learn
  - C. Faster editing ✓
  - D. No electricity needed
5. A plotter is used for:
- A. Scanning drawings
  - B. Printing large-format drawings ✓
  - C. Storing drawings
  - D. Editing drawings
6. RAM stands for:
- A. Read Access Memory
  - B. Random Access Memory ✓
  - C. Rapid Application Memory
  - D. Remote Access Module
7. Which CAD software is commonly used for architecture?
- A. MATLAB
  - B. AutoCAD ✓
  - C. Photoshop
  - D. Excel
8. The "brain" of the computer is the:
- A. Monitor
  - B. Hard drive
  - C. CPU ✓
  - D. RAM

ARCHES (Questions 9-14)

9. The topmost central stone of an arch is called:
- A. Springer
  - B. Voussoir
  - C. Keystone ✓
  - D. Crown
10. The inner curve of an arch is called:
- A. Extrados
  - B. Intrados ✓
  - C. Soffit
  - D. Haunch
11. Which arch type was used in Roman architecture?
- A. Pointed arch
  - B. Semicircular arch ✓
  - C. Tudor arch
  - D. Ogee arch

12. The horizontal distance across an arch is the:

- A. Rise
- B. Run
- C. Span ✓
- D. Pitch

13. Arches work by converting vertical loads to:

- A. Upward thrust
- B. Horizontal thrust ✓
- C. Circular motion
- D. No change in direction

14. Gothic arches are characterized by:

- A. Semicircular shape
- B. Flat top
- C. Pointed shape ✓
- D. Horseshoe shape

#### ROOFS (Questions 15-22)

15. The sloping member of a roof is called:

- A. Purlin
- B. Rafter ✓
- C. Ridge
- D. Joist

16. Roof pitch is calculated as:

- A.  $\text{Span} \div \text{Rise}$
- B.  $\text{Run} \div \text{Rise}$
- C.  $\text{Rise} \div \text{Run}$  ✓
- D.  $\text{Rise} \times \text{Span}$

17. A roof with four sloping sides is a:

- A. Gable roof
- B. Hip roof ✓
- C. Flat roof
- D. Shed roof

18. In a King Post Truss, the king post is in:

- A. Tension
- B. Compression ✓
- C. Neither
- D. Both

19. The horizontal member at the top of a roof is:

- A. Wall plate
- B. Tie beam

- C. Ridge ✓
- D. Purlin

20. Which roof type is simplest to construct?

- A. Mansard
- B. Hip
- C. Gable ✓
- D. Gambrel

21. The tie beam in a truss is in:

- A. Compression
- B. Tension ✓
- C. Shear
- D. Torsion

22. A roof with two different slopes on each side is:

- A. Gable
- B. Hip
- C. Mansard ✓
- D. Flat

SYMBOLS (Questions 23-30)

23. In section, concrete is shown with:

- A. Horizontal lines
- B. Triangles and dots ✓
- C. Wavy lines
- D. Solid black

24. The symbol ⊗ represents:

- A. Socket outlet
- B. Ceiling light ✓
- C. Floor drain
- D. Window

25. In plan, a WC is shown as:

- A. Circle only
- B. Rectangle only
- C. Cistern and bowl ✓
- D. Letter "W"

26. A two-way switch is labeled:

- A. S
- B. S<sub>2</sub> ✓
- C. 2S
- D. SW

27. Wood in section is shown with:

- A. Evenly spaced 45° lines
  - B. Flowing grain lines ✓
  - C. Dots
  - D. Solid fill
28. The symbol  $\parallel\square$  represents:
- A. Light switch
  - B. Socket outlet ✓
  - C. Door
  - D. Window
29. Steel in section is shown with:
- A. Random dots
  - B. Triangles
  - C. Evenly spaced thin lines at 45° ✓
  - D. Wavy lines
30. A dotted line on a floor plan between S and  $\otimes$  shows:
- A. Hidden line
  - B. Control connection ✓
  - C. Water pipe
  - D. Section line

#### BLUEPRINT & PROFESSIONALS (Questions 31-35)

31. Traditionally, blueprints had:
- A. Black lines on white paper
  - B. White lines on blue paper ✓
  - C. Blue lines on white paper
  - D. Red lines on blue paper
32. A Quantity Surveyor uses drawings to:
- A. Design buildings
  - B. Calculate material quantities ✓
  - C. Install electrical systems
  - D. Construct foundations
33. The process of measuring quantities from drawings is:
- A. Scaling
  - B. Taking off ✓
  - C. Plotting
  - D. Drafting
34. Which professional needs to know fixture locations?
- A. Architect
  - B. Structural engineer
  - C. Plumber ✓
  - D. Quantity surveyor

35. Modern large-format drawings are printed using:
- A. Laser printer
  - B. Inkjet printer
  - C. Plotter ✓
  - D. Photocopier

CAD COMMANDS (Questions 36-40)

36. The shortcut for LINE command is:
- A. LI
  - B. L ✓
  - C. LN
  - D. LINE
37. Which command creates parallel lines?
- A. COPY
  - B. ARRAY
  - C. OFFSET ✓
  - D. MIRROR
38. To cut a line at intersection, use:
- A. TRIM ✓
  - B. EXTEND
  - C. BREAK
  - D. FILLET
39. Layers in CAD are used for:
- A. 3D modeling
  - B. Color printing
  - C. Organization and control ✓
  - D. File compression
40. A block in CAD is:
- A. A solid 3D object
  - B. A locked layer
  - C. A group of objects saved as one unit ✓
  - D. A command shortcut

---

ANSWERS TO MULTIPLE CHOICE

---

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 1.C  | 2.B  | 3.B  | 4.C  | 5.B  | 6.B  | 7.B  | 8.C  |
| 9.C  | 10.B | 11.B | 12.C | 13.B | 14.C | 15.B | 16.C |
| 17.B | 18.B | 19.C | 20.C | 21.B | 22.C | 23.B | 24.B |

25.C 26.B 27.B 28.B 29.C 30.B 31.B 32.B  
33.B 34.C 35.C 36.B 37.C 38.A 39.C 40.C

---

## SECTION B: THEORY (Answer 5 out of 7)

---

### QUESTION 1 (Arches)

- a) Define the following arch terms:
- i) Keystone (2 marks)
  - ii) Voussoir (2 marks)
  - iii) Intrados (2 marks)
- b) Explain how arches work structurally. (4 marks)
- c) Name TWO types of arches used in:
- i) Roman architecture (2 marks)
  - ii) Gothic architecture (2 marks)

Total: 14 marks

### QUESTION 2 (Roofs)

- a) Define the following roof terms:
- i) Pitch (2 marks)
  - ii) Span (2 marks)
  - iii) Rise (2 marks)
- b) A building has:
- Span = 8 metres
  - Rise = 2.5 metres
- Calculate:
- i) The run (2 marks)
  - ii) The pitch (express as ratio) (3 marks)
- c) List THREE members found in a King Post Truss. (3 marks)

Total: 14 marks

### QUESTION 3 (Symbols)

- a) Draw the sectional symbol (hatching pattern) for:

- i) Concrete (2 marks)
- ii) Wood (2 marks)
- iii) Steel (2 marks)
- iv) Brickwork (2 marks)

- b) Draw the floor plan symbol for:
- i) Ceiling light (1 mark)
  - ii) One-way switch (1 mark)
  - iii) Socket outlet (2 marks)
  - iv) Water closet (WC) (2 marks)

Total: 14 marks

#### QUESTION 4 (CAD)

- a) What does CAD stand for? (1 mark)
- b) List FIVE advantages of CAD over manual drawing. (5 marks)
- c) Name THREE popular CAD software programs. (3 marks)
- d) Explain what a "Layer" is in CAD and give TWO reasons why layers are important. (5 marks)

Total: 14 marks

#### QUESTION 5 (Professionals)

- a) Explain how a Quantity Surveyor uses building drawings differently from an Architect. (4 marks)
- b) Describe the process a contractor follows to set out a building on site using floor plans. (6 marks)
- c) What specific information does an electrician extract from a floor plan? Give FOUR examples. (4 marks)

Total: 14 marks

#### QUESTION 6 (Blueprint Services)

- a) What is a "blueprint" and why are technical drawings still called blueprints today? (4 marks)
- b) List FIVE equipment/items needed to start a blueprint

reproduction service. (5 marks)

- c) Describe THREE challenges of running a blueprint business in Nigeria and suggest solutions. (5 marks)

Total: 14 marks

#### QUESTION 7 (Advanced CAD)

- a) What is a "Block" in CAD? Explain THREE advantages of using blocks. (6 marks)
- b) What does XREF stand for? Explain how XREFs help with team collaboration. (4 marks)
- c) What is BIM? Give TWO differences between traditional CAD and BIM. (4 marks)

Total: 14 marks

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#### SECTION C: PRACTICAL DRAWING (Compulsory)

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---

#### QUESTION 8 (30 marks)

Draw a simple BATHROOM FLOOR PLAN with the following specifications:

Dimensions: 2.5m × 2.0m

Scale: 1:50

Wall thickness: 225mm

Include:

- Walls (using proper line weight)
- One door (900mm with swing) - 5 marks
- Water Closet (WC) with proper symbol - 5 marks
- Wash Hand Basin (WHB) with proper symbol - 5 marks
- Bath tub (1700mm) with proper symbol - 5 marks
- One ceiling light with switch - 3 marks
- Floor drain - 2 marks

Add:

- Overall dimensions - 2 marks
- Title: "BATHROOM FLOOR PLAN" - 1 mark



- Scale notation: "SCALE 1:50" - 1 mark
- Your name and date - 1 mark

Use proper drawing techniques:

- Neat work
- Correct symbols
- Appropriate line weights
- Good lettering

Total: 30 marks

#### QUESTION 9 (20 marks)

Draw a KING POST TRUSS with the following specifications:

Span: 8m (draw at scale 1:100 = 80mm on paper)

Rise: 2.5m (draw at 25mm on paper)

Label ALL members:

- Ridge
- Rafters (both sides)
- King Post
- Struts (both sides)
- Tie Beam
- Wall Plates

Indicate forces:

- Mark compression members with (C)
- Mark tension members with (T)

Add dimensions for:

- Total span
- Rise height

Include:

- Title block
- Scale notation
- Your name

Total: 20 marks

---

## 11.4 STUDY TECHNIQUES

### Diagram 133: Effective Revision Strategies

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## STUDY PLAN - 7 DAYS TO EXAM

---

### DAY 1: HARDWARE & CAD BASICS

---

#### Morning (2 hours):

- Review computer hardware
- List all input/output devices
- Practice definitions

#### Afternoon (2 hours):

- Review CAD advantages
- Study CAD software types
- Make flashcards

#### Practice:

- Draw and label 5 input devices
- Draw and label 5 output devices
- Write 10 advantages of CAD

### DAY 2: ARCHES

---

#### Morning (2 hours):

- Review all arch terminology
- Study arch types
- Understand structural function

#### Afternoon (2 hours):

- Practice drawing arches
- Label all parts
- Make comparison table of arch types

#### Practice:

- Draw 3 arch types with labels
- Write definitions from memory
- Explain how arches work

### DAY 3: ROOFS

---

#### Morning (2 hours):

- Review roof types
- Study roof terminology
- Practice pitch calculations

#### Afternoon (2 hours):

- Draw King Post Truss
- Draw Queen Post Truss
- Label all members
- Mark forces (C/T)

Practice:

- 5 pitch calculation problems
- Draw trusses from memory
- List all roof members

#### DAY 4: SYMBOLS

---

Morning (2 hours):

- Practice material symbols
- Draw all hatching patterns
- Material symbols quiz

Afternoon (2 hours):

- Practice floor plan symbols
- Plumbing symbols (WC, Basin, etc.)
- Electrical symbols (Lights, Switches)

Practice:

- Draw 10 material symbols
- Draw 10 plumbing symbols
- Draw 10 electrical symbols
- Do from MEMORY!

#### DAY 5: PROFESSIONALS & BLUEPRINT

---

Morning (2 hours):

- Review how each professional uses drawings
- Study QS quantity takeoff process
- Blueprint reading techniques

Afternoon (2 hours):

- Practice quantity calculations
- Study blueprint service business
- Review construction applications

Practice:

- Calculate quantities for sample building
- Explain each professional's role
- Draw simple site set-out

#### DAY 6: CAD COMMANDS & LAYERS

---

Morning (2 hours):

- Review all drawing commands
- Study modify commands
- Practice command shortcuts

Afternoon (2 hours):

- Layer management
- Block concepts
- XREF basics

Practice:

- Make command shortcut list
- Design layer structure for project
- Explain blocks and XREFs

## DAY 7: PRACTICE & REVIEW

---

Morning (2 hours):

- Take practice test (multiple choice)
- Time yourself
- Mark and review mistakes

Afternoon (2 hours):

- Practice drawing questions
- Bathroom floor plan
- King Post Truss
- Timed practice

Evening (1 hour):

- Quick review of weak areas
- Read through notes
- Get materials ready
- Early to bed!

---

## MEMORY TECHNIQUES

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### 1. ACRONYMS

---

Arch parts (bottom to top):

"Some Intruders Have Key Extrados"

Springer - Intrados - Haunch - Keystone - Extrados

### 2. MNEMONICS

---

Drawing commands:  
"Lazy Cats Always Rest Peacefully"  
Line - Circle - Arc - Rectangle - Polyline

### 3. VISUAL MEMORY

---

Draw diagrams repeatedly  
Label from memory  
Color code different parts

### 4. TEACH SOMEONE

---

Explain concepts to friend/parent  
If you can teach it, you know it!

### 5. PRACTICE TESTS

---

Take multiple practice tests  
Time yourself  
Simulate exam conditions

### 6. FLASHCARDS

---

Front: Term/Question  
Back: Definition/Answer

Example:  
Front: "What is a Keystone?"  
Back: "The topmost central voussoir  
that locks the arch in place"

### 7. STUDY GROUPS

---

Study with classmates  
Quiz each other  
Share notes  
Explain concepts to each other

---

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## EXAM DAY TIPS

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### BEFORE EXAM:

- ✓ Good night's sleep (8 hours)
- ✓ Healthy breakfast
- ✓ Arrive 30 minutes early

- ✓ Bring all materials:
  - Pencils (2H, HB, 2B)
  - Eraser
  - Ruler
  - Set squares
  - Compass
  - Protractor
  - Calculator
  - Sharpener

#### DURING EXAM:

- ✓ Read instructions carefully
- ✓ Read ALL questions first
- ✓ Answer easiest questions first
- ✓ Manage your time:
  - Multiple choice: 30 minutes
  - Theory: 45 minutes
  - Drawing: 45 minutes
- ✓ Check your work
- ✓ Don't leave blanks - attempt everything!

#### DRAWING TIPS:

- ✓ Use 2H for construction lines (light)
- ✓ Use HB for final lines (dark)
- ✓ Draw faintly first, then darken
- ✓ Use instruments for all lines
- ✓ Label clearly and neatly
- ✓ Add title block
- ✓ Check dimensions

#### THEORY TIPS:

- ✓ Answer the question asked
- ✓ Use technical terminology
- ✓ Be specific - not vague
- ✓ Show calculations
- ✓ Draw diagrams if helpful
- ✓ Write neatly

---

#### COMMON MISTAKES TO AVOID

---

- X Not reading questions carefully
- X Forgetting to label drawings
- X Missing title blocks
- X Wrong scale usage
- X Incorrect symbols

- X Poor time management
- X Messy work
- X Not showing calculations
- X Leaving questions blank
- X Rushing through easy questions

---

## 11.5 REVISION EXERCISES

### Exercise Set 1: Quick Fire Questions

Answer these in 30 seconds each:

1. What does RAM stand for?
2. Name 3 input devices
3. What is the keystone of an arch?
4. Formula for roof pitch?
5. Symbol for ceiling light?
6. What is a plotter?
7. Concrete sectional symbol?
8. CAD command to trim?
9. What is a layer?
10. What does BIM stand for?

[Answers at end of document]

### Exercise Set 2: Calculations

1. Building span = 12m, rise = 3.5m  
Calculate: a) Run b) Pitch
2. Wall dimensions: 25m long × 3m high  
Wall thickness: 225mm (9-inch blocks)  
Openings: 3 doors (2m × 1m each)  
4 windows (1.5m × 1.2m each)

Calculate:

- a) Gross wall area
  - b) Opening area
  - c) Net wall area
  - d) Number of blocks (10 per m<sup>2</sup>)
  - e) Add 5% wastage
3. Roof area = 180m<sup>2</sup>  
Aluminum sheets = 3m × 0.75m each  
Calculate sheets needed (add 10% overlap)

[Show all working]

### Exercise Set 3: Drawing Practice

Practice these drawings (20 minutes each):

1. Draw semicircular arch (span 100mm)  
Label: All parts
2. Draw King Post Truss (span 80mm, rise 25mm)  
Label: All members  
Show: Forces (C/T)
3. Draw bathroom floor plan (2m × 2.5m at 1:50)  
Include: WC, Basin, Bath, Door, Light, Switch
4. Draw these symbols (actual size):
  - 5 material symbols in section
  - 5 plumbing symbols
  - 5 electrical symbols

---

## 11.6 FINAL EXAM PREPARATION CHECKLIST

### Diagram 134: Are You Ready?

---

#### KNOWLEDGE CHECKLIST

---

Rate yourself: 1 (weak) to 5 (strong)

| TOPIC                                    | RATING |
|------------------------------------------|--------|
| Computer Hardware                        | [ ]    |
| CAD Fundamentals                         | [ ]    |
| Arches (terminology, types, drawing)     | [ ]    |
| Roofs (types, pitch, trusses)            | [ ]    |
| Material Symbols                         | [ ]    |
| Plumbing Symbols                         | [ ]    |
| Electrical Symbols                       | [ ]    |
| Blueprint Reading                        | [ ]    |
| Professional Roles (QS, Contractor, etc) | [ ]    |
| Quantity Takeoff                         | [ ]    |
| CAD Interface                            | [ ]    |
| CAD Drawing Commands                     | [ ]    |
| CAD Modify Commands                      | [ ]    |
| CAD Layers                               | [ ]    |
| Blocks & XREFs                           | [ ]    |
| BIM Basics                               | [ ]    |



Focus extra study on topics rated 1-3!

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---

## SKILLS CHECKLIST

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---

Can you do these WITHOUT HELP?

- ☐ Draw and label a semicircular arch
- ☐ Draw and label a King Post Truss
- ☐ Draw bathroom floor plan with symbols
- ☐ Calculate roof pitch from dimensions
- ☐ Calculate wall quantities (blocks needed)
- ☐ Draw 10 material section symbols
- ☐ Draw 10 plumbing plan symbols
- ☐ Draw 10 electrical plan symbols
- ☐ List all CAD drawing commands with shortcuts
- ☐ List all CAD modify commands with shortcuts
- ☐ Explain layers and their importance
- ☐ Compare manual drawing vs CAD vs BIM
- ☐ Explain how different professionals use drawings

---

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## MATERIALS CHECKLIST

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Do you have everything?

### DRAWING INSTRUMENTS:

- ☐ Drawing board or A3/A2 paper
- ☐ T-square or parallel ruler
- ☐ Set squares (45° and 30°/60°)
- ☐ Compass (with pencil attachment)
- ☐ Divider
- ☐ Protractor
- ☐ Scale ruler (1:50, 1:100, etc.)
- ☐ French curves (optional)

### DRAWING MATERIALS:

- ☐ Pencils: 2H (construction), HB (final), 2B (lettering)
- ☐ Eraser (soft plastic)
- ☐ Eraser shield
- ☐ Sharpener
- ☐ Masking tape

☐ Drawing paper/sheets

**OTHER:**

- ☐ Calculator (for pitch/quantity calculations)
  - ☐ Pen (for writing theory)
  - ☐ Spare pencils (backup!)
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**MENTAL PREPARATION**

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- ☐ Reviewed all notes
  - ☐ Completed practice questions
  - ☐ Practiced timed drawings
  - ☐ Studied with classmates
  - ☐ Asked teacher about unclear topics
  - ☐ Got good sleep before exam
  - ☐ Ate proper breakfast
  - ☐ Feeling confident!
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**FINAL REMINDERS**

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- ★ Read ALL questions before starting
- ★ Answer EASIEST questions first
- ★ Manage your TIME carefully
- ★ Use proper DRAWING TECHNIQUES
- ★ LABEL everything clearly
- ★ Add TITLE BLOCKS to all drawings
- ★ SHOW ALL CALCULATIONS
- ★ CHECK your work before submitting
- ★ Don't leave ANY question blank
- ★ BELIEVE in yourself - You've prepared well!

GOOD LUCK!

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**ANSWERS TO QUICK FIRE QUESTIONS**

1. Random Access Memory
2. Keyboard, Mouse, Scanner (any 3 of: trackpad, microphone, webcam, stylus, joystick)
3. The topmost central voussoir that locks the arch

4. Pitch = Rise ÷ Run
  5. ⊗ (Circle with X)
  6. Large-format printer for technical drawings
  7. Triangles and dots mixed (aggregate pattern)
  8. TRIM (shortcut: TR)
  9. Transparent organizational sheet for grouping objects
  10. Building Information Modeling
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## EVALUATION

### Final Self-Assessment Questions:

1. Can you confidently explain the difference between **RAM** and **storage**?
  2. Can you draw and label **five different arch types** from memory?
  3. Can you calculate **roof pitch** given span and rise?
  4. Can you draw **all material section symbols** correctly?
  5. Can you draw a **complete bathroom floor plan** with all correct symbols in 30 minutes?
  6. Can you list **ten CAD commands** with their shortcuts and functions?
  7. Can you explain what **layers** are and why they're the most important CAD feature?
  8. Can you calculate **how many blocks** are needed for a wall given its dimensions?
  9. Can you explain the difference between **CAD** and **BIM**?
  10. Are you ready for the exam? **Yes or No?**
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## ASSIGNMENT

### FINAL EXAM PREPARATION PORTFOLIO

Create a comprehensive **study portfolio** containing:

#### 1. Summary Sheets (5 pages minimum)

- One page for each major topic
- Key definitions
- Important diagrams
- Formulas and calculations
- Common mistakes to avoid

#### 2. Symbol Reference Sheet (1 page)

- All material symbols
- All plumbing symbols
- All electrical symbols
- Drawn neatly and labeled

### **3. Practice Drawings (3 minimum)**

- Semicircular arch with labels
- King Post Truss with labels
- Bathroom floor plan with symbols

### **4. Command Reference (1 page)**

- List of 20 CAD commands
- Shortcuts
- Brief description of each
- When to use

### **5. Self-Made Practice Test**

- Create 20 multiple choice questions
- Create 3 theory questions
- Create 2 drawing questions
- Include answer key
- Exchange with classmate and test each other

### **6. Reflection Essay (1 page) Write about:**

- What you found most difficult this term
- What you enjoyed most
- How you've improved
- What you'll work on next term
- Your goals for using CAD in the future

**Submit this portfolio the day before the exam for bonus marks!**

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